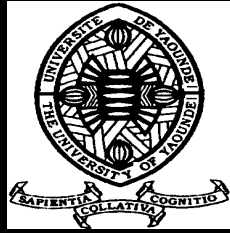


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ÉCOLE NATIONALE SUPÉRIEURE
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DÉPARTEMENT DE GÉNIE
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THE UNIVERSITY OF YAOUNDE I

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MALARIA PREDICTION BASED ON ENVIRONMENTAL DATA COLLECTED THROUGH INTERNET OF THINGS (IoT) SENSORS

MASTER'S THESIS

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Submitted in Partial Fulfilment of the Requirements for the:

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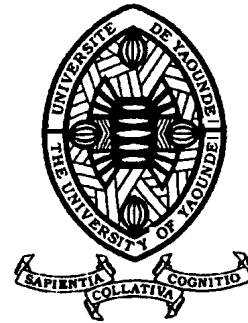
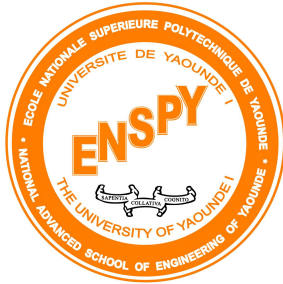
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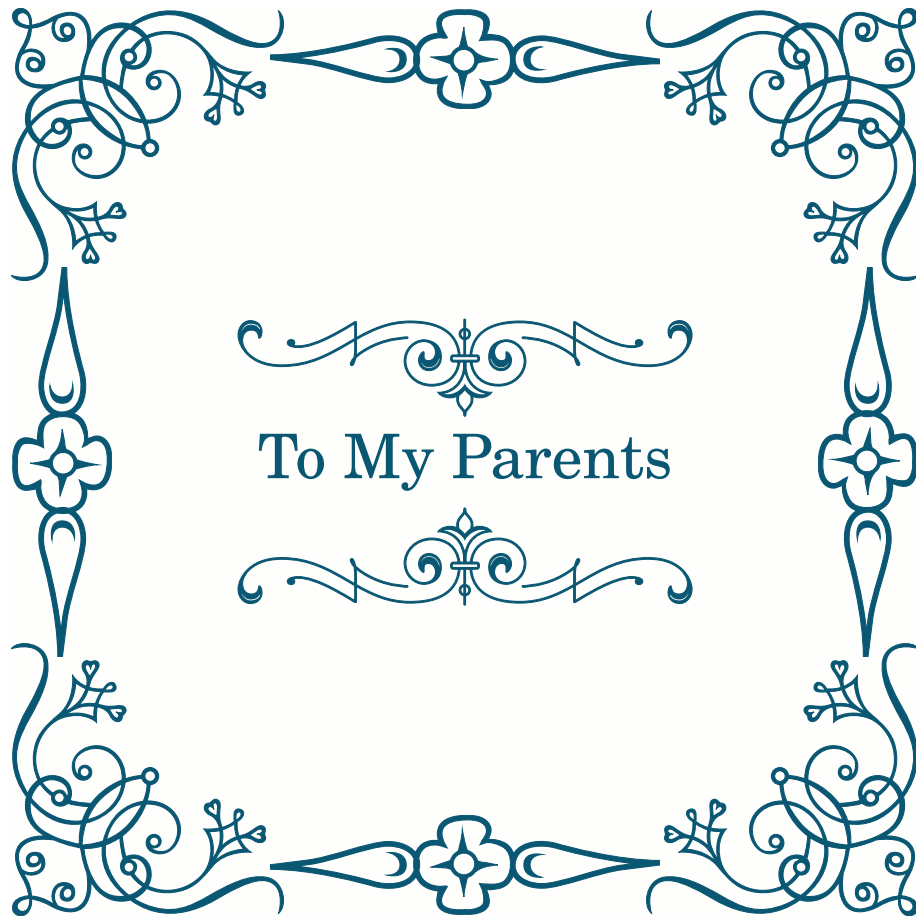
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DEDICATION



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ABBREVIATIONS

WHO	World Health Organisation
PNLP	Programme National de Lutte contre le Paludisme (National Malaria Control Program, Cameroon)
SSS	Stratégie Sectorielle de la Santé (Health Sector Strategy, Cameroon)
HBHI	High Burden to High Impact
IRS	Indoor Residual Spraying
ITN	Insecticide-Treated Net
XGBoost	eXtreme Gradient Boosting
LightGBM	Light Gradient Boosting Machine
RF	Random Forest
SVM	Support Vector Machine
LSTM	Long Short-Term Memory
SHAP	SHapley Additive exPlanations
AUC	Area Under the (ROC) Curve
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
IoT	Internet of Things
LoRa	Long Range (radio technology)
MQTT	Message Queuing Telemetry Transport
API	Application Programming Interface
CSV	Comma-Separated Values
IPCC	Intergovernmental Panel on Climate Change
NDVI	Normalised Difference Vegetation Index
NDWI	Normalised Difference Water Index
RCP	Representative Concentration Pathway

GLOSSARY

Incidence rate : The number of malaria cases in a district-month normalised by the district population and expressed per one thousand inhabitants, comparable across districts of different sizes and the quantity on which the risk class is defined.

Risk class : A binary label, high or low, assigned to each district-month according to whether its incidence rate exceeds a threshold set at the seventy-fifth percentile of the incidence distribution within each region.

Reanalysis : A historical climate product reconstructing past meteorological conditions by combining observations with a numerical weather model, used here as a proxy for the sensor measurements over the 2019–2022 training period.

Deployment substrate : The operational infrastructure on which a trained model runs in service, here the IoT sensor network, which supplies live measurements at inference but contributed no training data.

Proxy : A measurable quantity used in place of another to which it stands in a known relationship. The reanalysis variables serve as a proxy for the sensor measurements, being the same physical quantities from a different source.

Feature engineering : The construction of new input variables from the raw data to expose structure to the learning algorithm, here the seasonal encoding, the geographic descriptors, and the lagged climate and case-history variables.

Lag feature : A variable formed from the value of another variable at an earlier time step, capturing the delayed effect of environmental change or recent transmission on present malaria risk.

Gradient-boosted trees : An ensemble method building a model as a sequence of decision trees, each correcting its predecessors' errors; XGBoost and LightGBM are the implementations used here.

Temporal leakage : The optimistic bias arising when a model is evaluated on time periods overlapping its training data, avoided here through time-aware validation.

Time-aware cross-validation : A validation scheme whose folds respect chronological order, so the model is trained on past months and evaluated on subsequent ones, giving an honest estimate of forecasting performance.

Leave-one-region-out : A spatial validation scheme training on all regions but one and evaluating on the held-out region, in turn for each, measuring prediction of an area absent from training.



Spatial non-transferability : The finding that a model accurate on observed districts fails on entirely unseen regions because it anchors on geographic identity; the principal limitation of the system.

Recursive forecasting : A procedure feeding a model's own predictions back as the lagged inputs for subsequent predictions, enabling multi-step forecasting when the true recent case counts are unavailable at inference.

Bootstrap model : The model variant trained without case-history features, which initiates a recursive forecast from environmental and spatial inputs alone; its score is a persistence-free estimate of the environmental signal.

Oracle : An idealized evaluation supplying a model with true values it would not possess in operation, here the real case lags, giving an upper bound against which the recursive score is compared.

SHAP attribution : A feature-attribution method assigning each input its signed contribution to a prediction, the contributions summing exactly to the difference between the prediction and a baseline, used to explain and corroborate the drivers.

Ingestion layer : The component transforming raw environmental data, from the reanalysis service or the live sensor stream, into the single canonical schema the models consume.

ABSTRACT

Malaria remains a leading public-health burden in Cameroon, where passive surveillance documents outbreaks only after they occur. This thesis designs, implements, and evaluates an explainable system that predicts district-level malaria risk from environmental conditions; the task is framed as a binary classification of elevated monthly transmission, with a case-count regression as cross-check. The system is intended to run on an Internet-of-Things (IoT) sensor network deployed in 2026; no sensor data therefore trains or validates the models. The models are trained on district data for 2019–2022 that pair confirmed cases from the Programme National de Lutte contre le Paludisme (PNLP) with temperature, humidity, pressure, and precipitation from the Open-Meteo reanalysis, the quantities the sensors measure. Gradient-boosted tree ensembles (XGBoost) were evaluated under four validation regimes separating in-sample interpolation, temporal and spatial generalisation, and forecasting over an unseen period. Under time-aware cross-validation, the engineered seasonal, geographic, and lagged-climate features raised the classification $F1$ from 0.466 with raw climate alone to 0.666, yet a persistence model ($R^2 = 0.892$) outperformed the environment-only regressor (0.836). Leave-one-region-out evaluation exposed weak spatial transfer (mean $F1$ of 0.372), while the two-model Recursive Chain made the case-history model deployable without real-time case reporting ($R^2 = 0.875$ against 0.927 for the oracle). On unseen 2023–2025 data, the models over-predicted incidence at Foumban by 28 percent, but ranked malaria burden across 194 districts at a Spearman correlation of 0.859. SHAP analysis identified geography, transmission history, and lagged climate as the dominant drivers. The system is deployed in a Presentation Layer that runs the Recursive Chain within a FastAPI service, explains each prediction with SHAP, and displays district risk on a cartographic dashboard, while an autonomous MLOps pipeline schedules the monthly runs.

Keywords: malaria prediction; Internet of Things; XGBoost; SHAP; temporal and spatial cross-validation.

RÉSUMÉ

Le paludisme demeure un fardeau majeur de santé publique au Cameroun, où la surveillance passive ne documente les flambées qu'après leur survenue. Ce mémoire conçoit, met en œuvre et évalue un système explicable qui prédit le risque de paludisme à l'échelle du district à partir des conditions environnementales ; la tâche est formulée comme une classification binaire d'une transmission mensuelle élevée, avec une régression sur le nombre de cas comme vérification croisée. Le système est destiné à fonctionner sur un réseau de capteurs de l'Internet des objets déployé en 2026 ; aucune donnée de capteur n'entraîne ni ne valide donc les modèles. Ceux-ci sont entraînés sur des données de district de 2019–2022 qui associent les cas confirmés du Programme National de Lutte contre le Paludisme (PNLP) à la température, à l'humidité, à la pression et aux précipitations de la réanalyse Open-Meteo, grandeurs mesurées par les capteurs. Des ensembles d'arbres à gradient boosté (XGBoost) ont été évalués selon quatre régimes de validation séparant l'interpolation en échantillon, les généralisations temporelle et spatiale, et la prévision sur une période non vue. Sous validation temporelle, les variables saisonnières, géographiques et climatiques retardées portent le $F1$ de la classification de 0,466 avec le seul climat brut à 0,666, mais un modèle de persistance ($R^2 = 0,892$) surpasse le régresseur fondé sur le seul environnement (0,836). La validation par exclusion d'une région révèle un faible transfert spatial ($F1$ moyen de 0,372), tandis que la chaîne récursive à deux modèles rend le modèle à historique de cas déployable sans déclaration de cas en temps réel ($R^2 = 0,875$ contre 0,927 pour l'oracle). Sur des données de 2023–2025 non vues, les modèles surestiment l'incidence de Fouban de 28 %, mais classent le fardeau de 194 districts avec une corrélation de Spearman de 0,859. L'analyse SHAP désigne la géographie, l'historique de transmission et le climat retardé comme les facteurs dominants. Le système est déployé dans une couche de présentation qui exécute la chaîne récursive au sein d'une API FastAPI, explique chaque prédiction par SHAP et affiche le risque par district sur un tableau de bord cartographique, tandis qu'un pipeline MLOps autonome planifie les exécutions mensuelles.

Mots-clés : prédiction du paludisme ; Internet des objets ; XGBoost ; SHAP ; validation croisée temporelle et spatiale.

GENERAL INTRODUCTION

GENERAL CONTEXT

Despite decades of global investment and scientific progress, malaria remains one of the most devastating infectious diseases affecting humanity. According to the World Health Organisation (WHO), an estimated 282 million malaria cases were recorded globally in 2024, resulting in approximately 610,000 deaths. The WHO African Region bears the overwhelming majority of this burden, accounting for 94% of global cases and 95% of global deaths, with children under five representing more than 75% of all malaria-related fatalities on the continent [1]. Equally concerning is the reversal of global progress since 2015: the 8.5% increase in incidence recorded by 2024 has placed the 2030 targets of the WHO Global Technical Strategy well beyond reach [1].

Within this global context, Cameroon occupies a particularly critical position. The country ranks seventh globally in estimated malaria case count, with approximately 7.586 million cases in 2024, representing 2.7% of global cases, and belongs to the eleven High Burden to High Impact (HBHI) countries collectively responsible for 64% of global cases and 68% of global deaths [1]. At the national level, malaria accounts for approximately 46% of hospital admissions and 21% of infant deaths among children aged zero to five years [2]. Cameroon's trajectory since 2015 reveals a concerning duality: while malaria mortality has declined by approximately 25–70%, reflecting improvements in case management, incidence has remained largely stagnant. Bridging this gap requires a structural shift from reactive case documentation toward predictive, environmentally-informed early-warning systems capable of anticipating transmission risk before outbreaks emerge.

The transmission dynamics of malaria are fundamentally linked to environmental conditions. Temperature, relative humidity, and rainfall directly influence the breeding, survival, and biting activity of *Anopheles* mosquitoes, the primary vectors of *Plasmodium falciparum*. Previous studies have demonstrated that transmission intensity peaks at moderate temperatures, near 25 °C, and declines as temperatures rise further, an optimum substantially lower than earlier models had predicted, and that transmission is further favoured by relative humidity above 60%, while rainfall intensity determines the formation of larval breeding habitats [3]. ecological landscape, spanning equatorial forests, highland savannas, and Sudano-Sahelian zones, makes this environmental dependency particularly significant, as transmission intensity and seasonality vary substantially across its health districts [2].

Despite this well-established environmental epidemiology, existing health surveillance systems in Cameroon remain predominantly passive and retrospective. The *Stratégie Sectorielle de la Santé (SSS) 2016–2027* explicitly identifies weaknesses in the national electronic disease surveillance infrastructure, noting that routine health data are insufficient to support timely and evidence-based decision-making [2]. Health facility reports are typically delayed and spatially coarse, thereby limiting their ability to capture the fine-grained environmental variability that most strongly predicts imminent transmission risk.



TECHNOLOGICAL CONTEXT

The application of machine learning to malaria outbreak prediction constitutes a recognised and growing subdiscipline of computational epidemiology. Model families in active use include classical ensemble methods, namely Random Forest (RF), Support Vector Machines (SVM), and gradient-boosted variants such as XGBoost and LightGBM as well as sequential deep learning architectures, notably Long Short-Term Memory (LSTM) networks, all sharing the premise that epidemiological incidence can be modelled as a function of environmental, climatic, and demographic covariates.

A converging body of evidence from sub-Saharan African contexts supports this premise while delimiting its scope. Dennis Muriithi et al. [4] demonstrated, on national survey data from Kenya, that Random Forest achieved a classification accuracy of 97.33% and an AUC of 98.3%, with SHAP (SHapley Additive Explanations) feature importance identifying the presence of *Plasmodium falciparum* as the single most important predictor, ahead of the clinical symptom variables and of any environmental covariate [4, 5]. Almamy Youssouf Ly et al. [6] found that XGBoost achieved the lowest forecasting error (RMSE: 12.74%, $R^2 = 0.90$) across thirteen health districts in Senegal within a rigorous time-series cross-validation framework, and further applied SHAP to produce operationally interpretable feature attributions [6]. At continental scale, Nkiruka et al. [7] demonstrated across six sub-Saharan African countries that precipitation, temperature, and surface radiation constitute significant predictors of malaria outbreaks, while Elliot Mbunge et al. [8] showed that Random Forest achieved 87% precision in ward-level malaria prediction in Zimbabwe using remote sensing variables, with outputs directly linked to the operational deployment of indoor residual spraying teams.

The Internet of Things (IoT) has emerged concurrently as a practical infrastructure for continuous, automated environmental data acquisition at spatial and temporal resolutions unattainable through manual or satellite-based methods. In practice, IoT architectures integrate heterogeneous sensor modules, namely, temperature, relative humidity, atmospheric pressure, and rainfall, into low-cost microcontroller-based nodes capable of autonomous operation in remote environments [9, 10]. LoRaWAN communication provides substantially lower power consumption and greater transmission range than Wi-Fi, making it the preferred protocol for rural or off-grid deployments [11, 9]. By publishing sensor readings to cloud brokers via MQTT over TLS, these systems ensure data security and enable direct integration with machine learning pipelines [11]. Crucially, whereas existing predictive studies have relied predominantly on monthly or annual averages derived from reanalysis or satellite products, IoT sensor networks enable continuous, real-time collection of high-resolution ground-level meteorological data, providing the granular, localised inputs that predictive models require for accurate sub-district risk estimation [12].

A further dimension that such systems must address is model interpretability. Machine learning models, and deep learning architectures in particular, are inherently opaque: their predictions provide no direct explanation of which environmental factors drove a given risk estimate [13]. In public health contexts, this opacity constitutes a significant barrier to operational adoption, as district-level decision-makers require not only accurate predictions but transparent, actionable explanations to justify resource allocation [14]. SHAP, grounded in cooperative game theory and demonstrated to produce explanations more consistent with human intuition than alternative methods [15], has emerged as the standard post-hoc interpretability framework in this domain, with applications in African malaria prediction already established [4, 6].

A review of the existing literature reveals that no study builds a malaria prediction pipeline whose feature space is defined in advance by what a ground-level sensor network can supply at the moment of



inference, validates it under temporal and spatial generalisation as well as on unseen years, and delivers it as a running service. This research addresses that gap in the context of Cameroon, the seventh highest-burden malaria country globally, and a country whose national health strategy has explicitly identified the weakness of its electronic surveillance infrastructure as a structural priority [2, 1].

PROBLEM STATEMENT

Malaria surveillance in Cameroon operates principally through passive, facility-based reporting: health centres submit aggregated case counts on a weekly or monthly basis to district health information systems, from which national estimates are compiled. This architecture is inherently retrospective: it documents outbreaks after they have occurred rather than anticipating them. Routine surveillance reporting delays, well-documented across sub-Saharan African health systems [2], render this infrastructure structurally incapable of informing timely preventive action.

Compounding this temporal limitation is a data granularity failure. Malaria transmission is driven by localised environmental conditions that vary significantly at sub-district scale and change within days. Existing machine learning-based prediction approaches have relied predominantly on satellite-derived or reanalysis datasets that provide monthly averages at coarse spatial resolutions, smoothing precisely the fine-grained variability that most strongly predicts imminent transmission risk [8]. The antecedent environmental signal that precedes a malaria outbreak by two to four weeks is therefore poorly captured by the data inputs on which current predictive systems depend.

A third dimension concerns the usability of model outputs. Even where machine learning models achieve strong predictive performance, their outputs remain largely inaccessible to the public health practitioners who must act on them. Without interpretable, feature-level explanations, prediction outputs cannot be integrated into operational decision-making with the confidence required to justify resource mobilisation [13, 14]. The cumulative consequence of these three failures (reactive surveillance, coarse environmental inputs, and opaque model outputs) is a persistent gap between the predictive potential of available technologies and their actual deployment in Cameroon's malaria control infrastructure.

Central Research Question

How can real-time climate observations collected from a ground-level environmental sensor network be leveraged, at deployment, to accurately and interpretably predict district-level malaria risk in Cameroon, and to support proactive public health decision-making under operational deployment constraints?

RESEARCH QUESTIONS

The central research question is investigated through four specific sub-questions:

Environmental sensor data acquisition and integration: Can climate observations from the ground-level IoT sensor network deployed in Cameroon (temperature, humidity, pressure, and rainfall) be ingested and preprocessed into a form schema-compatible with models trained on historical reanal-



ysis data, so that they serve as valid deployment-time input to a district-level malaria prediction pipeline?

Predictive features and deployability: Which feature families best predict district-level malaria risk, and how does performance change when the model is restricted to deployment-realistic environmental features?

Model interpretability: To what extent do SHAP-based feature attributions, applied to the final selected model, identify environmental drivers consistent with established epidemiological theory regarding *Anopheles* breeding and *Plasmodium* transmission dynamics?

Operational integration: Can the resulting end-to-end pipeline be deployed within a compatible architecture and produce district-level risk alerts with sufficient lead time to support practical public health decision-making in Cameroon?

HYPOTHESES

The hypotheses of this study address the predictive and interpretability dimensions of the proposed malaria prediction framework, corresponding to the sub-questions on predictive features and on model interpretability. The remaining sub-questions are addressed through the design, implementation, and experimental validation of the proposed system.

- ❖ **Contextual and lagged features:** Incorporating geographic context, seasonal structure, and lagged climate variables improves district-level malaria prediction compared with using instantaneous climate variables alone, under both classification and regression formulations.
- ❖ **Epidemiological coherence of the attributions:** SHAP-based feature attributions of the selected prediction model identify environmental drivers and lagged effects that are consistent with established epidemiological knowledge of *Anopheles* breeding and *Plasmodium falciparum* transmission dynamics.

RESEARCH OBJECTIVES

General Objective

The general objective of this research is to design, implement, and evaluate an integrated IoT-driven and machine learning-based malaria prediction system, enhanced with explainable artificial intelligence methods, capable of supporting proactive district-level public health decision-making in Cameroon through accurate, interpretable, and operationally deployable risk predictions.

Specific Objectives

End-to-end deployment: To develop and deploy an end-to-end malaria risk prediction pipeline that integrates ground-level environmental measurements from the deployed IoT sensor network [16], transforms them into the model input schema, performs district-level risk inference, and presents the resulting alerts through a cartographic dashboard.



Predictive modelling and validation: To construct a district-level dataset by integrating historical malaria case data with climate variables corresponding to those measured by the IoT sensors, identify relevant seasonal and lagged relationships, and develop and evaluate machine learning models for malaria risk classification and incidence regression under temporal, spatial, and unseen-year validation.

Model explainability: To apply feature-importance and SHAP-based analyses to the selected model in order to identify the environmental and temporal factors driving district-level malaria risk predictions and assess the consistency of these drivers across classification and regression formulations.

THESIS STRUCTURE

The remainder of this thesis is organised into three chapters that build sequentially upon one another:

Chapter 1 General Concepts and State of the Art:

A structured synthesis of the literature covering malaria epidemiology and environmental determinants, machine learning and ensemble methods, IoT-based environmental monitoring, and explainable artificial intelligence in public health. This chapter *defines* the conceptual foundations used throughout the thesis, including supervised learning, gradient boosting, ensemble methods, and the Shapley-value basis of SHAP, and concludes with the gap analysis motivating this research.

Chapter 2 Materials and Methodology:

Building on Chapter 1, this chapter describes how the proposed approach is *applied* to the problem. It defines the contribution relative to the IoT sensor network, constructs the integrated district-level dataset from historical reanalysis climate and PNL case records, and specifies the feature-engineering pipeline, model configuration and training, four-regime validation protocol, recursive forecasting procedure, and SHAP-based attribution approach.

Chapter 3 Implementation, Results and Discussion:

The integrated dataset is constructed and analysed, with exploratory analyses characterising temporal patterns, lag relationships, and ecological heterogeneity. The modelling pipeline is evaluated across model families, feature regimes, and validation regimes, including external tests on unseen 2023–2025 records at district and national scale. SHAP attributions are interpreted against epidemiological theory, and the end-to-end pipeline, including IoT ingestion and the cartographic dashboard, is demonstrated on the operational deployment window. A closing discussion explains the results, verifies the hypotheses, compares the work with the state of the art, and states its limitations. This chapter fulfils the three specific objectives and tests the two hypotheses.

The thesis closes with a **Conclusion and Perspectives** chapter that synthesises the principal findings, recalls the main limitations, and outlines directions for future work.

GENERAL CONCEPTS AND STATE OF THE ART

T*his chapter establishes the scientific and technical foundations of this research along three complementary axes. The first axis surveys the epidemiological, environmental, and technological concepts that constitute the theoretical substrate of the proposed system. The second axis reviews the state of the art in machine learning-based malaria prediction, IoT-based environmental monitoring, and explainable artificial intelligence in public health, with particular emphasis on sub-Saharan African (SSA) contexts. The third axis synthesises this body of evidence into a structured gap analysis that motivates and positions the original contributions of this thesis.*



1.1 General Concepts

1.1.1 Malaria: Epidemiology and Environmental Determinants

This subsection establishes the epidemiological and biological foundations of this research. It introduces the core concepts of epidemiology as applied to infectious disease surveillance, surveys the aetiology, lifecycle, and clinical dimensions of malaria, and examines the mechanistic relationships between environmental and climatic conditions and malaria transmission intensity, relationships that motivate the IoT-based environmental monitoring and machine learning prediction approach developed in subsequent chapters.

1.1.1.1 Introduction to Epidemiology

Epidemiology constitutes the scientific foundation upon which contemporary public health practice is built. The term derives from the Greek words *epi* (upon), *demos* (people), and *logos* (study), and is formally defined as *the study of the distribution and determinants of health-related states or events in specified populations, and the application of this study to the control of health problems* [17]. This definition captures two inseparable dimensions of the discipline: the analytical, concerned with measuring and explaining disease patterns, and the operational, concerned with translating that understanding into interventions. It is this dual character that makes epidemiology indispensable to the design of predictive public health systems of the kind proposed in this work.

Distribution across Person, Place, and Time: A fundamental task of epidemiology is to describe the *distribution* of disease within populations, that is, to characterise how health events vary according to who is affected, where they occur, and when they occur [18]. These three axes, conventionally termed *person*, *place*, and *time*, constitute the descriptive framework of the discipline. Understanding variation along the temporal axis is of particular relevance to this research: malaria incidence in sub-Saharan Africa is not randomly distributed in time but follows seasonal cycles tightly coupled to climatic fluctuations (rainfall patterns, temperature regimes, and humidity levels) that govern the breeding dynamics of *Anopheles* mosquitoes [7]. Characterising this temporal structure is therefore a prerequisite for any forecasting system intended to anticipate rather than merely document outbreak risk.

Determinants as Risk and Protective Factors: Beyond description, epidemiology seeks to identify the *determinants* of disease, namely the biological, environmental, social, and behavioural factors that systematically increase or reduce disease risk [19]. In the context of malaria, these determinants operate along multiple causal pathways. Environmental risk factors, including elevated ambient temperature, high relative humidity, and cumulative rainfall, create conditions favourable to mosquito survival, reproduction, and biting activity; conversely, prolonged drought or extreme flooding may temporarily suppress vector populations by reducing or destroying larval breeding habitats [7]. Protective factors encompass both individual-level interventions, such as insecticide-treated nets (ITNs) and chemoprophylaxis, and population-level measures, such as indoor residual spraying (IRS) and environmental management. The identification of modifiable determinants is precisely what makes predictive modelling actionable: a model capable of quantifying the contribution of each environmental covariate to predicted risk directly informs the prioritisation of preventive interventions [8].



Key Epidemiological Concepts: Three terms are central to the framing of this research [18]. An **epidemic** refers to the occurrence of a disease at a frequency that substantially exceeds what is normally expected within a defined population and period. A **pandemic** is an epidemic that extends across national boundaries to affect multiple countries or continents a designation applied to malaria historically and relevant to the scale of the modelling problem. An **endemic** disease is one whose transmission is sustained at a relatively constant level within a given population or geographic area without the need for external importation; this is the predominant epidemiological condition in Cameroon, where malaria transmission is quasi-permanent in the equatorial zones of the Centre, South, and Littoral regions, and highly seasonal in the Sudano-Sahelian North [2]. This endemic-with-seasonal-peaks profile is directly relevant to the design of the prediction system developed in this work, which must capture both the baseline transmission level and the amplitude of seasonal surges.

From Descriptive to Computational Epidemiology: Classical epidemiological surveillance relied on passive, facility-based reporting systems that produce population-level estimates with considerable temporal delay. While these systems remain the backbone of national surveillance, their retrospective character limits their utility for early-warning applications [2]. The emergence of computational and data-driven approaches initially statistical time-series models, and more recently machine learning methods capable of modelling complex non-linear relationships between environmental covariates and disease incidence has extended the epidemiological toolkit toward predictive inference [6, 7].

1.1.1.2 Overview of Malaria

Malaria is a vector-borne parasitic infectious disease caused by protozoan parasites of the genus *Plasmodium*, transmitted to humans exclusively through the bites of infected female *Anopheles* mosquitoes [20, 21]. The name derives from the Italian *mal'aria* meaning “bad air” reflecting a historical association with marshy environments that predated the identification of its causative agent [21]. Towards the end of the nineteenth century, Charles Louis Alphonse Laveran identified parasites in the blood of malaria patients, Ronald Ross demonstrated the mosquito as the transmission vector, and Giovanni Battista Grassi confirmed that human malaria is transmitted exclusively by *Anopheles* species [20]. Of the more than 400 *Anopheles* species identified worldwide, only approximately 30 are considered major malaria vectors under natural conditions [20, 21].

Environmental dependence and predictive implications: The transmission dynamics of malaria are fundamentally conditioned by prevailing environmental and climatic variables, a relationship whose mechanistic basis and epidemiological consequences are examined in detail in the following subsection, and which constitutes the scientific rationale for the IoT-driven predictive approach developed in this work.

1.1.1.3 Environmental and Climatic Determinants of Malaria Transmission

The relationship between environmental conditions and malaria transmission is one of the most extensively documented associations in infectious disease epidemiology. Unlike pathogens transmitted directly between human hosts, *Plasmodium* parasites are obligately dependent on a biological intermediary the female *Anopheles* mosquito whose entire life cycle, from larval development through adult



survival and biting behaviour, is acutely sensitive to prevailing climatic conditions [22, 23]. Understanding these climate vector parasite relationships is not merely an academic exercise: it constitutes the mechanistic foundation upon which any environmental data-driven malaria prediction system must be built.

Temperature: Temperature is the primary climatic determinant of malaria transmission intensity, governing multiple interacting biological processes simultaneously. The duration of the extrinsic incubation period of *P. falciparum* within the mosquito vector the time required for ingested gametocytes to develop into infective sporozoites is strongly temperature-dependent: at 28 °C, sporogony is completed in 9–10 days, while at 20 °C the same process requires 22 days [22]. *P. falciparum* development ceases entirely below 16 °C and above 32–40 °C [22], while for *P. vivax* the minimum threshold is lower, at 15 °C [24]. Within the viable transmission window of 15–40 °C, optimal parasite development occurs between 20 °C and 25 °C [24]. Beyond parasite biology, temperature also governs larval development rate, adult mosquito longevity, and blood-meal digestion time; a 1 °C change in the 18–26 °C range can alter mosquito lifespan by more than a week [22]. A mean daily temperature exceeding 21.5 °C has been identified as a practical threshold above which mosquito incubation is effectively accelerated, resulting in increased malaria incidence [24, 23].

Empirical evidence confirms these biological mechanisms at the population level. Across nine sub-Saharan African countries studied from 2000 to 2018, temperature fluctuations were found to precede malaria incidence rate variations by approximately one year; in Uganda, a peak mean temperature of 23.47 °C in 2016 preceded a peak incidence of 294.9 per 1,000 population in 2017 [25]. In Ethiopia, periods of decreasing incidence coincided with declining temperatures, while incidence rebounds aligned with temperature increases [25]. A critical complication, however, is that temperature is not the sole determinant: inverse temperature incidence relationships have been documented in Ghana and Nigeria during specific periods, underscoring the importance of multivariate modelling and the limitations of single-variable approaches [25].

Precipitation: Rainfall exerts a dual and non-linear influence on malaria transmission. Its primary mechanism is the creation of surface water bodies temporary pools, puddles, irrigation canals, and flooded agricultural land that serve as larval breeding habitats for *Anopheles* mosquitoes [23, 26]. Malaria transmission in West Africa requires a sustained minimum of approximately 80 mm of rainfall over four consecutive months, and transmission intensity is markedly elevated during the rainy season, with peak incidence across the region typically occurring between August and October [23]. In Burkina Faso, observational studies have documented a 9-week lag between peak rainfall and peak malaria incidence, representing the cumulative time required for larval development, adult emergence, parasite development within the vector, and the onset of clinical cases in the human population [7].

This relationship is, however, characterised by a critical non-linearity: heavy or extreme rainfall can simultaneously flush away existing larval sites and disrupt vector development, thereby temporarily reducing transmission even as it creates new habitat [23, 25]. Similarly, rainfall encourages vegetation growth that provides resting and breeding shade for adult mosquitoes, while drought eliminates standing water and suppresses vector populations [7]. These non-linear and context-dependent dynamics constitute a fundamental argument for machine learning approaches over linear statistical models in rainfall malaria forecasting: they demand methods capable of capturing complex, conditional relationships between covariates and outcomes [6].



Relative humidity: Relative humidity conditions the survival and activity of adult *Anopheles* mosquitoes independently of temperature. A relative humidity range of 60–86% has been identified as optimal for *Anopheles* thriving and *Plasmodium* transmission [23]; below approximately 60%, desiccation reduces mosquito longevity and therefore the probability of a mosquito surviving the full extrinsic incubation period long enough to become infective [26]. Empirically, relative humidity sustained above 60% consistently defines the meteorological envelope within which active malaria transmission persists in West and Central African settings, even during periods of reduced rainfall [26, 23]. Critically, sustained ambient humidity can maintain transmission activity through dry-season periods when breeding sites are reduced, contributing to the year-round endemic transmission pattern observed in equatorial zones of Cameroon [2].

Wind speed and atmospheric pressure: The role of wind speed in malaria transmission is indirect and context-dependent. While no significant direct statistical relationship between wind speed and malaria incidence has been established, *Anopheles* mosquitoes can be passively displaced by wind over distances of up to 10 km, contributing to spatial spread beyond the typical active flight range of 2 km [23]. Mean wind speeds below 1.8 m/s have been associated with higher malaria incidence, as calmer conditions favour localised mosquito congregation near breeding sites and human habitations [23]. Atmospheric pressure, while a secondary predictor compared to temperature, humidity, and rainfall, serves as a useful covariate in IoT-based environmental monitoring systems where it is collected continuously alongside primary variables, and has been incorporated as a supplementary feature in multi-variable malaria prediction models [7].

Elevation, vegetation, and land surface conditions: Beyond the primary meteorological variables, malaria transmission is further shaped by topographic and land-cover factors that mediate climatic effects at the local scale. Elevation reduces transmission risk through its inverse relationship with temperature: areas above 1,800–2,000 m altitude are generally free of endemic *P. falciparum* transmission, which explains why Cameroon's western highlands experience lower and more seasonal transmission compared to the equatorial lowlands [22, 2]. Vegetation indices, including the Normalised Difference Vegetation Index (NDVI), are positively associated with malaria incidence in savannah environments, reflecting the role of vegetation in providing mosquito resting habitat and moisture retention [23, 8]. Proximity to forests within 500 m has been identified as a hotspot risk factor when combined with weekly rainfall exceeding 10–15 mm [23]. Soil moisture and surface water indices (NDWI) similarly amplify transmission risk by extending the persistence of larval breeding sites beyond the immediate rainfall period [23].

Climate change and the trajectory of malaria risk: The environmental relationships described above are not static. Under ongoing climate change, the geographic boundaries and seasonal windows of malaria transmission are shifting in ways that directly threaten the control gains achieved since 2000. The IPCC Sixth Assessment Report (2021) confirmed with high confidence that climate extremes will increasingly challenge human health in sub-Saharan Africa, including through the expansion of vector-competent zones [24]. In East Africa, *Anopheles* range expansion has been documented at rates of up to 4 m per year in elevation and 18 km per year in latitude; under RCP 8.5 scenarios, approximately 100 million individuals in East Africa could be exposed to malaria for the first time [24]. Highland areas previously protected by cooler temperatures are increasingly experiencing endemic transmission as



warming breaches biological thresholds [22, 26]. Across Africa as a whole, the 40% reduction in malaria incidence and 60% reduction in mortality achieved between 2000 and 2022 are at risk of reversal as environmental conditions become increasingly favourable for vector expansion and transmission intensification [26].

These projections reinforce the operational urgency of developing climate-informed early warning systems for malaria in sub-Saharan Africa. Ayanlade et al. [24] explicitly identify the necessary components of an effective climate-based early warning system as: vulnerability assessment, seasonality forecasting, continuous environmental monitoring, and real-time integration of observed malaria morbidity data, a specification that precisely describes the architecture proposed in the present study. The capacity to collect high-resolution, real-time environmental data through IoT sensor networks, and to translate those data into district-level transmission risk forecasts through machine learning models, directly addresses the monitoring and forecasting requirements that climate-sensitive malaria surveillance demands.

1.1.2 Machine Learning

This subsection establishes the machine learning foundations on which the predictive system of this work is built. It defines machine learning and its principal learning paradigms, examines the technical and structural limitations that constrain predictive modelling together with the techniques that mitigate them, and reviews the metrics used to evaluate models. It then introduces the family of ensemble methods adopted in this work. The application of these methods to malaria and vector-borne disease prediction is surveyed separately in the state of the art (Section 1.2.1).

1.1.2.1 Fundamentals of Machine Learning

Definition: **Machine learning** is a subfield of computer science that gives computers the ability to learn without being explicitly programmed. By using algorithms and mathematical models, it allows computers to identify patterns in data and make autonomous decisions based on those patterns [27].

Types of machine learning: Machine learning is conventionally divided into supervised, unsupervised, semi-supervised and reinforcement paradigms, according to whether and how the training data carry labels [27].

This work operates entirely within the *supervised* learning setting: the historical malaria records provide the labels (case counts and derived risk categories) against which the environmental features are trained. It is worth noting that no single algorithm is optimal across all problems: the *No Free Lunch* theorem establishes that an algorithm performing well on one class of problems necessarily performs worse on another, which makes the empirical comparison of several candidate models indispensable [28, 29]. A complementary panorama of these paradigms is given by Cornuéjols and Miclet [30].

Challenges and limitations: Despite its successes, machine learning is subject to a number of limitations operating at two levels: a technical–statistical level and a deeper structural level. At the technical level, the central tension is the *bias–variance trade-off*: enlarging the hypothesis space lowers the approximation error (bias) but raises the estimation error (variance), so that reducing one term generally increases the other [29]. A model that is too complex relative to the signal *overfits*, its empirical error is low while its generalisation error is large, whereas a model that is too simple *underfits* [29, 31].



At a more structural level, Malik [32] argues that these limitations are downstream of deeper commitments: quantification operates on proxies rather than constructs, modelling centres on a *central tendency* that penalises outliers, and *prediction is not explanation*, correlations are fragile, can be gamed (Goodhart’s law), and provide an unreliable basis for intervention.

Techniques to address the limitations: Of the mitigations the literature proposes, three bear on this work and are described where they are applied in Chapter 2: regularisation, which penalises model complexity; blocked cross-validation, which respects the temporal structure of the data; and class weighting, which corrects the imbalance of the risk label.

The regularisation coefficient λ is a hyperparameter, typically tuned by cross-validation over a logarithmic grid [29]; data-quality issues are handled upstream by imputation and by binning or smoothing of noisy attributes [33]. A recurring caution is that several of these techniques may themselves introduce failure modes: repeated hyperparameter tuning can overfit the validation set, and feature transformations applied before splitting cause data leakage, so each remedy must in turn be validated [31].

Evaluation of machine learning models: Model evaluation distinguishes better from worse models by comparing the ground-truth value with the predicted value [31]. Two choices govern the metrics used throughout this work, and both follow from the structure of the problem rather than from convention. For the classification task the positive class is a minority, roughly a quarter of district-months, so accuracy is misleading: a model that never raises an alert would still score highly. The $F1$ score, the harmonic mean of precision and recall, is reported instead, since an early-warning system is penalised both for missing outbreaks and for raising false alarms [29, 31]. For the regression task the coefficient of determination R^2 is reported, because it is scaled against a mean-predicting baseline and therefore states directly whether the model is worth more than that baseline, a comparison that matters here because several evaluation regimes turn out to fall below it; the mean absolute error is reported alongside it, in the units of the target, wherever an operationally interpretable magnitude is required.

Finally, the literature stresses that every in-sample estimate including cross-validation is *optimistic* when observations are not independent. Using Efron’s optimism framework [34], the gap between the true error Err and the training error err is

$$\text{Opt}(\hat{\mu}) = \text{Err}(\hat{\mu}) - \text{err}(\hat{\mu}) = \frac{2}{n} \text{tr Cov}_f(Y, \hat{Y}), \quad (1.1)$$

which is strictly positive whenever held-out points are correlated with the training data [32]. Consequently, for temporally or spatially structured data, a random partition leads to an under-estimation of the error, and dependency-aware schemes such as blocked (temporal) cross-validation must be used instead [35, 32]. This consideration is directly relevant to the present work, whose district-level malaria series are both temporally and spatially structured; cross-validated scores are therefore treated as preliminary, with genuine out-of-sample testing reserved as the ultimate criterion of model success [32].

1.1.2.2 Ensemble Methods

Ensemble methods combine several base learners into a single, stronger predictor. Two complementary strategies are distinguished [33]: *bagging*, which trains learners in parallel on resampled data and aggregates them to reduce *variance*, and *boosting*, which trains learners sequentially so that each corrects its predecessor, thereby reducing *bias*. Both strategies are particularly effective on structured



(tabular) data, where they remain state of the art for many tasks [31]. This property is directly relevant to the present work, whose integrated district-level dataset is tabular and combines heterogeneous feature types continuous climate measurements, and cyclical temporal encodings for which tree ensembles are particularly well suited.

Bagging and Random Forests: *Bagging* (bootstrap aggregating) trains B models on B bootstrap resamples of the training set and averages their predictions (regression) or takes a majority vote (classification), which lowers the variance of unstable, high-variance learners such as decision trees [33]. A **Random Forest** extends bagging by adding feature randomisation: at each split, only a random subset of the features is considered as a candidate, which decorrelates the trees and further reduces variance [36].

A key practical advantage of random forests is the **feature importance** they provide [36]. Two measures are common:

- the *mean decrease in impurity* (Gini importance), which sums, for each feature, the impurity reduction it produces over all splits where it is used, averaged across the forest;
- the *permutation importance*, which measures the increase in prediction error when the values of a feature are randomly permuted, breaking its association with the target.

These scores give the random forest a degree of **interpretability**: although the full ensemble is a black box, feature-importance rankings expose which variables drive the predictions, and individual trees remain inspectable. This interpretability must nonetheless be read with caution, since importance scores are correlational and can be biased toward high-cardinality or correlated features [32]. A further benefit is the *out-of-bag* estimate of generalisation error, computed on the samples left out of each bootstrap, which provides validation without a separate holdout [33].

Gradient Boosting, XGBoost and LightGBM: *Boosting* builds an additive model in a stagewise manner, each new learner being fitted to the errors of the current ensemble [33]. In *gradient boosting* [37], the model after M stages is

$$F_M(\vec{x}) = \sum_{m=1}^M \nu h_m(\vec{x}), \quad (1.2)$$

where M is the number of boosting stages, $\nu \in (0, 1]$ is the learning rate (shrinkage), and each base learner h_m (typically a shallow tree) is fitted to the *pseudo-residuals*, that is, to the negative gradient of the loss evaluated at the current model. Boosting therefore performs gradient descent in function space, progressively reducing the bias of the ensemble.

XGBoost (eXtreme Gradient Boosting) is a scalable implementation of gradient-boosted trees that adds an explicit regularisation term on tree complexity to the objective, uses a second-order (Newton) approximation of the loss, and incorporates a sparsity-aware split-finding procedure that handles missing values natively [38]. **LightGBM** further improves training speed and memory use on large tabular datasets through histogram-based split finding, a leaf-wise (best-first) tree-growth strategy, gradient-based one-side sampling and exclusive feature bundling, together with built-in support for categorical features [39]. Their efficiency and accuracy on heterogeneous tabular data explain their wide adoption, and Ye and Wang [31] take gradient-boosted decision trees as the benchmark against which modern deep tabular models are compared. Table 1.1 contrasts the two families, drawing on [36, 33, 37, 38, 39].

Table 1.1: Comparison of bagging and boosting ensembles

Property	Random Forest (Bagging)	Gradient Boosting (XGBoost / LightGBM)
Training	Trees trained in parallel on bootstrap samples.	Trees trained sequentially on residuals.
Main effect	Reduces variance.	Reduces bias.
Overfitting	Robust; little tuning needed.	Sensitive to noise; requires careful tuning.
Key hyperparameters	Number of trees, features per split.	Learning rate, depth, number of estimators, regularisation.
Interpretability	Feature importance, out-of-bag error.	Feature importance; regularised objective.

The feature-importance mechanisms common to both ensemble families provide the foundation on which the model-agnostic explainability analysis of this work is built, as developed in the following subsection on SHAP.

On the choice of ensemble methods over deep learning: Deep sequential architectures, such as Long Short-Term Memory (LSTM) networks and the Temporal Fusion Transformer, constitute an alternative model family for time-series forecasting. They were considered but not adopted in this work for two reasons grounded in the specific regime of the present problem. The first concerns the data regime. The integrated dataset is tabular and modest in size, comprising a few thousand district-month observations with a few dozen monthly points per district, which is precisely the regime in which gradient-boosted tree ensembles have been shown to retain a systematic advantage over deep architectures. In a controlled benchmark across a large collection of tabular datasets, Léo Grinsztajn et al. [40] established that tree-based models outperform deep networks on typical tabular data of this scale, attributing the gap to the deep models' sensitivity to uninformative features and their bias toward overly smooth decision boundaries, neither of which suits the irregular, interaction-rich structure of tabular climate and case data. The data requirements of recurrent and attention-based models, by contrast, are met only at sample sizes well beyond the few dozen observations per series available here, as also noted by Tianqi Ye and Bo Du [31].

The second reason concerns the nature of the interpretability the operational early-warning context requires. It is not that deep architectures are uninterpretable: attention weights and gradient-based attribution methods do provide post-hoc explanations of their predictions. The distinction is in the kind of explanation available. Gradient-boosted tree ensembles admit, through the tree-specific Shapley computation adopted in this work, an attribution that is exact, additive, and computed from the predictive model itself rather than from an approximate surrogate or a saliency heuristic. For a public-health early-warning system, whose forecasts must be auditable and whose explanations must be defensible to those who act on them, an exact and additive attribution is preferable to an approximate one, and it is obtained without the additional computational and architectural cost that the deep models would impose for no gain in predictive performance in this regime. For these reasons, the modelling strategy of this work is built on gradient-boosted ensemble methods, while recurrent and attention-based architectures are identified as a direction to revisit should substantially longer series become available.



1.1.3 Internet of Things

1.1.3.1 Definition and Conceptual Evolution

The Internet of Things (IoT) denotes a network of physical objects devices, instruments, vehicles, buildings and other items embedded with electronics, sensors, software and network connectivity that enables these objects to collect and exchange data and to be sensed and controlled remotely across existing network infrastructure [41]. The term was introduced by Kevin Ashton in 1999 to describe uniquely identifiable interconnected objects based on radio-frequency identification (RFID) technology; although no single canonical definition has been universally adopted, the IoT is most often characterised as a *dynamic global network infrastructure with self-configuring capabilities based on standard and interoperable communication protocols, in which physical and virtual “things” possess identities and attributes and are capable of using intelligent interfaces* [42, 43]. A defining property that recurs across the literature is the capacity of IoT systems to operate autonomously, exchanging and synchronising data in a device-to-device manner without requiring human intervention [9].

Conceptually, the IoT is positioned as the latest phase in the evolution of the Internet, succeeding the Internet of Documents, Commerce, Applications and People [41]. Abdul-Qawy et al. [43] trace its lineage to machine-to-machine (M2M) communications, arguing that the IoT is essentially the IP-based generalisation of M2M, extending connectivity from “any-time, any-place for any-one” to “any-time, any-place for any-thing”. The technological convergence enabling this shift includes wireless sensor networks (WSNs), barcodes and RFID, intelligent sensing, low-energy wireless communication and cloud computing [42]. Environmental monitoring is repeatedly identified as one of the most prominent and beneficial application domains of this paradigm [10, 43].

1.1.3.2 The Environmental Sensor Node

Only one element of the IoT stack bears directly on this work, namely the environmental sensor node and the link that carries its readings, and it is described here at the level of detail the later chapters require.

An environmental node is a microcontroller carrying a small set of transducers, selected against cost, size, energy consumption, accuracy, calibration and suitability to local atmospheric conditions [42, 10], together with a radio and a power source. Low-cost open-source microcontrollers of the Arduino and ESP families recur throughout the literature as the computational core of such a node [10, 12, 11]. Nodes report to a *gateway*, which aggregates their packets and forwards them to a server; shifting computation from the constrained node to the more capable gateway permits optimisation the individual nodes could not perform [42].

Because such nodes are battery-powered and deployed in the field, the binding constraint is energy rather than bandwidth, and the networks they form are characterised as low-power and lossy [43]. This is what motivates the long-range, low-bitrate radio used in the deployment of the present work. LoRa trades throughput for range and power, which suits readings that are small and infrequent, and organising the nodes as a *mesh* rather than a star, as the Meshtastic firmware does, allows multi-hop relaying between them. That property is decisive in the dense urban fabric of Yaoundé, where buildings attenuate radio signals and a single-hop link to a distant gateway would fail; the network architecture actually deployed is described in Chapter 2.



1.2 State of the Art

1.2.1 Machine Learning Applied to Malaria and Vector-Borne Disease Prediction

This subsection surveys the application of machine learning to malaria and vector-borne disease prediction. It traces the transition from classical statistical time-series models to ensemble learning methods, then reviews the empirical evidence for their use in sub-Saharan African settings, identifying the methodological principles, notably rigorous temporal validation and multi-model benchmarking, that the present study builds upon.

1.2.1.1 From Statistical to Machine Learning Approaches in Epidemiological Forecasting

Quantitative approaches to epidemiological forecasting have evolved substantially over the past three decades, driven by the increasing availability of surveillance data, advances in computational methods, and growing recognition that the non-linear, multi-factorial dynamics of vector-borne disease transmission strain the assumptions of classical statistical models. Tracing this evolution, from linear time-series methods to machine learning, is essential both for contextualising the methodological choices made in the present study and for understanding why classical models, despite their enduring value, are best regarded as conceptual reference points rather than as the primary predictive instrument for the problem addressed here.

The Box–Jenkins paradigm: The dominant framework of classical statistical forecasting is the family of Auto-Regressive Integrated Moving Average (ARIMA) models and their seasonal extension (SARIMA), formalised in the work of Box and Jenkins and now universally referred to as the Box–Jenkins methodology [44, 45]. An ARIMA(p, d, q) model represents an observed series as a linear filter driven by a sequence of uncorrelated random shocks: the autoregressive component (p) expresses the current value as a linear function of its own past values, the integration component (d) differences the series to remove non-stationarity, and the moving-average component (q) models the disturbance as a function of past forecast errors [44]. The SARIMA extension augments this structure with seasonal autoregressive, differencing, and moving-average terms operating at a fixed periodicity s (typically $s = 12$ for monthly data with annual seasonality), enabling the model to capture recurring seasonal patterns in addition to short-term dependence [44, 45]. The framework is governed by the principle of parsimony, which favours the model with the fewest parameters adequate to represent the data, and by an iterative cycle of identification, estimation, and diagnostic checking [44, 45].

Application to malaria forecasting: ARIMA-family models have been applied with documented success in malaria surveillance. Mohammad Y. Anwar et al. [46] applied ARIMA to monthly malaria incidence in Afghanistan, correctly identifying the annual transmission peak between July and September and associating outbreak periods with high rainfall and humidity. Rajasekhar Mopuri et al. [47] deployed a SARIMA model for climate-based malaria forecasting in Andhra Pradesh, India, capturing seasonal autocorrelation in case data and identifying rainfall as the primary driver. Within the sub-Saharan African context most relevant to this work, Odu Nkiruka et al. [7] reviewed ARIMA and SARIMA among existing statistical approaches to malaria prediction, noting that while such models extract seasonal structure effectively, they remain limited to univariate, seasonally driven forecasting, a limitation that motivated their own adoption of a gradient-boosting approach. These applications establish ARIMA and



SARIMA as the long-standing statistical reference point in malaria forecasting, against which more recent machine learning approaches are positioned.

Climate-informed statistical models: The Box–Jenkins family is not confined to the univariate case. Its SARIMAX extension admits exogenous regressors, and this climate-informed form remains the contemporary statistical standard for malaria forecasting in sub-Saharan Africa. Emmanuel Afolabi Bakare et al. [48] modelled monthly incidence in Adamawa State, Nigeria, from January 2015 to April 2024, selecting among candidate specifications by the Akaike and Bayesian information criteria and evaluating them by RMSE, MAE, and MAPE; a SARIMAX(1,0,1)(1,1,1)₁₂ model incorporating lagged rainfall and temperature, the lags chosen to represent the biologically plausible delay between environmental conditions and transmission, reproduced both the annual seasonality and the pronounced June-to-October surge. Tesfaye Taye Gelaw and Meseret Addisu Abera [49] applied the same framework to Bahir Dar, Ethiopia, fitting a SARIMAX(1,2,2)(1,2,2)₁₂ model to ninety monthly observations from January 2018 to June 2025 under a chronological eighty-twenty split, with nine weather variables entered as exogenous regressors. Both studies frame their output as the basis for a localised early-warning capability, the same operational objective pursued in the present work.

Structural limitations for the present problem: Despite their parsimony, interpretability, and mature inferential theory, ARIMA and SARIMA rest on assumptions that the malaria prediction problem addressed in this work systematically violates. Three limitations are decisive. First, even in the SARIMAX form the exogenous channel is additive and linear: each covariate contributes through a single coefficient whose functional form is fixed in advance rather than learned [44]. The cost of that constraint is visible in the studies just cited. At Bahir Dar none of the nine weather regressors attained statistical significance, and although the SARIMAX specification improved the information criteria markedly over its SARIMA counterpart, it returned exactly the same predictive error, a MAPE of 28.3% in both cases [49]: admitting climate as a linear exogenous term sharpened the fit without improving forecast accuracy. Second, the framework assumes linearity, and therefore cannot natively represent the non-linear, threshold-dependent climate–incidence relationships documented in Section 1.1.1.3 such as the dual effect of rainfall, in which moderate rainfall creates breeding habitat while extreme rainfall destroys it [44, 45]. Where such structure is present it must be transformed away rather than represented: the Bahir Dar series required a square-root transformation followed by second-order seasonal differencing before a heteroscedastic, non-linear trend could be rendered stationary [49]. Third, malaria dynamics are spatially heterogeneous: identical climatic conditions produce markedly different transmission intensities across districts of differing ecology, altitude, and intervention history [8], making the fitting and continuous updating of a separate model per district both computationally impractical and statistically fragile. Both recent SARIMAX studies are accordingly estimated for a single administrative unit, one state and one city respectively [48, 49], and the widening of their prediction intervals with the forecast horizon [48] bounds the lead time over which one fitted model stays operationally useful.

The transition to machine learning: These limitations motivated a progressive shift toward machine learning methods, accelerated by the growing availability of multi-source environmental and epidemiological data at sub-national resolution [7]. The distinction is architectural: where ARIMA assumes a fixed linear functional form and estimates a small number of parameters, machine learning models make no prior assumption about the input–output relationship, learning complex, non-linear,



high-dimensional mappings directly from data [4]. This yields three concrete advantages for malaria forecasting. Machine learning models natively accommodate multiple heterogeneous covariates (meteorological, spatial, and temporal) within a single framework without a pre-specified interaction structure [7]; they capture lagged and threshold-dependent relationships through mechanisms such as decision boundaries in ensemble methods [6]; and a single model trained across all districts can exploit spatial heterogeneity as a learnable feature rather than treating each district as an independent problem [8]. Odu Nkiruka et al. [7] demonstrated this advantage empirically across six sub-Saharan African countries including Cameroon, where a gradient-boosting model outperformed classical baselines while also quantifying each climate variable's contribution through feature importance, a capability linear models lack.

Crucially, the classical framework is not thereby rendered obsolete: it retains value as the conceptual source of principles, stationarity, differencing, autocorrelation, and seasonality, that continue to inform feature engineering in the data-driven paradigm, including the seasonal and lagged climate features constructed in this work. That the two SARIMAX studies discussed above are themselves recent confirms the point: the statistical and the learned approach are complementary, the former supplying the seasonal structure and the latter the non-linear, multi-covariate, and spatially pooled representation that structure alone cannot provide [48, 49]. The seasonal model accordingly survives in the present work as an explicit baseline against which the learned models are measured. The following subsection reviews the empirical evidence for ensemble machine learning in malaria and vector-borne disease prediction across sub-Saharan Africa.

1.2.1.2 Machine Learning for Climate-Driven Malaria Prediction

Building on the architectural distinction drawn in the previous subsection, this subsection reviews the empirical evidence for ensemble machine learning in malaria prediction, concentrating on sub-Saharan Africa and on analogous endemic settings whose data structures resemble the problem addressed in this work. The studies surveyed here share a common premise, namely that the climatic and environmental determinants of transmission detailed in Section 1.1.1.3 can be exploited as predictive features by models capable of learning non-linear, high-dimensional relationships directly from surveillance and environmental data. Rather than examining each study in isolation, the discussion is organised around five recurring dimensions: the choice of learning algorithm, the environmental predictors and their data sources, the validation strategy and performance metrics, the use of interpretability methods, and the methodological limitations that collectively define the gap this work seeks to address.

Convergence on ensemble tree methods: The most striking regularity across the literature is the convergence on tree-based ensemble methods, and on gradient boosting in particular, as the dominant predictive instrument. Gradient boosting, and XGBoost above all, is reported as the strongest performer across settings as varied as six sub-Saharan countries including Cameroon [7], district forecasting in Senegal [6] and rural Tanzania [50], with random forests preferred in some classification settings [8, 5], and the same family holding at national scale and against deep learning in Indonesia [51]. The study-by-study detail, including the setting, the target formulation, the predictors and the reported performance of each, is given in Table 1.2. This convergence is not incidental: ensemble trees natively accommodate heterogeneous meteorological, spatial, and temporal covariates within a single model, capture the

threshold-dependent climate effects discussed below, and expose a ranking of feature contributions that linear and classical time-series models cannot provide.

Environmental predictors and data sources: The reviewed studies draw their environmental predictors almost exclusively from satellite and reanalysis products rather than from ground-level measurement. Almamy Youssouf Ly et al. [6] sourced humidity, soil moisture, precipitation, solar radiation, and temperature from the NASA POWER reanalysis archive, augmenting these with demographic (population density) and intervention covariates from the national control programme. Elliot Mbunge et al. [8] assembled rainfall from CHIRPS at 5 km resolution and day and night land surface temperature, the normalised difference vegetation index, and the enhanced vegetation index from MODIS at 1 km resolution, all aggregated monthly to ward level. Jin-Xin Zheng et al. [50] similarly relied on the vegetation index, rainfall, and land surface temperature as their core climate inputs. Anjar Dimara Sakti et al. [51] broadened the feature space beyond climate to incorporate socioeconomic vulnerability (age and sex ratios) and healthcare capacity (access measures) within a combined susceptibility-vulnerability-capacity framework. Two observations follow. First, the environmental signal in this literature is overwhelmingly remote-sensed, captured at spatial resolutions of one to several kilometres and at monthly temporal cadence. Second, surveillance case counts, drawn from district health information systems or national control programmes, serve as the prediction target and, in several designs, as predictors in their own right. The near-total absence of in-situ environmental measurement is a point returned to at the close of this subsubsection. Table 1.2 synthesises these representative studies by setting, model family, prediction task, and headline result.

Table 1.2: Synthesis of representative machine learning studies for climate-driven and environment-based malaria prediction

Study	Setting	Models	Task (unit)	Headline result
Nkiruka et al. [7]	6 Sub-Saharan African countries (incl. Cameroon)	XGBoost (K-means, Pearson selection)	Incidence classification (country)	XGBoost best; precipitation, temperature, radiation dominant
Ly et al. [6]	13 districts, South-east Senegal	XGBoost, HistGB, LightGBM	Monthly incidence regression (district)	XGBoost $R^2 = 0.90$ (test), 0.93 (validation)
Zheng et al. [50]	3 districts, South-east Tanzania	XGBoost	Incidence vs. climate (district)	Non-linear rainfall threshold; seasonal alignment
Mbunge et al. [8]	Buhera, Zimbabwe (33 wards)	LR, SVM, DT, RF	Ward-level classification (ward)	LR and RF $\approx 83\%$ accuracy; RF F1 = 0.90
Muriithi et al. [5]	Kenya (synthetic, $n = 1000$)	RF, XGBoost (SHAP, DALEX)	Patient-level risk classification (patient)	Demonstrates SHAP and DALEX attribution; symptoms and parasite species dominate
Sakti et al. [51]	Indonesia (national)	GTB, CART, RF, DL-MLP	Spatial susceptibility mapping (grid)	National susceptibility map; static, non-temporal

Validation strategies and performance metrics: Performance evaluation in this literature falls into two families distinguished by the formulation of the prediction target. Regression-oriented studies report the coefficient of determination, root mean square error, and mean absolute error: Almamy Youssouf Ly et al. [6] achieved a coefficient of determination of 0.90 on the test set and 0.93 on the validation set, with errors expressed per mille of incidence, while Jin-Xin Zheng et al. [50] characterised the strength of the climate-incidence association directly. Classification-oriented studies report accuracy, precision, recall, the F1 score, and the area under the receiver operating characteristic curve: Elliot



Mbunge et al. [8] reported 83% accuracy and an F1 score of 0.90 for their best classifiers, and Dennis Kariuki Muriithi et al. [5] reported 98% accuracy for random forest. The headline figures, however, are not directly comparable, and their apparent strength must be read against the rigour of the underlying validation. Almamy Youssouf Ly et al. [6] are notable for adopting a five-split TimeSeriesSplit cross-validation strategy with Bayesian hyperparameter optimisation, a temporally honest protocol that respects the forward-looking nature of forecasting. Many of the remaining designs rely on random train-test partitions, which leak temporal and spatial structure into the evaluation, or on synthetic over-sampling through SMOTE [8] and class rebalancing [5], both of which can inflate reported performance relative to operational conditions. Consequently, the metric values themselves are less informative than the validation discipline that produced them.

Interpretability and the drivers of prediction: A further point of convergence is the now-standard use of post-hoc interpretability, principally Shapley additive explanations, to expose the features driving predictions. The substantive findings these methods yield are consistent and instructive. With respect to climate, the non-linear and threshold-dependent effect of rainfall recurs across settings: Jin-Xin Zheng et al. [50] found that the marginal effect of rainfall increases expected incidence up to a point but attenuates beyond roughly 0.4 to 0.5 m, where excessive rainfall may flush larval breeding sites, and Almamy Youssouf Ly et al. [6] reported the same dual mechanism alongside a modest and variable temperature effect and a positive association with humidity. With respect to non-climatic structure, geographic and static features carry substantial weight: Almamy Youssouf Ly et al. [6] found that latitude, longitude, and population density were among the most influential predictors, helping to identify spatial hotspots, while in patient-level designs the dominant contributors were clinical symptoms and parasite species rather than environmental variables [5, 4]. The collective message is that the climate signal, although real and mechanistically interpretable, is frequently secondary to spatial, demographic, and clinical structure, a finding with direct bearing on how a climate-driven model should be framed and on what it can realistically be expected to achieve.

Critical appraisal and research gaps: Read together, these studies expose several limitations that motivate the present work. The first concerns the heterogeneity, and occasional mismatch, of prediction targets. Patient-level diagnostic classification, as in Dennis Kariuki Muriithi et al. [5], where synthetic data and clinical symptoms drive a 98% accuracy, and in the precursor survey-based study of Dennis Muriithi et al. [4], where the presence of *Plasmodium falciparum* was the single most important feature, addresses a fundamentally different question from district-level incidence forecasting. Conflating the two inflates the apparent ceiling of climate-driven prediction and obscures the comparatively modest contribution that climate alone provides. The second limitation concerns spatial generalisation. Elliot Mbunge et al. [8] explicitly criticise the assumption of uniform transmission, yet, like the majority of the studies reviewed, validate their models within a single region or by pooling districts under random partitions. Leave-region-out evaluation, which alone reveals whether a model transfers to districts unseen during training, is almost never reported, leaving spatial transferability an open question. The third limitation concerns deployment honesty. Almamy Youssouf Ly et al. [6] include confirmed cases shifted by twelve months as a predictor, an autoregressive feature that materially improves in-sample performance but is operationally fragile, since the latency of monthly surveillance reporting renders such lagged case counts unavailable at the moment a forecast is required. Few studies separate oracle predictors, available only in hindsight, from genuinely deployable ones. The fourth limitation, and the



most consequential for this work, is the mismatch between the scale of transmission heterogeneity and the scale of environmental measurement. Jin-Xin Zheng et al. [50] emphasise the hyperlocal heterogeneity of transmission, yet their inputs, in common with the rest of the literature, are satellite and reanalysis products at kilometre-scale spatial resolution and monthly temporal cadence. Finally, several authors acknowledge that modest dataset sizes constrain the generalisability of their conclusions [6, 8].

Toward in-situ environmental sensing: The last of these limitations points directly toward the contribution of the present study. If transmission varies at a scale finer than that of the available environmental data, then improving prediction requires not only better models but better-resolved measurement. Ground-level sensing through Internet of Things devices offers precisely this: fine-grained, in-situ, and near-real-time observation of the microclimatic conditions (temperature, humidity, pressure, and rainfall) that govern vector ecology, at a spatial and temporal resolution unattainable from orbit. The deployment of such a sensing substrate, and its relationship to the modelling pipeline developed in this work, is treated in the following subsection on Internet of Things-based environmental health monitoring.

1.2.2 IoT-Based Environmental Monitoring in Public Health

1.2.2.1 The Role of IoT in Environmental Monitoring and Health Surveillance

Environmental monitoring is the systematic observation of the quality of an environment and of the factors and actions that influence it, with the dual aim of reducing adverse effects on the natural environment and safeguarding human health [10]. Conventional approaches particularly the manual collection of samples for laboratory analysis are widely criticised as labour-intensive, costly, time-consuming and incapable of furnishing the continuous, real-time information needed for timely public-health decisions [9]. Traditional WSN-based remote sensing, while cheaper than legacy instrumentation, remains limited in communication range, bandwidth and energy efficiency, and is therefore poorly suited to large or rural areas [9]. The IoT is consistently advanced as the solution to these limitations, providing accessible, real-time data regardless of time or place and acting, in the words of Narayana et al., as the “eyes, ears and mouthpiece” of an application that watches, reports and intervenes to prevent harm [10, 9].

In the specific context of environmental *health* surveillance, IoT systems function as early-warning infrastructures. Deteriorating water quality can trigger immediate preventive action against waterborne disease outbreaks, and in emergencies such as floods or industrial accidents the same systems can deliver the real-time information that response teams require for evacuation and remediation [9]. Ladeira et al. [11] frame their rural deployment explicitly around health-protective monitoring, motivated by the World Health Organisation’s estimate that air pollution was associated with approximately 4.2 million premature deaths in 2016, and by epidemiological evidence linking residential radon exposure to a roughly 16% increase in lung-cancer risk per 100 Bq/m³. Siddiqui et al. [12] extend the paradigm from environmental sensing to direct clinical surveillance, integrating physiological sensors with machine learning for remote disease prediction during and after the COVID-19 pandemic.

1.2.2.2 The Gap: Vector-Borne Disease is Absent from this Corpus

One gap in this literature is decisive for the present work. *Vector-borne disease surveillance*, and malaria in particular, is conspicuously absent from the environmental-IoT corpus, despite the fact that



the climatic variables driving vector ecology, namely temperature, humidity and rainfall, are precisely those that environmental IoT stations are built to capture [10, 9]. The reviewed deployments that do couple sensor streams to machine learning address clinical or generic environmental prediction rather than the explicit linkage of environmental observation to epidemiological outcome [9, 12, 11]. This gap motivates the integration of sensor-collected environmental observations with predictive epidemiological modelling that the remainder of this work develops.

1.2.3 Explainable Artificial Intelligence in Clinical and Epidemiological Decision-Making

The predictive value of a machine learning model in public health is realised only when its outputs can be understood, trusted, and acted upon by the decision-makers who must use them. This subsection establishes why interpretability is a decisive concern in epidemiological prediction, introduces the explainability methods that address it, narrows to Shapley additive explanations as the dominant framework in the reviewed literature, and surveys how that framework has been used, and where it has fallen short, in malaria and environmental health prediction.

Why interpretability matters in health decision-making: In clinical and epidemiological settings, a prediction that cannot be explained cannot easily be acted upon. The ensemble methods that dominate climate-driven malaria prediction are, despite their accuracy, opaque: they offer no direct account of which environmental factors drove a given risk estimate [13]. This opacity is not a merely technical inconvenience. Public-health decision-makers who must allocate scarce resources, namely insecticide-treated nets, indoor residual spraying teams, or diagnostic stockpiles, require justifiable grounds for their decisions, and a black-box alert that cannot be defended is unlikely to be trusted or adopted [14]. The cost of opacity is compounded in health contexts, where a confident but incorrect prediction can misdirect interventions away from the populations that need them most. Interpretability is therefore not an optional refinement but a precondition for the operational deployment of any predictive surveillance system.

The explainability landscape and the emergence of SHAP: Approaches to interpretability divide into intrinsically interpretable models, whose structure can be inspected directly but whose simplicity often limits predictive power, and post-hoc methods applied after training to expose the behaviour of an otherwise opaque model. Among the latter, Shapley additive explanations (SHAP) has become the dominant framework in the epidemiological literature. Introduced by Scott Lundberg and Su-In Lee [15], SHAP attributes a model's prediction to its input features by assigning each feature a contribution derived from cooperative game theory; the method is model-agnostic, applies naturally to tree ensembles, and yields both local explanations (why a specific prediction was made) and global explanations (which features matter across the dataset). It is this combination of theoretical grounding and practical generality that explains its near-ubiquity in recent malaria prediction studies, and it is why SHAP is the attribution method adopted in this work.

SHAP in malaria and epidemiological prediction: Across the reviewed malaria literature, SHAP is used principally to expose the environmental and structural drivers of model predictions, and two of its findings recur in a form directly relevant to the results of this work. Geographic descriptors



rank among the most influential predictors alongside the climate terms [6], and seasonality dominates the explanatory ranking ahead of lagged temperature and rainfall [50]; where individual-level clinical data are available, environmental variables are relegated to a secondary role [5]. Read together, these analyses demonstrate that SHAP not only ranks features but does so in a way that is broadly consistent with established malaria epidemiology, lending interpretive credibility to the learned relationships.

The gap from description to validation and deployment: Despite this growing adoption, the role of explainability in the reviewed literature remains largely descriptive. SHAP is applied, in most cases, to characterise a model after the fact, namely to report which features it relied upon, rather than to validate that the predictive signal is genuine or to drive a concrete operational decision. The convergence of feature attributions across different model formulations, which would offer internal evidence that the learned signal is robust rather than an artefact of a single model, is seldom examined. Moreover, in none of the reviewed studies is explainability coupled with real-time, ground-level environmental sensing; the explanations describe models trained on coarse satellite or reanalysis inputs. The opportunity to use explainability not merely to describe a model but to validate its signal and to inform operational, interpretable early warning, grounded in high-resolution sensor data, remains open. This gap is consolidated, alongside those identified in the preceding subsections, in the synthesis and gap analysis of Section 1.3.1.



1.3 Review and Positioning

1.3.1 Synthesis and Gap Analysis

The studies reviewed in this chapter share a common objective, namely to predict malaria risk or incidence from environmental and climatic data. Across this body of work, two broad methodological tendencies are discernible. The classical statistical approach, represented by the ARIMA and SARIMA family, offers the advantages of parsimony, interpretability, and a mature inferential theory; it remains, however, fundamentally linear, univariate, and dependent on stationarity, and is therefore unable to represent the non-linear, multivariate, and spatially heterogeneous dynamics that govern malaria transmission. The machine learning approach, dominated by tree-based ensembles such as Random Forest, XGBoost, and LightGBM, offers markedly greater accuracy and flexibility, natively accommodates heterogeneous covariates, and, when coupled with post-hoc explainability, exposes the drivers of its predictions; it does so, however, at the cost of greater data requirements and reduced transparency, and its reported performance is often inflated by validation protocols that leak temporal or spatial structure.

Beyond the strengths and weaknesses specific to each tendency, the reviewed literature exhibits a set of recurring limitations that cut across both. First, the environmental signal is drawn almost exclusively from satellite and reanalysis products at kilometre-scale resolution and monthly cadence, a granularity that smooths precisely the fine-grained variability most predictive of imminent transmission, and that mismatches the hyperlocal scale at which transmission actually varies. Second, several studies conflate fundamentally different prediction tasks, treating patient-level diagnostic classification, which leans heavily on clinical symptoms and parasite species, as though it were equivalent to district-level incidence forecasting, a conflation that inflates the apparent ceiling of climate-driven prediction. Third, spatial generalisation is almost never tested: leave-region-out evaluation, which alone reveals whether a model transfers to districts unseen during training, is conspicuously absent, leaving spatial transferability an open question. Fourth, few studies distinguish deployable predictors from oracle predictors available only in hindsight; the use of lagged case counts as features, for instance, improves in-sample performance but is operationally fragile, since the latency of monthly surveillance reporting renders such data unavailable at the moment a forecast is required. Fifth, where explainability is applied, it is used descriptively, to characterise a model after the fact, rather than to validate that its predictive signal is genuine or to drive an operational decision. These limitations are summarised, for the representative studies reviewed, in Table 1.3. In that table, a check mark (✓) denotes that a capability is present and a cross (✗) that it is absent or not reported. In the *Designed for IoT deployment* column, the check mark denotes design intent and deployment substrate, not the use of sensor-collected data as a training source.

Taken together, these studies reveal a critical gap in the literature. While climatic drivers of malaria, machine learning models, IoT-based sensing, and interpretability tools like SHAP have each been studied separately, their integration into a practical and reliable district-level prediction system for sub-Saharan Africa remains absent. This research addresses that gap by combining these elements into a unified, deployable framework.

1.3.2 Positioning of the Present Research

Taking account of the limitations identified above, this thesis constitutes a contribution at the intersection of environmental sensing, interpretable machine learning, and malaria surveillance in Cameroon, a high-burden setting whose national health strategy has explicitly identified the weakness of its elec-

Table 1.3: Positioning of the present research against representative studies in environmental malaria prediction

Study	Designed for IoT deployment	ML prediction	Explainability (SHAP)	Spatial generalisation	Deployment-honest	SSA context
Adeola et al. (2019) [52]	X	X	X	X	X	✓
Nkiruka et al. (2021) [7]	X	✓	X	X	✓	✓
Mbunge et al. (2022) [8]	X	✓	X	X	X	✓
Mwangungulu et al. (2023) [53]	X	X	X	X	X	✓
Ly et al. (2025) [6]	X	✓	✓	X	X	✓
Muriithi et al. (2025) [5]	X	✓	✓	X	X	✓
Zheng et al. (2025) [50]	X	✓	✓	X	X	✓
Sakti et al. (2025) [51]	X	✓	X	X	X	X
This work	✓	✓	✓	✓	✓	✓

tronic disease surveillance as a structural priority. Rather than adding another satellite-based model to an already crowded field, the present work is distinguished by four deliberate choices that respond directly to the gaps above.

First, the modelling is designed around fine-resolution, ground-level environmental sensing as its operational data source. Because the in-situ sensor network was deployed in Cameroon only in 2026, after the historical period under study, the models are trained and evaluated on matched district-level climate and malaria records for 2019–2022, with the climate variables (temperature, humidity, pressure, and rainfall) drawn from the Open-Meteo Historical Weather API as a faithful proxy for the quantities the sensor network measures. The Internet of Things network thus constitutes the deployment substrate through which the same variables are supplied in near-real time at inference, rather than the source of the training data, aligning the system with an operational data stream that satellite and reanalysis products cannot match at the scale of transmission. Second, the modelling strategy is built on interpretable ensemble methods, namely Random Forest, XGBoost, and LightGBM, coupled with SHAP analysis; here explainability is used not merely to describe the final model but to examine whether the drivers identified under the classification and regression formulations converge, thereby providing internal validation of the predictive signal. Third, the work confronts spatial generalisation directly, subjecting models to leave-region-out evaluation alongside temporal cross-validation, so that the divergence between temporal and spatial performance is measured rather than assumed. Fourth, the feature set is deliberately restricted to inputs that would be available at the moment of inference, treating the latency of case-reporting as a design constraint rather than an afterthought, so that reported performance reflects what an operational system could genuinely achieve.

MATERIALS AND METHODOLOGY

Building on the conceptual and theoretical foundations established in the previous chapter, this chapter sets out the materials and methodology through which the proposed malaria prediction system is constructed. It first presents the general principle of the approach, situating the modelling task and distinguishing the historical data on which the models are trained from the sensing infrastructure on which they are intended to operate. It then details the work in three sequential phases: the integration and preprocessing of district-level climatic and epidemiological data, the construction and validation of the predictive models, and their use for explainability analysis and district-level risk estimation. A closing summary draws the phases together and prepares the ground for the implementation and results reported in the following chapter.



2.1 General Principle of the Proposed Approach

This section presents the proposed system as a whole: the prediction task, the overall pipeline, the boundary of the contribution, and the distinction between training data and deployment substrate.

A clarification of the title is in order, because the study rests on a deliberate separation between the data source named in the title and the data on which the models are learned. The title refers to the *operational* data source of the deployed system: at inference the models consume environmental data from the IoT sensor network, deployed only in 2026 and therefore producing no historical training data. The models are consequently trained on a reanalysis proxy of the identical physical quantities the sensors measure, so that a model learned offline can be served, unchanged, on the live sensor stream. This separation is developed in the *Training Data and Deployment Substrate* subsection below.

2.1.1 The Prediction Task

The system is designed to anticipate malaria transmission risk at the district level in Cameroon from environmental and climatic conditions. The task is framed primarily as a binary classification problem, predicting whether a district will experience an elevated level of transmission in a given month, since an operational early-warning system issues categorical alerts on which preventive action can be taken. A regression formulation, predicting the continuous case count, is developed in parallel as a corroborative cross-check: the agreement of the environmental drivers identified under the two formulations provides supporting evidence, examined critically in Chapter 3, on whether the learned signal is a property of the phenomenon rather than of a single modelling choice. The classification task is therefore the operational objective, and the regression task serves a confirmatory, methodological role.

2.1.2 Overview of the Pipeline

The complete pipeline is shown in Figure 2.1. It proceeds through three sequential phases, which structure the remainder of this chapter. In **Phase 1**, heterogeneous data sources are integrated into a single district-level analytical dataset: climatic variables are ingested and harmonised, joined to malaria case records from the *Programme National de Lutte contre le Paludisme* (PNLP) through a fuzzy spatial join and district-level aggregation, and transformed into the engineered features on which the models depend. In **Phase 2**, regression and classification models are constructed, tuned, and subjected to a validation protocol that tests both temporal and spatial generalisation, and are compared against naive persistence and seasonal baselines. In **Phase 3**, the selected models are interpreted through SHAP analysis and used to produce district-level risk outputs. These three phases are framed by two further layers. Upstream, the **Acquisition Layer**, formed by the existing sensing infrastructure, collects the environmental data. Downstream, the trained models and their supporting artefacts are persisted as a deployment bundle, which feeds the **Presentation Layer**, implemented and demonstrated in Chapter 3: extending from the Recursive Chain to the end of the pipeline, it runs operational inference on the live monthly readings and serves and displays the resulting risk predictions. It comprises the Recursive Chain, a FastAPI backend serving the predictions and their SHAP explanations, an autonomous MLOps pipeline for the scheduled monthly runs, and a cartographic dashboard.

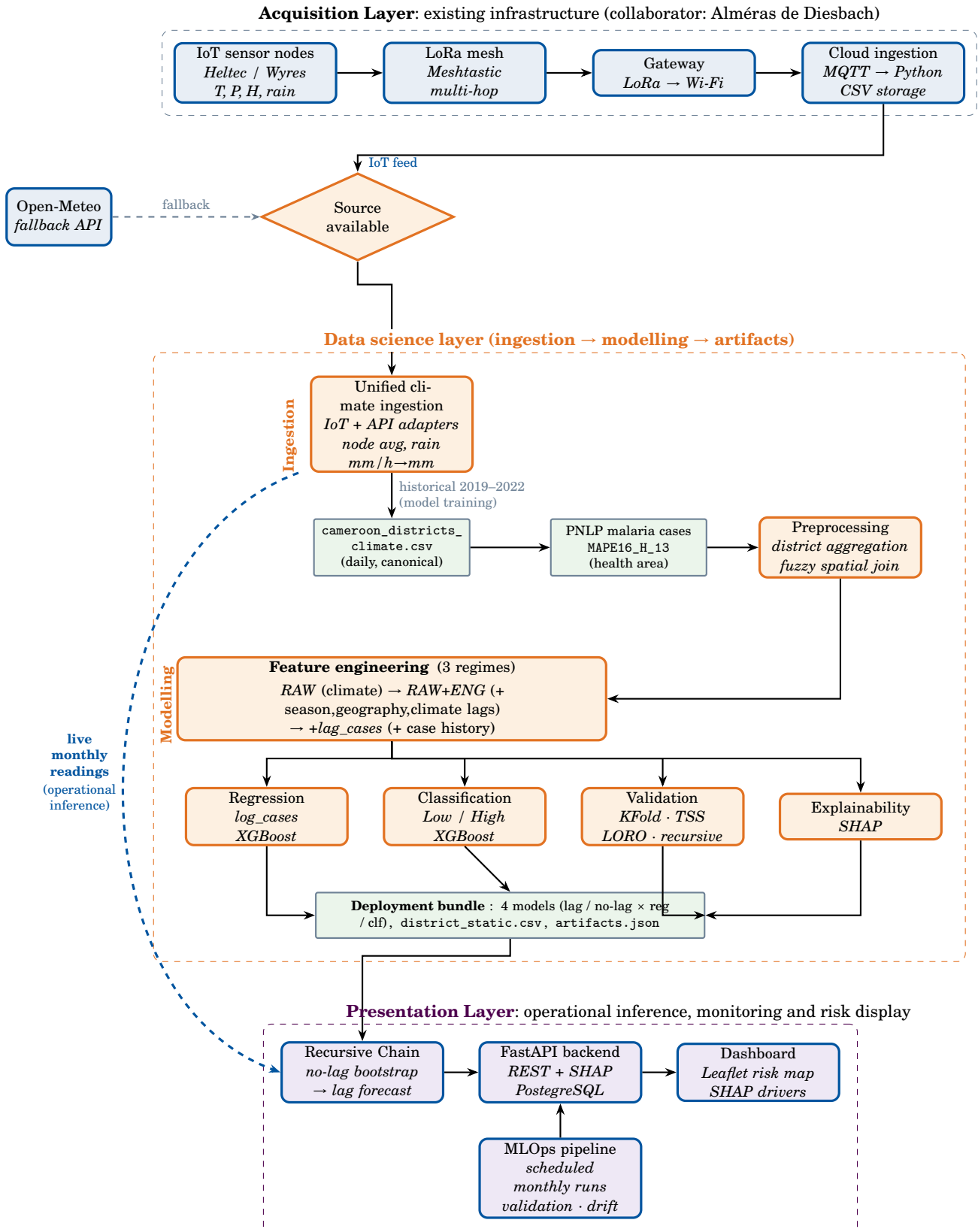


Figure 2.1: Overview of the proposed malaria prediction pipeline



2.1.3 Contribution Boundary

The pipeline integrates two distinct bodies of work, demarcated in Figure 2.1. The upper, blue-grey region is the Acquisition Layer, the existing sensing infrastructure that collects the environmental data: the IoT sensor nodes, the LoRa mesh network, the gateway, and the cloud-ingestion chain were designed and deployed by the collaborator Alm  ras de Diesbach [16], with the present author assisting in the field-deployment phase. The two lower regions are the contribution of this work. The central, orange region is the data-science layer, which begins at data integration: the unified ingestion and harmonisation of climatic data, the construction of the integrated district-level dataset, the feature engineering, the training and validation of the predictive models, the explainability analysis, and the persistence of the resulting artefacts. The lower, violet region is the Presentation Layer, which puts these models into operation: the Recursive Chain, the FastAPI inference service, the autonomous MLOps pipeline, and the cartographic dashboard. The description that follows covers these two layers; the sensing infrastructure is described only insofar as it constitutes the data source and the deployment substrate.

2.1.4 Training Data and Deployment Substrate

A central design consideration concerns the relationship between the data on which the models are trained and the infrastructure on which they are intended to run. The sensor network was deployed in Cameroon in 2026, after the historical period under study, and therefore produced no training data. The models are instead trained and evaluated on matched district-level records for the period 2019–2022, with malaria cases drawn from the PNLP and the corresponding climatic variables (temperature, humidity, pressure, and rainfall) obtained from the Open-Meteo Historical Weather API. These API variables are chosen precisely because they correspond to the quantities the sensor network measures, so that the historical data serve as a proxy for the operational data stream.

The Internet of Things network thus functions as the *deployment substrate* rather than as a training data source: at inference time, it supplies the same environmental variables in near-real time, at a spatial and temporal resolution that satellite and reanalysis products cannot match. This separation, training on a historical proxy and deploying on a live sensor stream, is deliberate, and the feature set is constructed throughout to respect the constraint that only quantities available at the moment of inference may be used as inputs. The implications of this constraint for feature engineering and for the treatment of historical case counts are developed in the phases that follow.

The two data streams therefore follow distinct paths, drawn separately in Figure 2.1. The *historical data* are used exclusively for model development: they pass through the canonical climate file, the preprocessing, and the feature engineering into the training and validation of the models, and end in the deployment bundle. The *live monthly readings*, delivered by the Acquisition Layer and aggregated to monthly resolution by the unified ingestion (with the Open-Meteo feed as a fallback), take no part in training: they are ingested directly into the Presentation Layer, where the Recursive Chain uses them as the environmental inputs of operational inference.

2.2 Data Integration and Preprocessing

The first phase of the methodology assembles the dataset on which all subsequent modelling rests. It first situates the IoT Acquisition Layer, and then describes the collection, retrieval, integration, cleaning, and transformation of the historical sources into a single machine learning dataset.

2.2.1 IoT-Based Environmental Data Acquisition

The environmental Acquisition Layer was designed, implemented, and experimentally validated by Alm  ras de Diesbach, who proposed a low-cost and low-power ground-based weather sensor network for malaria risk surveillance in dense urban environments [16]. It is presented here because it specifies the exact environmental quantities that the historical training data were required to reproduce.

2.2.1.1 Network Architecture

The Acquisition Layer is a wireless sensor network built on Long Range (LoRa) radio and organised as a mesh using the open-source Meshtastic protocol [16]; a mesh was preferred over a star topology because the dense urban fabric of Yaound   attenuates radio signals, and multi-hop relaying extends range and preserves connectivity when nodes fail. Three node roles compose the network: low-duty-cycle sensor nodes that emit telemetry over the mesh at a fixed sub-hourly cadence, solar-powered router nodes that relay traffic toward the gateway, and a gateway node that bridges the LoRa mesh to the Internet domain. The system thus spans two communication domains, a LoRa Meshtastic domain for in-field acquisition and an Internet domain for storage and processing, joined at the gateway, as reproduced in Figure 2.2 (Stream A the meteorological uplink, Stream B the downlink of instructions).

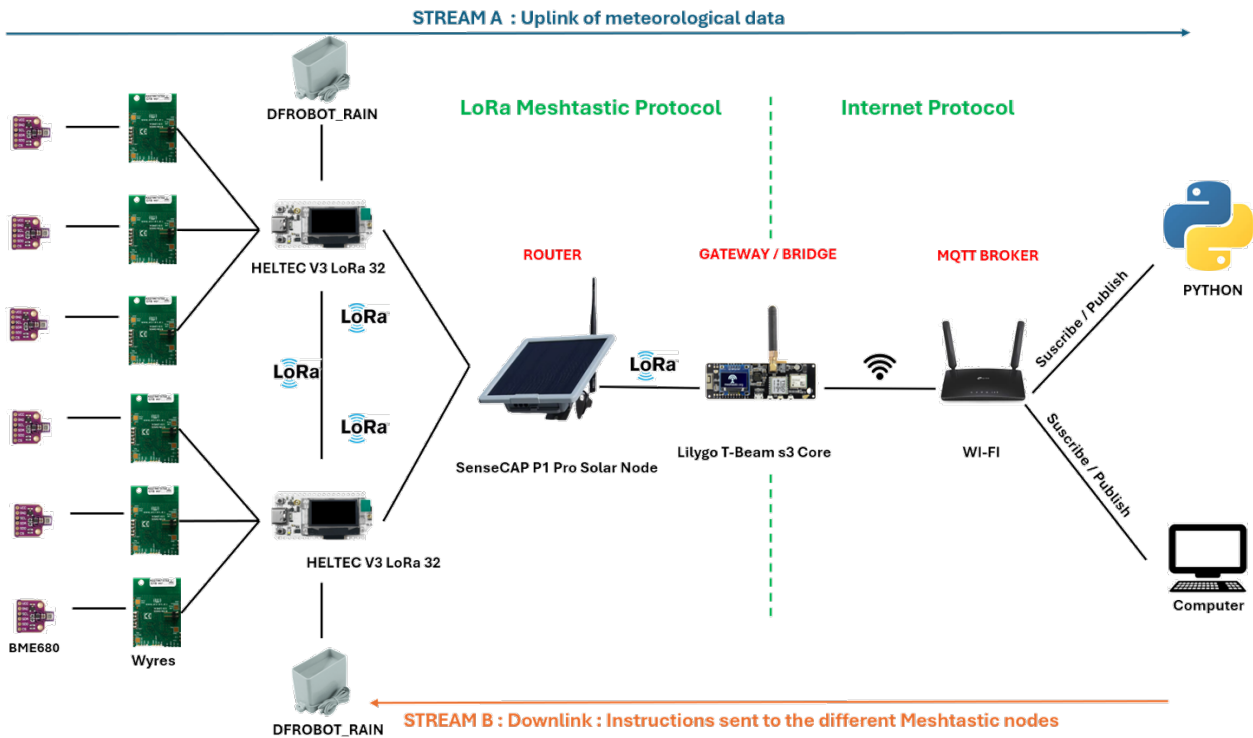


Figure 2.2: Architecture of the IoT Acquisition Layer, adapted from Alm  ras de Diesbach [16]

2.2.1.2 Deployed Sensors and Environmental Variables

Each sensor node integrates three environmental sensors over the I^2C bus, enclosed in an IP67-rated housing [16]: a BME680 for temperature, relative humidity, and atmospheric pressure; a DFRobot tipping-bucket rain gauge whose firmware derives rainfall intensity (mm/h) from tip counts; and a VEML7700 for ambient light converted to surface irradiance. Table 2.1 summarises the sensors and the variables they produce, as reported by Alm  ras de Diesbach [16].

Table 2.1: Sensors of the IoT nodes and the environmental variables they acquire

Sensor	Bus / principle	Variable(s) acquired
BME680	I^2C	Temperature, relative humidity, atmospheric pressure
DFRobot rain gauge	Tipping bucket (reed switch)	Rainfall intensity (mm/h)
VEML7700	I^2C	Ambient light $\rightarrow$ surface irradiance

Of the quantities the network produces, four are common to widely available historical weather sources, namely temperature, relative humidity, atmospheric pressure, and precipitation. As explained in Subsection 2.2.3, these four variables define the environmental feature space used for training, which guarantees that the operational sensors and the historical training data describe the same physical quantities. Surface irradiance is measured by the network but was not retained as a model input, because the historical source used for training does not provide a directly matching variable; retaining it would have created a feature available at inference but absent during training.

2.2.1.3 Transmission, Storage, and Role within the Prediction System

At the gateway the LoRa traffic is bridged to a Message Queuing Telemetry Transport (MQTT) broker following a publish/subscribe model [16]: the gateway publishes JSON telemetry, a subscribing client persists the readings to a store, and instructions are returned on a separate topic. Validated as a functional prototype in 2026, the infrastructure is dimensioned for an eventual city-scale deployment across Yaoundé, with sensor nodes grouped under solar-powered routers, each supplying local measurements at a temporal granularity that satellite and multi-model climate products cannot reach. Its purpose within the present work is operational rather than empirical: it is the substrate on which the trained model is intended to run once in service.

It should be noted that the spatial granularity of this network and that of the predictive model differ by design. The sensor network resolves environmental conditions at the intra-urban scale of individual nodes, whereas the models of this work operate at the health-district level, which is the unit at which the historical malaria labels are reported. The two are reconciled at inference by aggregating the local sensor measurements to the district unit expected by the model, in the same way that the historical climate descriptors were aggregated to the district level during training (Subsection 2.2.3). The dense urban network therefore demonstrates the feasibility of the operational acquisition substrate at high resolution, while the model consumes its measurements at the spatial unit for which it was trained.

2.2.2 Historical Malaria Case Data

The epidemiological ground truth was provided by the Cameroon National Malaria Control Program (Programme National de Lutte contre le Paludisme, PNLP), which compiles clinical malaria case reports from health facilities across the national territory. The records used in this work span a 38-month window from January 2019 to February 2022 inclusive, and are reported at the level of the health area, the finest administrative unit of the national health information system. The window is defined by the extent of the case reporting delivered by the PNLP: continuous monthly reporting across the national territory is available from January 2019, and the delivered extract terminates at February 2022, so the year 2022 is represented only by its first two months. The window is referred to as 2019–2022 for brevity, but the operative interval is these 38 contiguous months, and it is this interval, rather than three full calendar years, that defines the dataset and every figure derived from it. The delivered dataset



comprises 1,795 health areas, each carrying monthly malaria case counts together with population and geographic attributes.

The case records are organised in a wide format in which each column encodes a single month of observations through a composite naming convention that combines the month, the year, and the indicator code, with the target indicator denoted MAPE16_H_13. This indicator corresponds to the monthly malaria case count recorded by the PNLP, and is distinct from the separately coded indicators for suspected uncomplicated and suspected severe malaria; it serves both as the regression target and, after thresholding in a later phase, as the basis for the binary risk classification target. Because the operational unit of malaria control decision-making is the health district rather than the health area, the health-area records are aggregated to the district level during integration, as detailed in Subsection 2.2.4.

Two properties of the delivered extract must be stated, because both bear on how the results should be read. The first concerns completeness. The extract arrives as a complete rectangle, without a single missing or zero-valued cell across its 1,795 health areas and 38 months, and this completeness is not a property of the original surveillance returns. The PNLP collection is affected by intermittent connectivity and manual recording, and the delivered file has already been completed upstream: missing monthly values were estimated by a regression combining each indicator's own preceding three months, the corresponding values of the five neighbouring health areas, and the population and surface area of the target area and its neighbours weighted by inverse distance, with indicators retained only above a completeness threshold of 92 percent. A minority of the target values modelled in this work are therefore themselves estimates rather than direct observations, and the perfect rectangularity of the panel is an inherited property of that procedure rather than evidence of complete reporting. The exact proportion affecting this indicator is not recoverable from the delivered file, which carries no imputation flag.

The second concerns the accounting of months, which is easily confused because three distinct counts appear in this work. The delivered extract spans **38** months, January 2019 to February 2022. Forming the one- and two-month climate lags leaves the first two months of every district undefined, which yields the **36**-month modelling frame running from March 2019 to February 2022, of $197 \text{ districts} \times 36 \text{ months} = 7,092 \text{ district-months}$. Forming the case-history lags, which reach back three months, removes one further month and yields the **35**-month frame running from April 2019 to February 2022, of 6,895 district-months. No month is lost to data quality: all three counts follow from the depth of the lags alone.

2.2.2.1 Ethics and Data Authorisation

The records analysed here are routine monthly malaria surveillance returns of the PNLP, extracted from the District Health Information System (DHIS2) of the Ministry of Public Health of Cameroon [54]. Access to the platform and the delivery of the extract were authorised by the Head of the Cellule des Informations Sanitaires, who administers the DHIS2 instance, in the course of the collaborative work through which the dataset used here was assembled [55].

The records are aggregate counts reported at the level of the health area. They contain no individual-level information, no patient identifiers, and no data by which any person could be identified. The study involves no intervention, no contact with patients, and no collection of primary clinical data, and it therefore did not require review by an institutional ethics committee for research involving human subjects. The environmental covariates are drawn from a publicly accessible reanalysis service and carry no personal data; the sensor readings presented in Chapter 3 record physical quantities of the ambient environment only. The data are used solely for the research purposes described in this document, and the



complete pipeline is archived alongside the thesis so that the analysis can be inspected and reproduced.

2.2.3 Historical Climate Data Retrieval Using the Weather API

Because the IoT sensor network was not yet available during the 2019–2022 study period, the environmental conditions linked to historical malaria cases had to be reconstructed using an external climate source. For this purpose, historical weather data were collected from Open-Meteo Historical Weather API, which provides daily meteorological records through reanalysis data [56]. The variables retrieved included temperature, relative humidity, atmospheric pressure, and precipitation.

For each district, a single climate time series was extracted from a representative geographic point, defined as the centroid of the district based on the coordinates of its health areas. These records were stored in the district-level dataset (`cameroon_districts_climate.csv`).

To maintain consistency between training and deployment, the same centroid-based spatial reference is also used to aggregate live measurements from the IoT sensor network during inference. The four retrieved variables were not chosen arbitrarily. They are precisely the quantities that the deployed sensor network also measures (Table 2.1), which is what allows a model trained on the historical reanalysis to be served, unchanged, on the live sensor stream at deployment time. Retrieving variables that the network could not later supply would have produced a model that is not deployable on its intended substrate; conversely, the surface irradiance measured by the sensors was excluded from training because the historical source offers no matching reanalysis variable. This mutual alignment between the historical training source and the operational sensors is a deliberate design constraint of the pipeline rather than an incidental choice.

The constraint is worth stating in its full extent, because it also accounts for what is *absent* from the feature space. Chapter 1 reviews a wider set of environmental determinants of transmission, among them the vegetation and surface-water indices and terrain elevation, several of which are well attested in the literature. None of them appears in the models developed here, and their absence is not an oversight: none can be produced by a ground-level sensor node, so admitting them would have bought predictive performance at the price of a model that could not be served on the substrate the system is built for. The four-variable environmental space is what such a node can supply, and this work accepted that ceiling deliberately. The exclusion of surface irradiance is the sharpest illustration, since there the constraint cost a measurement the network already makes.

To match the monthly granularity of the malaria records, the daily climate series were aggregated to monthly summaries per district, yielding for each district and each month a set of environmental descriptors temporally aligned with the corresponding malaria case count. The state variables (temperature, humidity, pressure) are aggregated by their monthly mean and precipitation by monthly accumulation.

2.2.4 Spatiotemporal Data Integration

Integration joins the malaria records and the climate descriptors into a single tabular dataset indexed by district and month. Three operations were required: temporal parsing of the malaria column convention, spatial reconciliation of district names across the two sources, and aggregation of the malaria records from the health-area level to the district level.

The wide-format malaria columns were first reshaped into a long format. A dedicated parser, `col_to_date()`, decodes the composite `{Month}_{Year}_MAPE16_H_13` column names into explicit calendar dates, con-



verting each monthly indicator column into a dated observation. This produces a tidy table in which every row is a single (health area, month) observation carrying the malaria case count.

The malaria source and the climate table refer to districts through textual names that do not always coincide in spelling, accentuation, or formatting. To reconcile them, an approximate string-matching procedure based on the `rapidfuzz` library was applied, scoring each malaria district name against the climate-table names with the token-sort ratio and accepting the highest-scoring candidate above a similarity threshold, fixed at a token-sort ratio of 55. Matches flagged as ambiguous were resolved through a curated mapping (`FUZZY_MAP`).

Of the 197 districts, 118 carry an exact name match to a climate location, 43 are matched automatically by the approximate matcher at or above the threshold, ten are resolved by the curated mapping, and 26 match nothing at all and receive the region-and-month mean in place of a measured series. Four of the ten curated entries assign a district the climate series of a neighbouring district, because its own centroid series is absent from the climate table, which introduces a bounded source of climate-label noise on those four districts and is recorded here rather than presented as an exact match.

Every curated correspondence was checked against the administrative district list, and the join was verified to confirm that no malaria record was dropped for want of a match; the twenty-six unmatched districts are retained in the frame with imputed rather than measured climate, so completeness of the panel is preserved at the cost of climate fidelity on those districts.

Finally, the malaria records were aggregated from the 1,795 health areas to the health district level, the operational decision unit. Aggregation respects the nature of each attribute: extensive quantities, namely case counts and population, are summed across the constituent health areas; geographic coordinates and area are averaged to obtain a representative district centroid and extent; and the categorical cluster label is taken as the modal value across the constituent areas. The aggregated malaria records were then joined to the monthly district climate descriptors on the (district, month) key, producing the integrated dataset.

2.2.5 Data Cleaning and Preprocessing

Several cleaning operations were necessary before the integrated dataset could be used for modelling. The source files are semicolon-separated, and the geographic columns encode decimal values with a comma rather than a point, following the French locale convention. These columns were parsed accordingly and coerced to floating-point types, so that coordinates and area are interpreted numerically rather than as text.

Dates produced by the `col_to_date()` parser were normalised to a consistent chronological representation, which allows the records to be ordered in time and is a prerequisite for the temporally aware validation and the lag-based features introduced later. Missing values were handled according to their origin. Records for which the target indicator was absent were treated as non-observations and excluded from the supervised table, since no label exists to learn from. Isolated gaps in the explanatory climate variables, by contrast, were filled rather than discarded, so as not to lose an otherwise complete district-month. These gaps were imputed in two ordered stages that both respect temporal causality: first by the mean of the same region and month across districts, which substitutes a spatially local climatic value without drawing on the target series, and, for any residual gap, by the variable's overall mean. Both stages use only cross-sectional or global information and introduce no leakage from a district's own



future months into a given observation. The resulting dataset is internally consistent in type, encoding, and temporal indexing, and is free of the locale-specific artefacts of the raw sources.

2.2.6 Feature Engineering

The cleaned dataset was enriched with engineered features designed to expose the temporal, seasonal, and spatial structure of malaria transmission to the learning algorithms. Three families of features were constructed.

First, the calendar month, which is inherently cyclical, was encoded through its sine and cosine transforms, so that December and January are represented as adjacent rather than maximally distant, preserving the continuity of the seasonal cycle. Second, the geographic attributes of each district, namely latitude, longitude, area, and the categorical cluster label, were retained as spatial descriptors, since the spatial structure of transmission is strong in this setting. Third, temporal lag features were derived to capture the delayed response of malaria dynamics to environmental conditions and to recent transmission: lagged values of the climate variables (temperature, precipitation, and humidity) at one and two months account for the lag between environmental change and vector dynamics, while lagged malaria case counts at one, two, and three months, together with a three-month rolling mean, summarise recent transmission history. The case count itself was transformed with the $\log(1 + x)$ function to stabilise its variance and to support recursive forecasting at deployment time, where each predicted value re-enters the model as a lagged input for the following month.

These engineered features are organised into nested feature sets of increasing richness, so that later phases can quantify the marginal contribution of each family. A minimal raw set retains only the month and the four raw climate variables; an engineered set adds the cyclical month encoding, the geographic descriptors, and the climate lags; and a third set further incorporates the lagged and smoothed case-history features. Table 2.2 summarises the three feature sets. Their comparative evaluation, and the distinct handling that the case-history features require at deployment, are treated in the modelling phase.

Table 2.2: Nested feature sets constructed during Phase 1 and exploited in the modelling phase

Feature set	# features	Composition
Raw	5	Month and the four raw climate variables (temperature, humidity, pressure, precipitation).
Raw + Engineered	16	Adds cyclical month (sine, cosine), geographic descriptors (latitude, longitude, area, cluster), and climate lags (temperature, precipitation, humidity at one and two months).
+ Case history	20	Further adds lagged case counts (one, two, three months) and the three-month rolling mean of cases.

2.3 Model Construction

The second phase constructs the predictive models on the integrated dataset assembled in the preceding phase. Whereas the first phase was concerned with what the models learn from, the present





phase is concerned with how they are learned: the formulation of the prediction tasks, the definition of the classification target, the choice of learning algorithm, the treatment of class imbalance, the validation architecture through which model quality is measured, the baseline comparators against which the models are judged, and the optimisation of the model hyperparameters.

This phase is guided by a single principle: the model should be evaluated under conditions that closely reflect how it will be used in practice. Rather than focusing only on explaining historical data, the objective is to develop a system that can reliably predict future malaria risk using new environmental observations. For this reason, the validation strategy distinguishes between two evaluation settings. The first measures performance when the training and testing data come from the same time period, providing an optimistic estimate of model accuracy. The second evaluates the model on data from different months and regions that were not seen during training, giving a more realistic assessment of its ability to generalise. The same principle is applied to the case-history models. Because previous malaria case counts are not available when the system is deployed, these models are evaluated under conditions that reflect this limitation, ensuring that their reported performance represents what can realistically be expected in operational use.

The resulting construction workflow is summarised in Figure 2.3, in which each step carries the name of the subsection that describes it. The three nested feature sets of the integrated dataset feed the two prediction tasks, every model being trained on every feature set. A benchmark of candidate algorithms retains the gradient-boosted ensembles, whose hyperparameters are then optimised. The recursive forecasting procedure pairs a bootstrap model with the case-history model so that the latter can operate without measured case counts. Every model is finally evaluated under the four regimes of the validation architecture and set against the two baseline comparators on the same time-aware folds.

2.3.1 Formulation of the Prediction Tasks

The malaria target is approached through two complementary supervised tasks defined on the same integrated dataset. The first is a regression task that predicts the monthly malaria case count of a district, learned on the variance-stabilised target $\log(1 + x)$ introduced in the preceding phase. The second is a binary classification task that predicts whether a district faces a high transmission risk in a given month. The two tasks are not run in competition but in parallel, because they answer different operational questions. The regression provides a graded, quantitative forecast of the expected case burden, which is informative for resource planning, whereas the classification provides a discrete risk signal that is directly actionable as an early warning. Constructing both also serves a methodological purpose: the degree to which the two tasks identify the same environmental and spatial drivers is used, with the caveats set out in Subsection 2.4.3, as a consistency check on the predictive signal in the explainability phase.

The regression task targets the case count rather than the incidence rate at the modelling stage, so that the predicted quantity can re-enter the model as a lagged input in the recursive forecasting procedure described in Subsection 2.3.8. The incidence rate, which normalises the case count by population, is recovered from the predicted count and the district population when the discrete risk class is derived, as set out next.

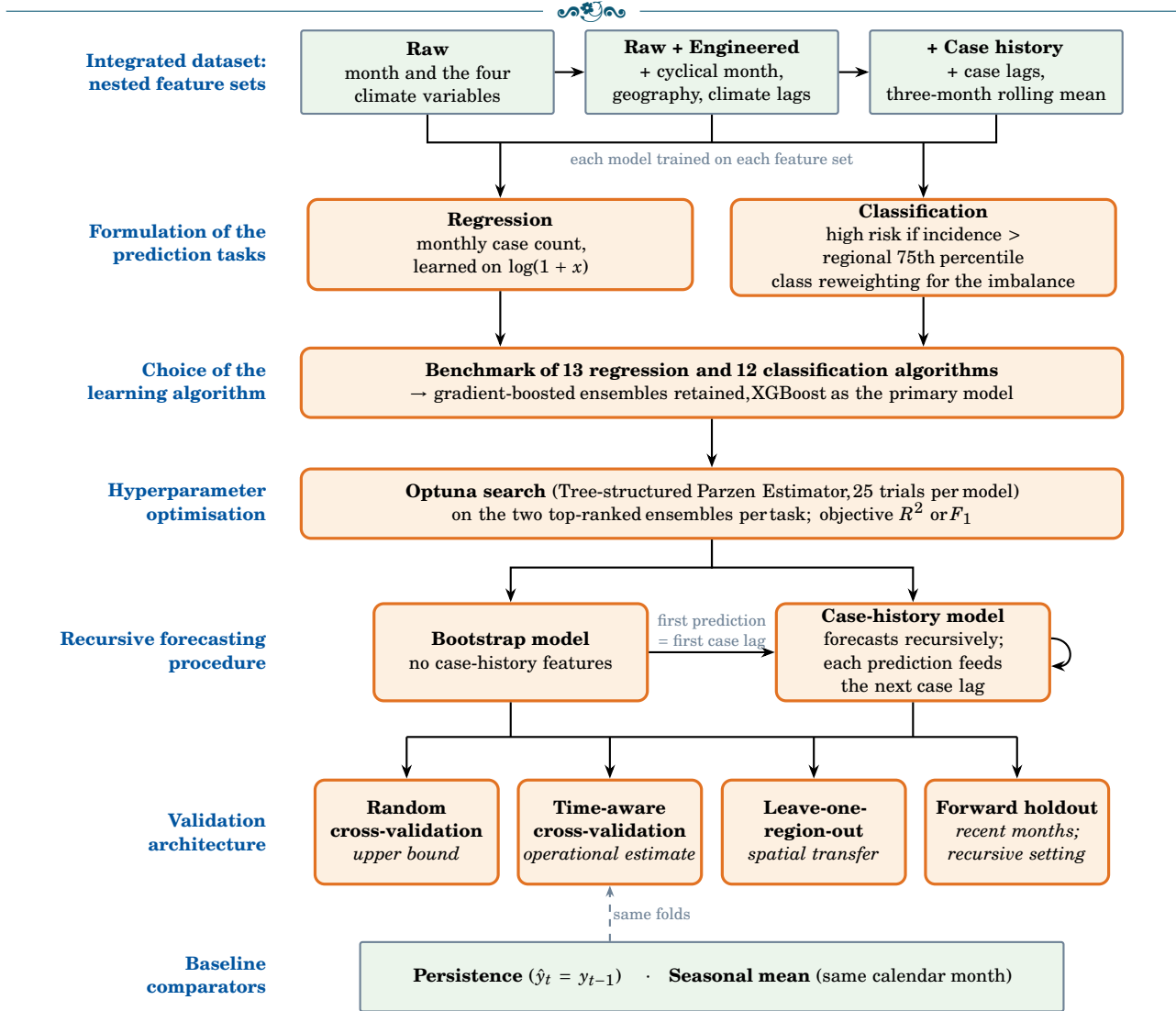


Figure 2.3: Model construction workflow

2.3.2 Definition of the Classification Target

The binary risk label is not an attribute of the raw data but a construction, and its definition is a methodological choice with direct consequences for the behaviour of the classifier. A district-month is labelled as high risk when its malaria incidence rate exceeds a threshold, and low risk otherwise. Two aspects of this definition required deliberate justification: the quantity to which the threshold is applied, and the value of the threshold itself.

The threshold is applied to the incidence rate, namely the case count normalised by the district population and expressed per one thousand inhabitants, rather than to the raw case count. A threshold on raw counts would conflate the burden of populous districts with genuinely elevated transmission, since a large district can report many cases while remaining at a moderate per-capita risk. Normalising by population makes the risk label comparable across districts of different sizes, which is the property required of an early-warning signal.

The denominator carries one limitation that must be stated. The source supplies a single population figure per health area and no annual series, so the district population used here is *constant across all 38 months* by construction, while the true population grows over the window. At a national growth rate of about 2.6 percent a year, a denominator fixed at its first-year value would overstate the incidence



of the final month by up to 8.5 percent, and a denominator representing the mid-period population would distribute an error of roughly plus or minus 4 percent across the window. The consequence for the classification task is smaller than this suggests, because the threshold is itself a percentile of the same quantity and the drift is common to all districts within a region: recomputing the labels against a denominator growing at that rate changes 96 of the 7,092 district-month labels, or 1.4 percent. The reported incidence *levels* should therefore be read as carrying a bias of this order, while the risk classification is substantially robust to it.

The threshold value is defined as the seventy-fifth percentile of the incidence rate computed separately within each region. A regional rather than a national threshold was adopted because the baseline level of malaria transmission varies substantially across the ecological zones of the country, so that a single national cut-off would label entire high-burden regions as permanently at risk and entire low-burden regions as permanently safe, collapsing the temporal discrimination that the early-warning task requires. An earlier definition based on the WHO-style fixed rule, which places the threshold two standard deviations above the long-term mean, as used in prior malaria classification work by Odu Nkiruka et al. [7], was discarded because at the monthly district resolution of the present data it produced a severely imbalanced label in which only a small minority of district-months were flagged as high risk; under such imbalance the classifier degenerates toward the trivial majority prediction and conveys no usable warning. The per-region seventy-fifth-percentile definition yields a label whose positive class is substantial enough to be learned while retaining a clear epidemiological interpretation, namely the upper quartile of a region's own transmission distribution.

The construction of these thresholds requires one explicit qualification, because it conditions how the spatial-generalisation results of Chapter 3 must be read. The per-region percentiles are computed once, over the whole 2019–2022 panel, before any partition into training and evaluation data. Under the random and time-aware regimes this is of little consequence, since every region contributes to training in either case and the threshold is a stable property of its distribution. Under leave-one-region-out it is material: the held-out region's labels are defined by that region's own historical distribution, which fixes its positive class at one quarter of its district-months by construction, and which an agency deploying into a genuinely new region would not possess. The leave-one-region-out score reported later must therefore be read as measuring transfer *given local calibration* rather than cold-start deployment. Substituting a threshold estimated only from the nine training regions lowers the mean held-out *F1* by approximately 0.07, which is the magnitude of the optimism this choice introduces.

2.3.3 Choice of the Learning Algorithm

Gradient-boosted decision tree ensembles were retained as the principal learning algorithm for both tasks, with the extreme gradient boosting implementation as the primary model. In all, thirteen regression and twelve classification algorithms were benchmarked, spanning linear, distance-based, and tree-ensemble families, so that the choice of learner rests on a broad empirical comparison rather than on assertion. This family was selected for properties that match the structure of the present problem. The integrated dataset is tabular and of moderate size, a regime in which gradient-boosted trees are consistently competitive and typically outperform both linear models and deep neural architectures, which require larger samples to generalise. The relationship between the environmental descriptors and malaria transmission is non-linear and involves interactions, for instance between season and geography, which tree ensembles capture natively through their hierarchical splits without manual specification.



The models are invariant to monotonic rescaling of the inputs, which is the property invoked in the preceding phase to justify the absence of global feature scaling. Finally, the tree structure yields feature-attribution quantities that the explainability phase exploits directly, so that the predictive model and its interpretation are drawn from the same object rather than requiring a separate surrogate.

Deep learning architectures were explored in preliminary experiments, using long short-term memory (LSTM) and gated recurrent unit (GRU) networks over sliding-window sequences, but did not match the gradient-boosted ensembles in this data regime, returning negative R^2 on both tasks. The available sample, comprising a few dozen monthly observations per district, is small relative to the data requirements of such architectures, so these models were not carried into the final protocol; the gradient-boosted ensembles provide both stronger performance in this regime and the native interpretability that the early-warning objective requires. A panel of simpler and scale-sensitive baselines, including linear and distance-based models, was nonetheless retained in the comparison so that the benefit of the chosen family could be quantified against meaningful references rather than asserted.

2.3.4 Treatment of Class Imbalance

The per-region seventy-fifth-percentile definition places roughly one quarter of district-months in the high-risk class, so that the negative class predominates without the extreme imbalance that the discarded definition produced. This residual imbalance was nonetheless addressed explicitly, because an early-warning system must not purchase a high overall accuracy by neglecting the minority class that carries the operational signal. During training, the classification loss was reweighted in inverse proportion to class frequency, so that errors on the scarce high-risk class are penalised more heavily than errors on the abundant low-risk class. This reweighting steers the model toward correctly identifying high-risk district-months rather than toward the trivial majority prediction. The effect of the reweighting, measured against an unweighted baseline, is reported in Chapter 3, where metrics appropriate to imbalanced classification, rather than overall accuracy alone, are used to assess it.

2.3.5 Validation Architecture

The validation architecture is the methodological core of the phase, because it determines whether the reported performance reflects the conditions the model will face in service or an optimistic artefact of the evaluation procedure. Four complementary regimes were constructed, each answering a distinct question about generalisation, and each retained because it exposes a limitation that the others conceal. The regimes are summarised in Table 2.3.

The first regime is a conventional cross-validation in which the dataset is partitioned at random into folds. Because the partition ignores time, the training and evaluation folds contain observations from the same calendar months, so that the model is implicitly allowed to see the period it is tested on. This regime is informative as an upper bound and for comparison with the literature, but it overstates the performance attainable when the model must forecast forward in time, a discrepancy known as temporal leakage.

The second regime is a time-aware cross-validation in which the folds respect the chronological order, so that the model is always trained on past months and evaluated on subsequent ones. This regime reproduces the operational situation in which next month is predicted from the months already observed, and it therefore provides the estimate of temporal generalisation. The difference between this



regime and the random one quantifies the temporal leakage and is reported as a diagnostic rather than concealed.

The third regime evaluates spatial generalisation through a leave-one-region-out procedure, in which the model is trained on all regions but one and evaluated on the held-out region, repeated until every region has served once as the test region. This regime answers the question of whether a model trained on the observed regions can predict a region absent from training, which is the situation that would arise if the operational network were extended to an area with no historical case records. Because the geographic descriptors carry a substantial part of the predictive signal, this regime is the most demanding of the four and exposes the limit of spatial transfer.

The fourth regime is a forward holdout in which the most recent months are withheld entirely and predicted after training on all earlier months. It serves as a single, interpretable out-of-time test that corroborates the time-aware cross-validation on a contiguous block of unseen months, and it is the setting in which the recursive forecasting procedure of Subsection 2.3.8 is exercised.

Every reported cross-validated score is the mean across its folds, and is accompanied in Chapter 3 by the fold-to-fold standard deviation, so that the closeness of two competing models is read against the dispersion of the estimate rather than from the point value alone.

Table 2.3: Validation regimes constructed in the modelling phase and the generalisation question each one measures

Validation regime	Generalisation question
Random cross-validation	In-sample interpolation; upper bound, subject to temporal leakage.
Time-aware cross-validation	Generalisation to future months in known districts; the operational estimate.
Leave-one-region-out	Generalisation to a region absent from training; the spatial-transfer limit.
Forward holdout (recent months)	Out-of-time corroboration on a contiguous block; setting for recursive forecasting.

2.3.6 Baseline Comparators

To situate the learned models against trivial alternatives, and in particular to isolate how much of the explained variance reflects a genuine environmental forecast rather than the strong month-to-month autocorrelation of transmission, two naive baselines are constructed and evaluated on the same time-aware holdout folds as the models. The persistence baseline predicts each district-month as that district's previous month, $\hat{y}_t = y_{t-1}$; it uses last month's observed value and is therefore a strong but non-deployable reference, since the latency of monthly case reporting renders the previous confirmed count unavailable at the moment a forecast is required. The seasonal baseline predicts the per-district mean of the same calendar month, fitted on the training fold alone, and captures pure seasonality with no environmental input.

The purpose of these comparators is diagnostic. Because malaria transmission is strongly autocorrelated, the persistence baseline is a demanding reference, and the margin by which the environment-only model exceeds the seasonal baseline, and falls short of persistence, is the appropriate measure of the environmental signal's operational value. This comparison is the methodological reason the recursive forecasting procedure of Subsection 2.3.8 is required: it is the mechanism by which the persistence sig-



nal, valuable but unavailable at inference, is reconstructed synthetically. The outcome of the comparison is reported in Chapter 3.

2.3.7 Construction across the Feature Sets

The three nested feature sets defined in the preceding phase, namely the raw set, the engineered set, and the set further enriched with case history, are not alternatives to be chosen between but a progression to be constructed and compared. Each model was trained on each feature set under the validation regimes above, so that the marginal contribution of each feature family could be isolated: the raw set establishes the baseline attainable from current climate alone, the engineered set quantifies the gain from seasonal encoding, geographic context, and climate lags, and the case-history set quantifies the further gain available when recent transmission history is supplied. Constructing the progression explicitly, rather than reporting only the richest model, is what allows the contribution of each engineering decision to be attributed rather than assumed.

The case-history set requires a distinction that the other two do not. Its lagged case features are strongly predictive, because malaria transmission is autocorrelated from one month to the next, but these features are not available at inference time in a system whose live inputs are environmental measurements alone. The performance attained when real case lags are supplied is therefore an upper bound that is not by itself deployable, and the construction of the case-history model is only complete once it is paired with the recursive procedure that supplies those lags from the model's own predictions, described next.

2.3.8 Construction of the Recursive Forecasting Procedure

To make the case-history model deployable under the constraint that no measured case counts are available at inference, a recursive forecasting procedure was constructed in which the model consumes its own predictions in place of the missing case lags. The procedure pairs two models that differ only in their feature set. A bootstrap model, trained without case-history features, produces the first prediction of a forecast sequence from environmental and spatial descriptors alone, since no case history yet exists to draw on. Its predicted case count then becomes the lagged case input for the following month, at which point the case-history model takes over and forecasts recursively, each prediction supplying the lag for the next. Where a measured case count does become available for a given month, it overrides the synthetic prediction and re-anchors the sequence, so that the procedure degrades gracefully from fully synthetic to partially observed operation.

The methodological significance of this procedure is that it transforms the case-history model from an upper bound that presupposes observed lags into a deployable forecaster whose lags are generated internally. Its evaluation is correspondingly distinguished from the upper-bound case: the case-history model is assessed both with the real case lags, which measures its ceiling, and under the recursive procedure on the forward holdout, which measures the realistic performance attainable when the lags are synthetic. The gap between the two, and the manner in which the recursive error accumulates over the forecast horizon as each prediction feeds the next, are reported in Chapter 3. The operational integration of the two-model chain into the inference service is treated in Chapter 3; the present phase establishes only its construction and its role in evaluation.



2.3.9 Hyperparameter Optimisation

The hyperparameters governing the complexity and the regularisation of the gradient-boosted ensembles, namely the number and depth of the trees, the learning rate, and the subsampling rates, were tuned by automated search rather than set by hand, so that the comparison between models and feature sets reflects each model near its attainable best rather than at an arbitrary configuration. A sequential model-based optimisation strategy was adopted through the Optuna framework, using a Tree-structured Parzen Estimator sampler over twenty-five trials per model, which proposes configurations, evaluates each by cross-validation, and concentrates subsequent proposals in the regions of the search space that performed well. The two top-ranked ensembles from the engineered-feature comparison were carried into the search for each task, the regression objective being the coefficient of determination and the classification objective the F_1 score.

2.4 Model Use, Explainability, and Output

The third phase concerns what the constructed models produce and how that production is made interpretable. The preceding phase established how the models are learned and validated; the present phase establishes how a learned model is used to generate a forecast, how that forecast is explained, and what the system ultimately emits for each district and month.

Three concerns organise the phase. The first is the use of the model in inference, namely the procedure by which a feature vector is turned into the quantities of interest, including the recovery of the incidence rate and the discrete risk class from the predicted count, and the operation of the recursive forecasting chain when no measured case history is available. The second is explainability, namely the attribution of each prediction to the variables that produced it, which the early-warning purpose requires so that a forecast can be justified rather than merely asserted. The third is the output of the system, namely the set of quantities emitted for each district-month, including the prediction, its associated risk signal, and its explanation.

2.4.1 Use of the Models in Inference

In inference, a trained model receives the feature vector of a district for a given month and returns a prediction on the variance-stabilised scale on which it was learned. The first operation in using the model is therefore to invert that transformation, recovering the predicted malaria case count from the $\log(1+x)$ -scaled output. The predicted count is the primary quantity returned by the regression model and the basis from which the remaining outputs are derived.

From the predicted count, the two operationally meaningful quantities are obtained. The incidence rate is recovered by normalising the predicted count by the district population and expressing it per one thousand inhabitants, which restores the population-comparable measure on which the risk label was defined in the construction phase. The discrete risk class is then obtained by comparing the predicted incidence rate to the district's regional threshold: a predicted incidence above the threshold yields a high-risk signal, and a value below it a low-risk signal. The threshold used at inference is the one fixed on the training data during construction, so that the decision boundary applied in service is identical to the one under which the model was validated. The classification model is used in parallel to return the risk class directly, together with a probability that expresses the confidence of the high-risk assignment; the probability is retained as part of the output because an early-warning signal is more actionable





when its certainty is quantified than when it is reduced to a bare label. The two paths to the risk class, the regression route through incidence and the direct classifier, can disagree near the threshold; that disagreement is not suppressed but retained as an uncertainty signal, as discussed in Chapter 3. Where the forecast runs ahead of the available case history, the inputs the case-history model requires are supplied by the Recursive Chain constructed in Subsection 2.3.8, which operates on observed counts wherever they exist and on its own predictions wherever they do not.

2.4.2 Explainability through Feature Attribution

A predictive model that issues a risk signal without justifying it is of limited use in a public-health setting, where a warning must be interpretable to those who act on it and auditable by those who oversee the system. The constructed models are ensembles of many decision trees, whose aggregate decision is not legible by inspection, and an explicit interpretation procedure is therefore required to render each prediction explainable. The attribution method adopted for this purpose is the computation of Shapley additive explanations, which assign to each input feature a contribution to the difference between a given prediction and a baseline expectation, with the property that the contributions sum to that difference and so account for the prediction exactly.

This attribution method was selected for two reasons that match the present setting. It is model-agnostic in principle yet admits an exact and efficient computation for tree ensembles specifically, so that the explanation is drawn from the same model that produced the prediction rather than from an approximate surrogate, and it is obtained without the sampling cost that a general-purpose estimator would incur. It also produces explanations at two complementary scales from a single procedure. At the local scale, the attribution decomposes an individual district-month prediction into the signed contribution of each feature, identifying which environmental, seasonal, spatial, or case-history variables raised the predicted risk and which lowered it. At the global scale, aggregating the local attributions across the dataset ranks the features by their overall influence, summarising which variables drive the model's behaviour as a whole.

2.4.3 Internal Consistency between the Two Tasks

The parallel construction of a regression and a classification model on the same dataset, motivated in the construction phase, is exploited here as a supplementary check on the predictive signal. The two models are trained on different targets, the continuous case count and the binary risk class, and they optimise different objectives; if both nonetheless attribute their predictions to substantially the same variables, that agreement is consistent with the drivers being a property of the phenomenon rather than of a single target formulation.

This check is deliberately framed as corroboration rather than proof; its limits, and the reasons agreement is partly expected by construction, are stated with the result in Chapter 3.

2.4.4 System Output

The output of the system, for each district and each month of a forecast, is the set of quantities that together constitute an actionable and explainable early-warning record. It comprises the predicted malaria case count, the predicted incidence rate derived from it, the discrete risk class with the probability that quantifies its confidence, and the local feature attribution that explains the prediction. The output also



records which of the two models in the Recursive Chain produced the prediction, so that a forecast obtained from observed history is distinguishable from one obtained from synthetic lags, preserving the provenance of each value.

The explanatory component of the output is derived from the local attribution by ranking, for the prediction in question, the features by the magnitude of their contribution and reporting for each its value, the signed contribution, and whether it raised or lowered the predicted risk. This ranked attribution is structured so that a concise, human-readable summary of the dominant drivers and the direction in which each acted can be presented alongside the numerical output, so that the explanation is legible to a user who is not a specialist in the underlying model. The output is thus structured to serve two audiences from the same content: the ranked numerical attribution for analytical scrutiny, and the summary reading for operational use.

This output is the terminal product of the methodology. It is the quantity that the inference service and the cartographic interface render, and it is the object whose quality, across the regression, the classification, the recursive forecasting, and the feature attributions, is evaluated in Chapter 3.

2.5 Chapter Summary

This chapter set out the methodology in three phases. The first integrated the historical malaria records of January 2019 to February 2022 with the corresponding environmental variables, producing the district-level dataset and the three feature regimes; the IoT network, deployed after that period, was designed to collect the same quantities, which is what allows the trained models to be served on it, subject to the proxy validation flagged as outstanding. The second developed the models for both tasks, addressed the class imbalance, introduced the persistence and seasonal baselines as diagnostic comparators, and adopted the four-regime validation strategy together with the recursive forecasting procedure that makes deployment possible without live case reports. The third specified how the models are used at inference, how their predictions are explained through feature attribution, and what the system emits. The next chapter presents the implementation of this framework, its experimental evaluation, and the results obtained.

IMPLEMENTATION, RESULTS AND DISCUSSION

T*his chapter reports the implementation of the methodology developed in the previous chapter and the experimental results it produces. It first describes the implementation: the tools and computational environment, the experimental protocol, and the study area and integrated dataset on which the experiments are conducted. It then presents the results in the order in which the methodology constructs them, namely the regression and classification performance, the behaviour of the models under the temporal and spatial validation regimes, the outcome of the recursive forecasting procedure, and the explainability analysis together with the cross-task consistency of the identified drivers. A discussion then interprets these findings against the objectives set for the system and the gaps identified in the state of the art, and states the limitations of the work, and a closing summary draws the chapter together.*



3.1 Presentation of the Data

3.1.1 Data Sources

The work draws on two epidemiological and environmental sources, together with a third operational source used at deployment, which must be kept distinct because they intervene at different moments in the lifetime of the system. The epidemiological ground truth was provided by the Cameroon National Malaria Control Program (Programme National de Lutte contre le Paludisme, PNLP), which compiles monthly confirmed malaria case counts from health facilities across the national territory over the 2019–2022 period, reported at the level of the health area and delivered for 1,795 health areas. The historical environmental conditions for the same period were retrieved from the Open-Meteo Historical Weather API [56], a reanalysis service supplying daily temperature, relative humidity, atmospheric pressure, and precipitation for each district. The third source is the IoT sensor network deployed in Yaoundé in 2026 [16], which supplies the live environmental measurements on which the trained model operates in service.

3.1.2 Organisation of the Data

The malaria records are organised in a wide tabular format in which each column encodes one month of observations through a composite naming convention combining the month, the year, and the indicator code, the target indicator being MAPE16_H_13, the monthly confirmed malaria case count. Each health area carries, in addition to the monthly case columns, its population and its geographic attributes, namely latitude, longitude, area, and an ecological cluster label. The historical climate is stored in a separate district-level table (`cameroon_districts_climate.csv`) holding the daily series of the four environmental variables for each district over the study period. The two are joined during integration on the district-and-month key, as described in the implementation section.

3.1.3 Exploratory Analysis of the Dataset

After aggregation of the health-area records to the health-district level, the operational decision unit, the integrated dataset comprises 7,092 district-month observations, spanning 197 health districts across the 10 regions of the national territory over a 36-month sequence, from March 2019 to February 2022. Of these district-months, 1,811, or about one quarter, fall in the high-risk class under the per-region threshold defined in the methodology. Descriptive properties of the integrated dataset are summarised in Table 3.1. The malaria case count is strongly right-skewed, a small number of district-months carrying very high counts, which motivates the $\log(1+x)$ transformation of the regression target introduced in the methodology and the definition of the risk class on the population-normalised incidence rate rather than on the raw count.

3.1.3.1 Climate Evolution in Yaoundé

Because the dataset spans many districts of differing ecological zones, the exploratory characterisation of the environmental signal is focused on Yaoundé, the district in which the IoT network was deployed and which is therefore the operational reference of the system. Yaoundé is presented here for this operational reason, and not as representative of the national climate: Cameroon spans from the Sudano-Sahelian regime of the far north to the equatorial regime of the south, so the bimodal equato-

Table 3.1: Descriptive summary of the integrated district-level dataset

Property	Value
Health areas in the source records	1,795
Aggregated health districts	197
Regions covered	10
District-month observations	7,092
Months covered	36 (2019-03 to 2022-02)
High-risk district-months	1,811 (25.5%)
Environmental variables	4
Feature sets constructed	3 (5, 16, 20 features)
Regression target	$\log(1 + \text{cases})$
Classification target	incidence > regional 75th percentile

rial cycle described below characterises the deployment site rather than the training set as a whole, a heterogeneity whose consequences for spatial generalisation are quantified later in this chapter.

Figure 3.1 presents the monthly evolution of temperature, relative humidity, atmospheric pressure, and precipitation in Yaoundé over the 2019–2022 study period, averaged across the three urban health districts. The shaded bands mark the annual rainy periods.

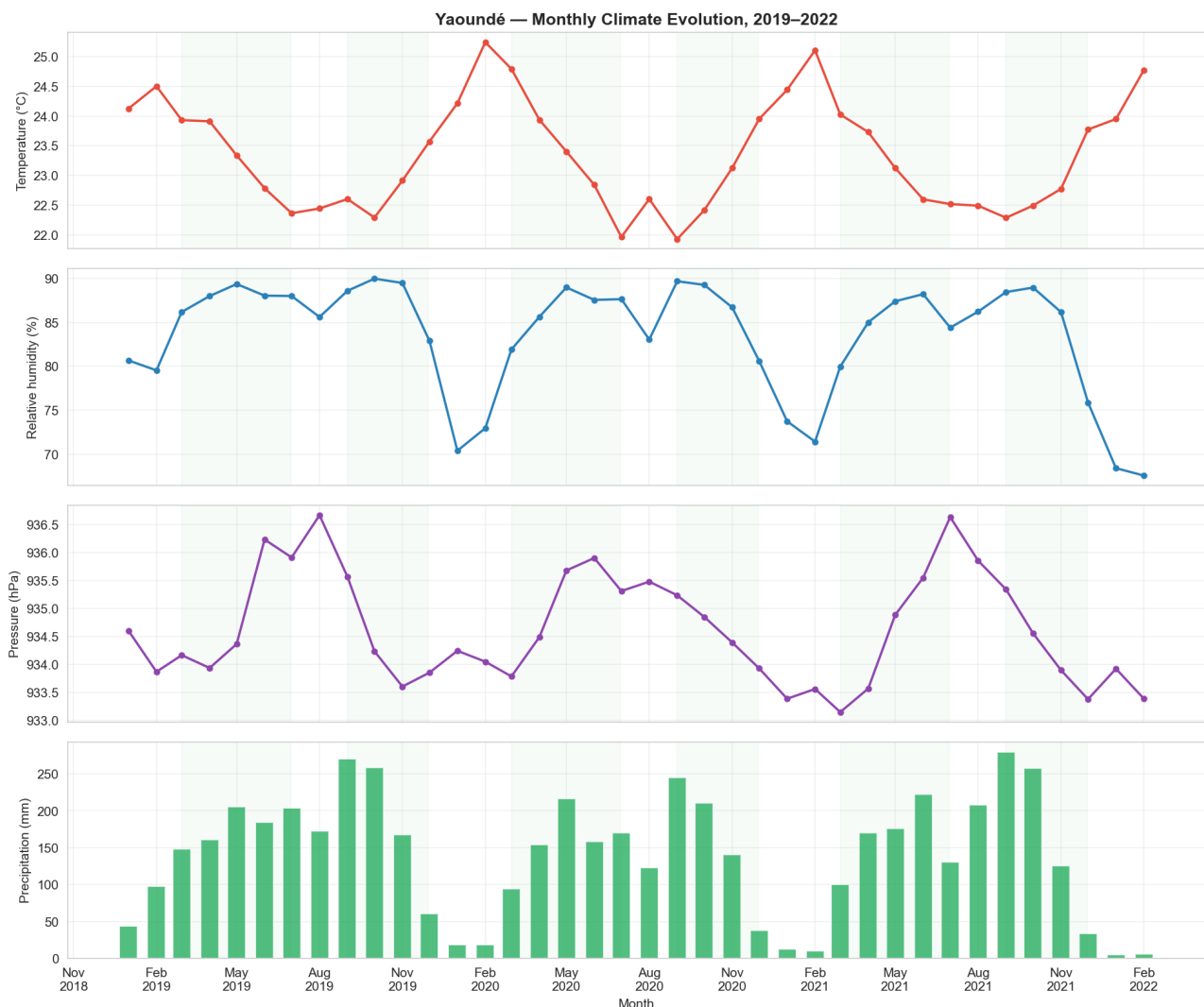


Figure 3.1: Monthly climate evolution in Yaoundé, 2019–2022



3.1.3.2 Malaria Target and the Risk Threshold in Yaoundé

Having characterised the environmental signal, the exploratory analysis now turns to the prediction target itself, again focused on Yaoundé. The three urban health districts of the capital, namely Biyem Assi, Efoulan, and Nkolndongo, are examined individually, since the per-district incidence, not the city aggregate, is the unit on which the model operates and on which the binary risk class is defined.

The definition of the binary risk label is illustrated in Figure 3.2, which draws the incidence of each district together with the regional risk threshold, the seventy-fifth percentile of the incidence distribution of the Centre region, and marks each district-month as low or high according to whether it lies below or above that threshold: the threshold is drawn as a dashed line, the high-risk district-months that the classifier predicts are shown in red, and the low-risk months in blue. The figure makes the structure of the classification target concrete: within Yaoundé, the high-risk months are concentrated almost entirely in the Biyem Assi district, which records eighteen high-risk months over the 38-month panel and sixteen over the 36-month modelling frame, whereas the incidence of Efoulan and Nkolndongo never reaches the threshold. This illustrates that the threshold operates at the district level rather than the city aggregate, and that the high-risk signal the classifier must learn is carried, within the capital, by the most affected district. It also illustrates, in miniature, the two regimes the target exhibits across the national dataset. Efoulan and Nkolndongo are *constant*: the regional threshold places them below it in every month, so no temporal discrimination is possible for them. Biyem Assi *crosses*: it lies above the threshold in some months and below it in others, and it is districts of this kind, which are the majority nationally, that carry the temporal signal the classifier is asked to learn. Subsection 3.4.5 quantifies that split over all 197 districts.

The threshold falls in the upper part of the incidence distribution, isolating the upper quartile of district-months rather than only the rare extreme values, which is the property that motivated the choice of the seventy-fifth percentile over a definition based on extreme deviations from the mean; the interquartile range of Biyem Assi straddles it, while those of Efoulan and Nkolndongo lie wholly below.

The relationship between the environment and the target is examined in Figure 3.3, which reports the correlation between malaria incidence and each climatic variable for lags ranging from zero to three months. A distinct pattern emerges. Temperature exhibits the strongest and most consistent relationship with malaria incidence, remaining positively correlated across all lags and reaching its highest value ($r \approx 0.41$) at a two-month lag. This delay is biologically plausible, as warmer conditions require time to influence mosquito population dynamics and the development of the malaria parasite before changes become visible in reported cases. Precipitation and relative humidity, by contrast, are near zero at short lags and turn weakly negative at two-to-three-month lags ($r \approx -0.25$ and -0.26). This pattern is coherent with the equatorial climate of Yaoundé, where rainfall and humidity are high throughout the year and are therefore not the factors that limit transmission; it is temperature, acting with a delay of one to two months, that provides the leading environmental signal in this setting, which is precisely the relationship the lagged climate features are designed to capture. These correlations are computed on the city-level Yaoundé series and are indicative rather than definitive; the feature attribution of Section 3.4 revisits the drivers at the full district scale.

The full description of the environmental data emitted by the operational IoT network, in both its raw and its preprocessed form, is presented later in this chapter, in Subsection 3.3.4, at the point where the ingestion layer that processes it is described.

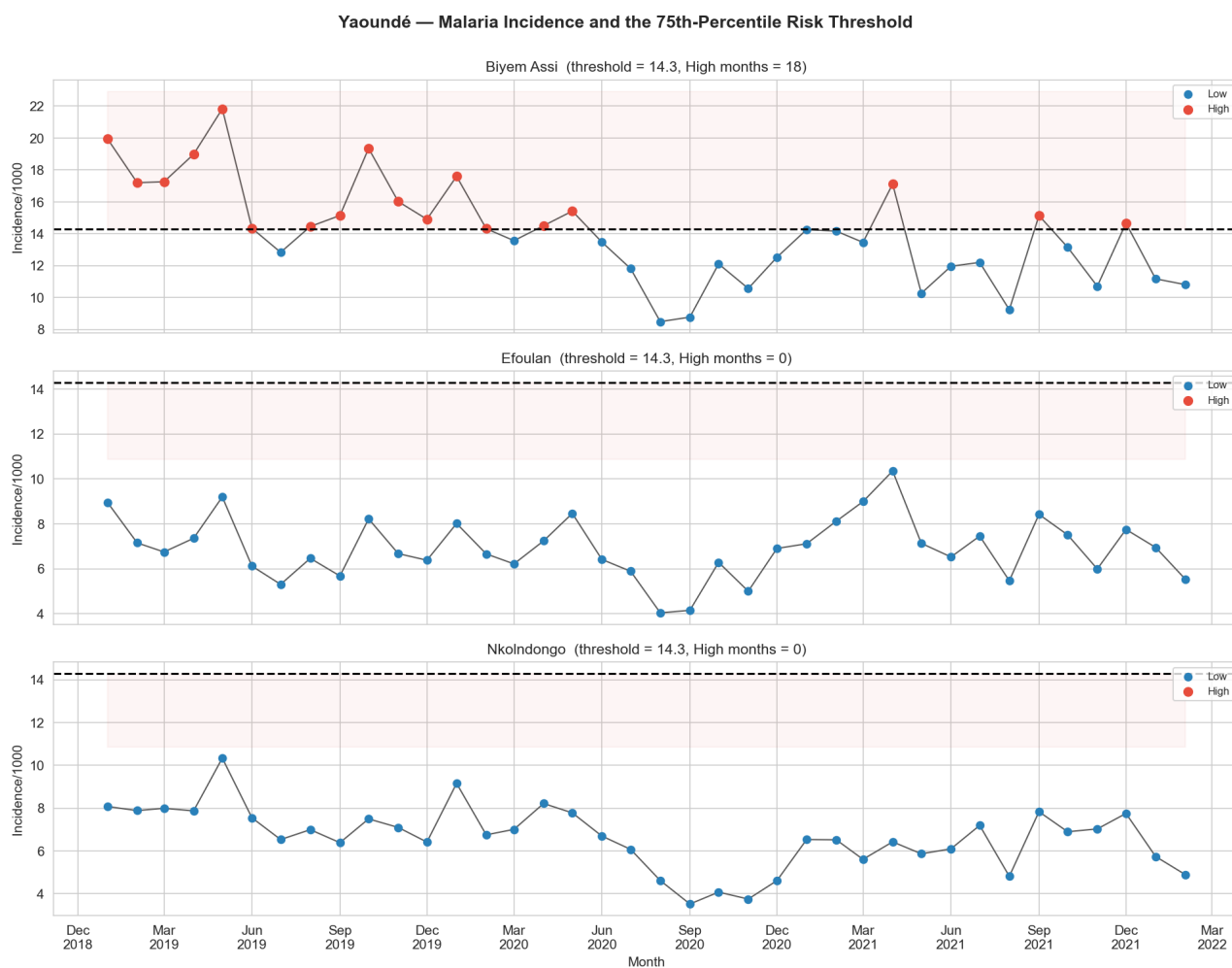


Figure 3.2: Malaria incidence of the three Yaoundé districts and the regional 75th-percentile risk threshold

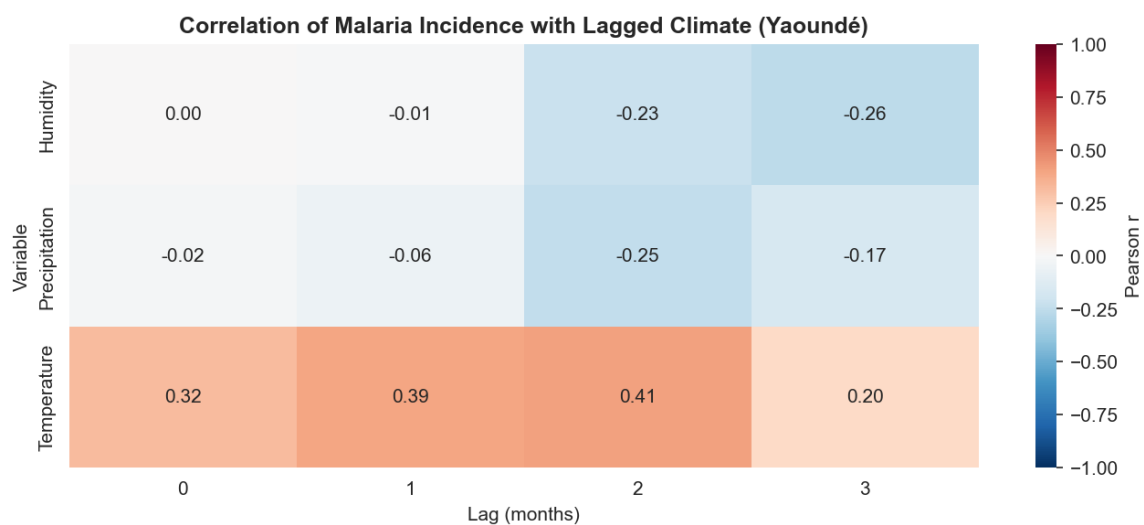


Figure 3.3: Correlation of Yaoundé malaria incidence with each climate variable at lags of zero to three months



3.2 Experimental Environment

The experiments were carried out on a consumer laptop with an Intel Core i7 processor and 8 GB of memory: the dataset is tabular and of moderate size, so no dedicated graphics acceleration is required, and the gradient-boosted ensembles train efficiently on the central processor. The implementation rests on the Python scientific ecosystem for data manipulation, model training, optimisation and explanation, and on a web framework for the operational service. The full hardware and software specifications, the library versions, the random seeds and the hyperparameter search space are given in Appendix A.

3.3 Implementation

The implementation realises, as running code, each phase of the methodology, from the construction of the supervised dataset to the assembly of the deployable inference service. It is presented in the order of the system flow, so that each component appears where it intervenes. The performance of the resulting models is reported in Section 3.4; the present section is concerned with how the system is built, and the figures it reports are descriptive properties of the data and the artefacts rather than measures of predictive quality.

3.3.1 Implementation of Model Training, Validation, and Tuning

The training and validation were implemented as a structured experimental protocol over the two tasks, the three feature sets, and the four validation regimes. The random and time-aware cross-validations were implemented through the partitioning utilities of scikit-learn, and the leave-one-region-out and forward-holdout regimes explicitly over the region and date indices of the dataset. The class imbalance of the classification task was handled by reweighting the loss in inverse proportion to class frequency. The hyperparameters of the principal ensembles were tuned with Optuna, the tuning objective used a random-split inner cross-validation, so the tuned configurations are optimised toward in-sample interpolation and the reported time-aware and spatial estimates are, if anything, conservative for those configurations. The recursive forecasting procedure was implemented as a two-model chain in which a bootstrap model, trained without case-history features, initiates a forecast sequence and the case-history model continues it, each prediction supplying the lagged case input of the next, a measured case count overriding the synthetic value whenever one is available.

3.3.2 The Deployment Artefacts

The experimental pipeline culminates in a deployment bundle that is the sole interface between the offline construction and the online service. The bundle comprises four serialised models, namely the regression and the classification models in their case-history and their bootstrap variants; a static district table holding, for each of the 197 districts, its geographic descriptors, its population, and its regional risk threshold; and a metadata file recording the feature lists of the two model variants, the construction-time medians used to fill any missing feature at inference, the source-reconciliation mapping, and the validated performance scores. Persisting these artefacts together allows the operational service to reproduce the exact feature space and decision threshold of training without re-executing the experimental pipeline.



3.3.3 Implementation of the Climate Ingestion Layer

The ingestion layer was implemented as a reusable component that produces the canonical climate table irrespective of the origin of the measurements, so that the same downstream pipeline serves the historical reanalysis used for training and the live sensor stream used at deployment. It defines a common daily schema behind which interchangeable source adapters are registered, so that the operational source is selected by configuration alone; the reanalysis services are wrapped directly, while the sensor network delivers its readings through the file-based adapter, into which a gateway writes the schema the pipeline expects. The transformations that the raw sensor output requires before it matches that schema are presented next.

3.3.4 Presentation of the IoT Sensor Data

To make the operational data used by the prediction pipeline concrete, the data emitted by the IoT sensor network are presented here in their raw form, as collected, and in the form they assume after preprocessing by the ingestion layer. Each reading contains the environmental measurements provided by the on-board sensors together with metadata associated with the mesh transmission. During ingestion, the environmental variables are extracted and retained in the canonical format required by the prediction pipeline. A representative sample of the raw environmental readings from the Yaoundé deployment is shown in Table 3.2.

The IoT records used in this illustration contain **1,567 environmental readings** collected from the sensor nodes. These readings are contained within **3,139 received packets**, which include both environmental measurements and communication-related information. The sensor operates at a nominal reporting interval of **180 seconds**, with each environmental record containing the measured temperature, relative humidity, atmospheric pressure, and rainfall variables. The resulting records constitute the operational input format from which the ingestion layer extracts the environmental features required for inference. Table 3.3 presents the corresponding preprocessed representation used by the downstream prediction pipeline.

Table 3.2: Sample of the raw environmental readings emitted by the IoT sensor node at the Yaoundé deployment site

Temp. (°C)	Hum. (%)	Press. (hPa)
29.07	56.48	928.15
29.00	56.43	928.15
28.91	56.42	927.90
28.84	56.59	927.80
28.76	56.85	927.73
28.72	57.02	927.76

Three characteristics of the raw stream require treatment before it can be used, and each is handled by the ingestion adapter. First, the readings are acquired at a sub-hourly cadence, whereas the models operate at the monthly resolution of the malaria records, so they are aggregated, the state variables by their mean and rainfall by accumulation, first to the daily resolution presented here and ultimately to the monthly resolution of training. Second, where several nodes report on the same topic the adapter averages them at each timestamp, so that a site is represented by a single time series irrespective of how many nodes serve it. Third, the sensor family reports rainfall as an instantaneous intensity in



millimetres per hour, whereas the historical training variable is an accumulated depth in millimetres; the adapter converts the intensity to an accumulation by multiplying each reading by the sampling interval and summing over the aggregation window, so that sensor rainfall becomes directly comparable to the historical precipitation. The rain gauge was not fitted to the node used for this demonstration, and precipitation is therefore supplied by the reanalysis feed in the interim; the channel is nevertheless retained in the canonical schema and in the adapter, because the operational deployment is to be instrumented for it. The result of applying these transformations is shown in Table 3.3, in which the sub-daily readings become daily records in the canonical schema expected by the model pipeline. Precipitation does not appear in that table: it is carried by the schema but supplied by the reanalysis feed until the gauge is fitted.

Table 3.3: Preprocessed IoT environmental data in the model-ready training format

Location	Temp. (°C)	Hum. (%)	Press. (hPa)
Yaoundé 1	30.46	56.14	927.41
Yaoundé 1	30.92	58.96	927.70

The preprocessed records carry the environmental variables that constitute the feature space of the trained models, expressed in the same units and at the same resolution as the historical training data. This is the concrete manifestation of the design constraint stated in the methodology: the live sensor stream, once preprocessed, is indistinguishable in form from the historical reanalysis on which the models were trained, which is what allows the trained models to be served on it without modification.

3.3.5 Implementation of the Inference Service

The operational inference service was implemented with the FastAPI framework, loading the deployment bundle once at start-up and exposing the prediction functionality over a set of endpoints. On receiving the environmental measurements for a district, it reconstructs the training feature vector, runs the two-model Recursive Chain, derives the incidence rate and the discrete risk class against the district threshold, and attaches the feature attribution of the prediction. A relational schema persists the measurements, the engineered feature vectors, the predictions and the attributions, so that any prediction can be reproduced and re-explained after the fact.

3.3.6 Operational Pipeline Structure

Beyond serving individual predictions, the operational layer implements the recurring workflow required to keep a deployed model current. Two properties of that layer are established. A source-abstraction layer decouples the pipeline from any single climate provider, so that the operational source can be switched by configuration alone, and the same feature-engineering code is shared between training and serving, which eliminates *training-serving skew*, the silent failure mode in which the features computed at inference diverge from those seen during training. A scheduler drives the pipeline autonomously, running a monthly forecasting cycle and polling for newly arrived data at a shorter interval. Predictions therefore reach the system by two deliberately different paths. The *interactive* path, triggered by an operator from the dashboard, runs the full two-model Recursive Chain for one district over the requested months and returns the SHAP explanation of each prediction, because a case history is then either supplied with the request or held in the database. The *autonomous* path, triggered by



the scheduler, scores every district for a single month with the bootstrap model alone and persists the results, because no reported case history exists at run time.

A drift-monitoring component is specified alongside these, separating data drift on the model inputs, detectable without waiting for ground truth, from performance drift, detectable only once cases are reported. It is *not* reported here as a result, because it was never exercised: the operational store contains served predictions, engineered feature vectors and attributions, but no pipeline run, no data-quality record and no drift report. The component is therefore described among the perspectives of the Conclusion, as specified and implemented but not yet demonstrated, rather than presented as a validated capability of the system.

3.4 Results

This section presents the performance of the developed models. The results are organised following the main steps of the modelling process: first, the impact of the different feature groups is examined; next, the models are evaluated under both temporal and spatial generalisation settings; then, the results of the regression and classification tasks are reported; finally, the explainability of the models and their practical deployment performance under recursive forecasting are analysed.

3.4.1 Contribution of the Feature Regimes

The first result quantifies the marginal contribution of each feature family by comparing the three nested regimes under random cross-validation, for both tasks. Figure 3.4 reports the progression. The raw climate variables alone are weak predictors, the regression explaining only about a fifth of the variance of the target ($R^2 = 0.183$) and the classification reaching an $F1$ of 0.503, only modestly above chance. Adding the engineered features, namely the seasonal encoding, the geographic descriptors, and the climate lags, produces the largest single improvement, lifting the regression R^2 to 0.774 and the classification $F1$ to 0.736, which establishes that the predictive signal is carried jointly by season, geography, and the delayed effect of climate rather than by instantaneous climate alone. Adding the case-history features improves the models further, to $R^2 = 0.843$ and $F1 = 0.759$, reflecting the strong month-to-month autocorrelation of malaria transmission.

These four figures are obtained under the random partition, which allows a model to be evaluated on the same period it was fitted on, and they are therefore leakage-prone and should be read as an upper bound rather than as an operating point. Recomputed under the time-aware partition on a single common frame, the same progression runs from $R^2 = 0.134$ to 0.641 and then 0.768, and from $F1 = 0.466$ to 0.666 and then 0.692. The ordering of the three regimes is unchanged, so the conclusion drawn above survives; the levels do not, and it is the time-aware figures that are carried into the Abstract. Under that partition the raw-climate regression has a fold-to-fold deviation as large as its own mean, which is to say that raw monthly climate carries no dependable signal out of time at all.

One point of arithmetic must be settled, because it otherwise reads as a contradiction with Table 3.4: the case-history regression reaches $R^2 = 0.843$ here and 0.941 there, under the same random partition. The two are computed on *different targets*, the variance-stabilised incidence rate here and the variance-stabilised case count there, the latter being the quantity the Recursive Chain feeds back into itself. Case counts are the easier target, since dividing by a district population adds between-district variance, and re-running both on one frame confirms the gap as an artefact of that choice.

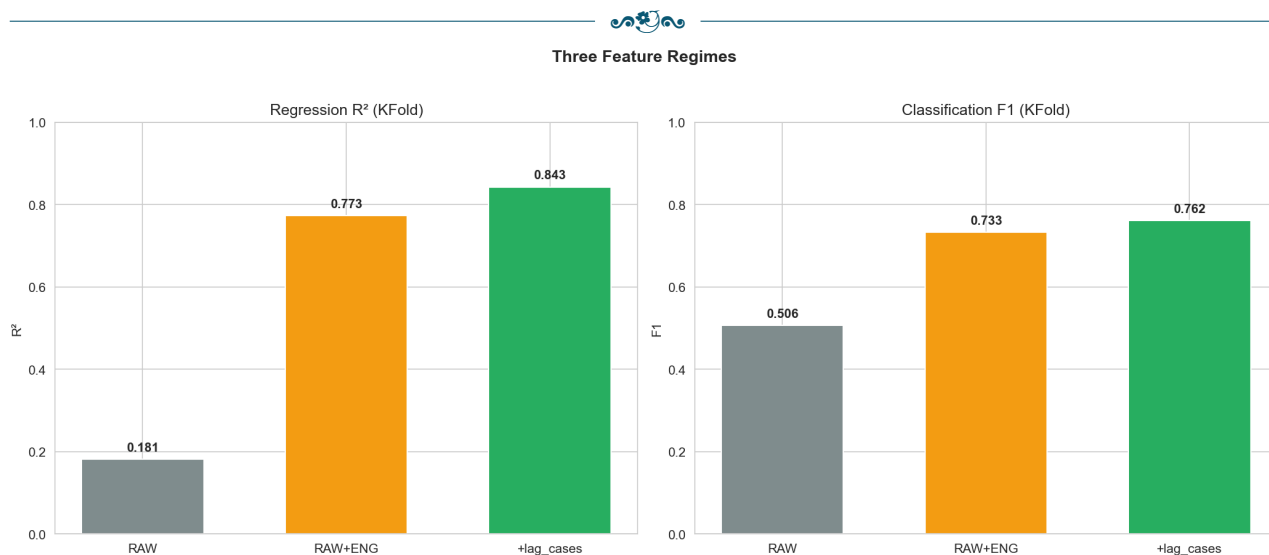


Figure 3.4: Performance progression across the three feature regimes, for the regression and the classification tasks

3.4.2 Temporal Generalisation and Leakage

Because the random cross-validation of the previous subsection allows the model to see the same time period it is evaluated on, its scores overstate the performance attainable when the model must forecast forward in time. The difference between the random and the time-aware cross-validation measures this temporal leakage, and is reported in Figure 3.5 for the full panel of classifiers under the engineered feature set. Every model scores higher under the random partition than under the time-aware one, and the gap is the amount by which a leakage-prone evaluation would flatter an operational forecaster. The gradient-boosted ensembles retain the leading positions under both regimes, which confirms the algorithm choice of the construction phase.

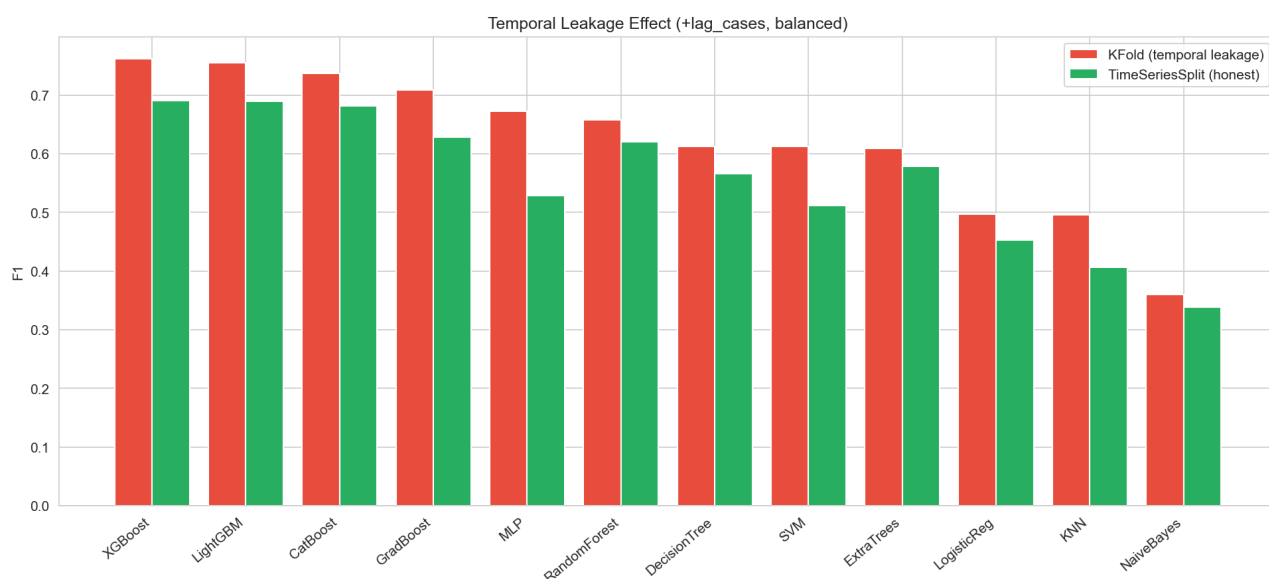


Figure 3.5: Temporal leakage: classification $F1$ under random and time-aware cross-validation

3.4.3 Performance Envelope across Validation Regimes

Reporting the four validation regimes together provides a complete performance range for the model, from its most optimistic in-sample behaviour to its more conservative performance on unseen time periods and unseen regions. Figure 3.6 illustrates this performance envelope for the classification task.

The envelope runs from the raw baseline under random cross-validation, the weakest starting point, through the engineered set under the same partition, which gives the most optimistic in-sample estimate, to the time-aware results and finally to leave-one-region-out, the most difficult setting. For practical early-warning deployment in districts already observed during training, the time-aware result, not the in-sample estimate, is the operating point of the system.

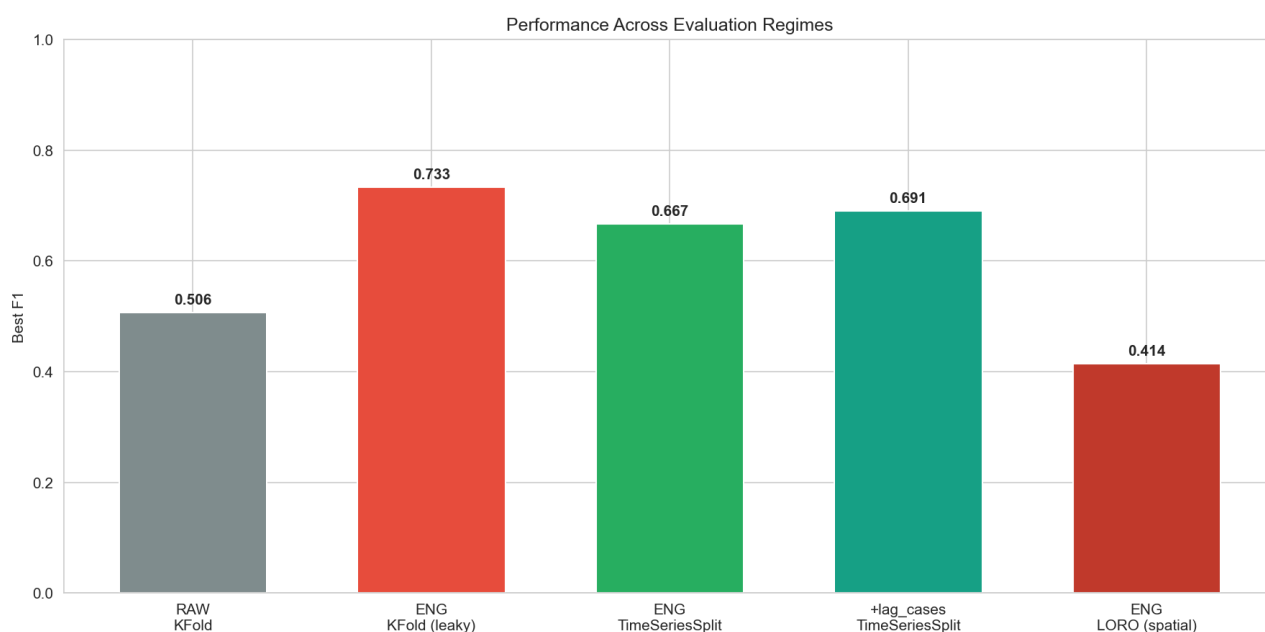


Figure 3.6: Performance envelope of the classification task across the four validation regimes

3.4.4 Performance of the Deployed Models

The performance of the final deployed models, in both their bootstrap and case-history variants, is summarised in Table 3.4 under random cross-validation (KFold) and the more realistic time-aware cross-validation (TimeSeriesSplit); each entry is reported as the mean across folds $\pm$ the fold-to-fold standard deviation. The classification models carry the operational task: they predict the binary risk level on which an alert is issued, and are evaluated using the $F1$ score, which is the appropriate measure given the imbalance of the target classes. The regression models estimate the variance-stabilised malaria case counts, are evaluated using the coefficient of determination (R^2), and serve two subsidiary purposes, namely as a corroborative cross-check on the classification result and as the component that the Recursive Chain of Subsection 3.4.8 feeds back into itself. They are reported here once, in full, and are referred to rather than re-narrated thereafter.

Interpretation: The regression models achieve strong predictive performance, and the drop from random cross-validation to the time-aware setting remains relatively small. This suggests that the models generalise well when predicting future months in districts already observed during training. The fold-to-fold dispersion, however, is not uniformly negligible, and it must be read alongside the means. Under the random partition it is small for every model, at most 0.014. Under the time-aware partition

Table 3.4: Performance of the final deployed models under random and time-aware cross-validation

Model	Random CV	Time-aware CV
<i>Classification — the operational task</i>		
Classifier, bootstrap (no case lags)	$F1 = 0.742 \pm 0.014$	$F1 = 0.668 \pm 0.076$
Classifier, case history	$F1 = 0.769 \pm 0.011$	$F1 = 0.697 \pm 0.059$
<i>Regression — cross-check and recursive component</i>		
Regressor, bootstrap (no case lags)	$R^2 = 0.898 \pm 0.008$	$R^2 = 0.839 \pm 0.059$
Regressor, case history	$R^2 = 0.941 \pm 0.004$	$R^2 = 0.915 \pm 0.021$

it is an order of magnitude larger, reaching 0.076 for the bootstrap classifier, which exceeds the 0.029 that separates the two classifiers from one another. The two classifiers are therefore *not* distinguishable under time-aware validation, and the apparent gain from case-history features in the classification task cannot be asserted on this evidence; for the regression task the between-model gap of 0.076 does exceed the largest fold deviation of 0.059, but only narrowly. What the table supports is that both tasks degrade modestly out of time, not that the models can be finely ranked against one another.

3.4.5 Is the Risk Label Time-Varying?

A classification score of this kind invites a specific objection, and it is serious enough to be answered directly rather than left to the reader. If the binary risk label were a near-static property of the district, so that a district sat permanently above or permanently below its regional threshold, then a model could attain a high $F1$ merely by learning *which* districts are high-burden, and the score would measure spatial sorting rather than early warning. Since geography dominates the feature attribution reported in Subsection 3.4.7, the objection cannot be dismissed on inspection.

Three measurements settle it. First, the label does vary in time for most districts: over the 36 months, **154 of the 197 districts**, or 78.2 percent, cross their regional threshold at least once, while 39 remain always low and only 4 remain always high. Second, a variance decomposition of the binary label attributes **36.7 percent** of its variance to district identity and **63.3 percent** to month-to-month variation within districts, an intra-class correlation of 0.350; the majority of the signal is temporal. Third, and most directly, a trivial baseline that predicts each district's modal class in every month, with the mode learned on the training fold alone and evaluated on the same time-aware folds as Table 3.4, attains $F1 = 0.480 \pm 0.079$, against 0.657 ± 0.086 for the classifier. The model's advantage over a per-district constant is therefore $+0.178 F1$, and restricting the evaluation to the 154 districts that actually vary leaves it at 0.627 against 0.422 for the same baseline.

The objection is thus not sustained in its strong form: the classifier is learning temporal structure, not only spatial position. It should nonetheless be recorded that the trivial per-district constant already recovers roughly three-quarters of the model's $F1$, so a substantial part of the headline score reflects knowing which districts are habitually at risk, and that this baseline had not previously been reported. An early-warning system whose advantage over "assume each district behaves as it usually does" is 0.18 of an $F1$ point is useful, but its value should be claimed at that magnitude and not above it.

Because malaria transmission is highly autocorrelated, and the case-history model directly uses lagged case information, the interpretation of its stronger performance is examined more carefully in Subsection 3.4.6, where it is compared against simple persistence and seasonal baselines. By contrast, the bootstrap model relies only on environmental and spatial information, making it the clearer measure of the predictive value of those features alone.

The classification task is naturally more challenging, since reducing malaria risk to a binary label removes much of the information retained in the regression setting. Even so, the models still achieve a practical and usable $F1$ score on future months. Across both regression and classification tasks, the case-history variants consistently outperform the bootstrap variants, confirming the importance of recent transmission history. The practical cost of making this advantage deployable, however, is addressed in the recursive forecasting analysis of Subsection 3.4.8.

The confusion matrix of the final classifier on a continuous block of unseen future months is shown in Figure 3.7, together with its receiver operating characteristic curve. The results show that the model is able to identify most of the truly high-risk district-months while maintaining a reasonable false-alarm rate. The ROC curve, which remains clearly above the diagonal, further confirms that the classifier retains meaningful discriminative power on out-of-time data.

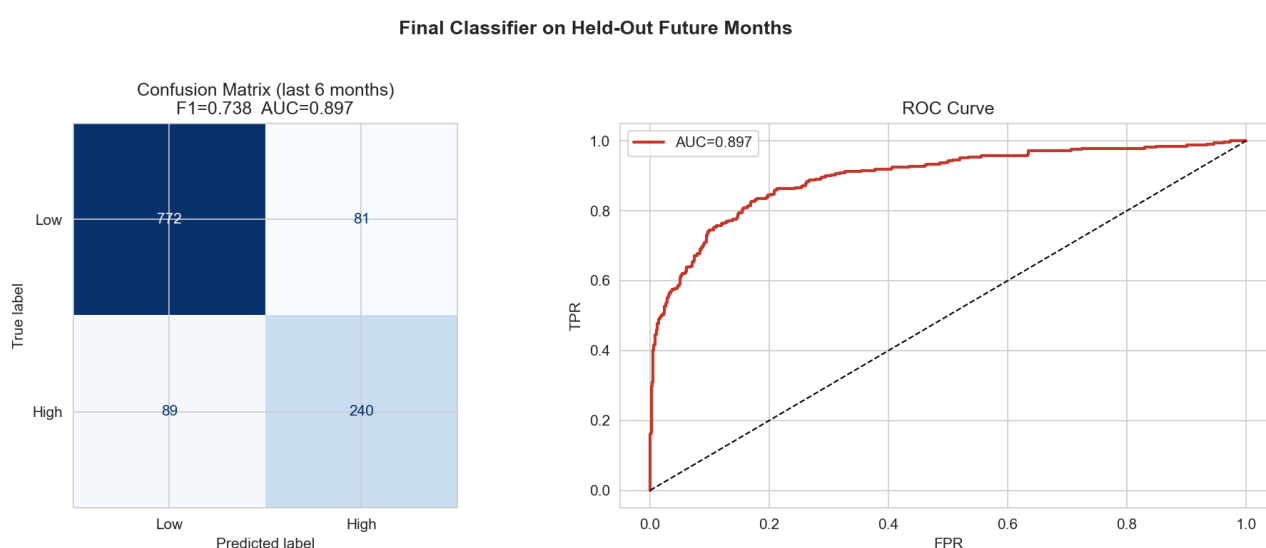


Figure 3.7: Confusion matrix and ROC curve of the final classifier on a block of held-out future months

3.4.6 Baselines: Persistence and the Environmental Signal

The high coefficients of determination reported above invite a necessary control. Because malaria transmission is strongly autocorrelated from one month to the next, a share of the explained variance may reflect mere persistence, namely the tendency of next month to resemble this month, rather than a genuine environmental forecast. To measure this directly, two naive baselines were evaluated on the same time-aware holdout as the models: a persistence baseline that predicts each district-month as its previous month's value, $\hat{y}_t = y_{t-1}$, and a seasonal baseline that predicts the mean of the same calendar month per district. Table 3.5 reports the two baselines against the environment-only model, on both the case-count and the incidence target, with performance measured by R^2 on the log scale.

Table 3.5: Environment-only model against naive persistence and seasonal baselines

Predictor	R^2 (log_cases)	R^2 (log_incidence)
Persistence ($\hat{y}_t = y_{t-1}$)	0.892	0.751
Seasonal mean (district $\times$ month)	0.799	0.533
Environment-only (engineered, no case lags)	0.836	0.623

Interpretation: The comparison highlights an important and transparent result. For predicting



case counts, the persistence baseline achieves an R^2 of 0.892, which is higher than the environment-only model at 0.836. A similar pattern appears for incidence prediction, where persistence reaches 0.751 compared with 0.623 for the environment-only model. However, the environment-only model still performs better than the seasonal baseline on both targets, improving by 0.037 for case counts and 0.090 for incidence.

A note on two closely related figures is warranted, since they concern the same model under different evaluation blocks. The environment-only regressor is the bootstrap regressor of Table 3.4; its time-aware score there is $R^2 = 0.839$, whereas the same model on the baseline holdout of Table 3.5 is $R^2 = 0.836$. The small difference reflects the distinct evaluation blocks of the two experiments, the tuned time-aware cross-validation in the one case and the fixed baseline holdout in the other, and not two different models; the two figures are consistent to within the dispersion of the estimate.

This shows that recent malaria transmission history is the strongest predictor of next-month malaria, carrying more predictive power than environmental and spatial features alone. In practice, this means that any model using real case-history lags is naturally difficult to outperform with environmental inputs only.

At the same time, this does not reduce the importance of environmental variables. Their performance above the seasonal baseline, together with the attribution analysis, shows that they provide meaningful epidemiological information and help explain how geographic and historical patterns shape transmission. It also carries a decisive operational implication that motivates the design of the deployed system: persistence, though the strongest predictor, is not available in a sensor-only setting, since it requires the previous month's confirmed case count, which the latency of case reporting withholds at the moment a forecast is needed. The environmental model, by contrast, depends only on quantities the sensor network supplies in near-real time. The value of the environmental signal is therefore not that it beats persistence in the abstract, but that it is deployable where persistence is not; and the recursive procedure of Subsection 3.4.8 is precisely the mechanism that recovers the persistence signal synthetically, from the model's own predictions, when the measured lag is unavailable.

The leave-one-region-out regime tests whether a model trained on the observed regions can predict a region absent from training, by training on nine of the ten regions and evaluating on the tenth, in turn for each region. The per-region results, the regression R^2 and the classification $F1$ of each held-out region, are reported in Table 3.6, where the regions are ordered by $F1$, and visualised in Figure 3.8. The pattern is clear: the northern, drier regions generalise best, the held-out Nord region reaching a classification $F1$ of 0.644 and a positive regression R^2 of 0.402, followed by Extreme Nord at an $F1$ of 0.633. The southern and coastal regions generalise worst, the held-out Littoral region falling to an $F1$ of 0.180 with a strongly negative R^2 of -0.931 , and the regression R^2 turning negative on eight of the ten held-out regions. Averaged across the regions, the leave-one-region-out classification $F1$ is 0.372 ± 0.158 and the regression R^2 is -0.461 ± 0.574 , both markedly below the time-aware estimates of the previous subsections.

One qualification governs how the classification figure should be read, and it makes the result more pessimistic rather than less. As stated in Subsection 2.3.2, the per-region risk threshold is computed over the whole panel, so the held-out region's labels are defined by that region's own historical incidence distribution. The evaluation therefore grants the model a quantity a genuinely new deployment would not have, namely the local distribution against which "high" is defined, and it fixes the held-out region's

positive class at a quarter of its district-months by construction. Recomputing the same experiment with a threshold estimated only from the nine training regions lowers the mean held-out $F1$ by approximately 0.07, and the effect is largest precisely where the regional baseline is most unusual: for the held-out Nord Ouest region, whose own threshold is far below the national level, the positive class falls from a quarter of its district-months to under a tenth and the score collapses. The figure of 0.372 should accordingly be understood as spatial transfer *given local calibration*, which is an upper bound on cold-start deployment into an uncharacterised region.

Table 3.6: Leave-one-region-out spatial generalisation per held-out region

Held-out region	# districts	R^2	$F1$
Nord	15	+0.402	0.644
Extreme Nord	32	+0.107	0.633
Sud Ouest	19	−0.528	0.399
Nord Ouest	20	−0.822	0.397
Est	15	−0.254	0.395
Ouest	20	−0.015	0.279
Sud	10	−1.584	0.277
Centre	32	−0.359	0.270
Adamaoua	10	−0.626	0.245
Littoral	24	−0.931	0.180
Mean $\pm$ SD		−0.461 $\pm$ 0.574	0.372 $\pm$ 0.158

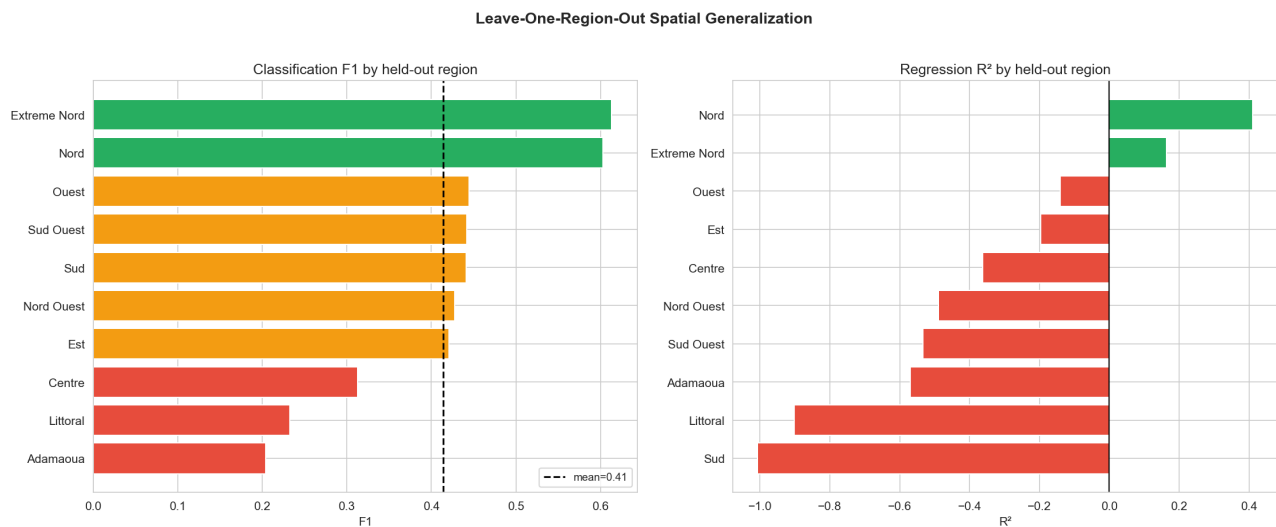


Figure 3.8: Leave-one-region-out spatial generalisation: per-region classification $F1$ and regression R^2

Interpretation: The difference between the time-aware evaluation, which predicts future months in districts already seen during training, and the leave-one-region-out evaluation, which tests the model on a completely unseen region, reveals one of the most important findings of this work. It shows that the model relies heavily on the geographic identity of each district and does not transfer reliably to regions with no historical case records.

The stronger performance in the northern regions, compared to the weaker results in the coastal and southern regions, aligns with the ecological differences observed during the exploratory analysis. Regions with drier climates and more regular seasonal patterns appear easier to predict from one another, while the equatorial coastal regions, which have distinct climatic conditions and more complex transmission



patterns, are harder to generalise from the rest of the country.

A methodological caveat qualifies this reading, and is stated so that the finding is not over-interpreted. The worst-generalising regions, Littoral and Sud, are also among the largest and most ecologically heterogeneous, and are therefore precisely those for which a single district centroid, from which the climate series was retrieved, represents the internal climate least faithfully. Part of the measured spatial failure may consequently reflect the coarseness of the centroid climate representation in those regions rather than an intrinsic non-transferability of the epidemiological signal. Disentangling the two would require multiple climate points per district, aggregated to the district unit, and re-running the leave-one-region-out evaluation on that finer climate; this is identified as outstanding work. The finding as reported should therefore be read as an upper bound on the spatial-transfer difficulty attributable to the model, part of which the climate resolution may explain.

From an operational perspective, this means that deploying the model in a new region cannot simply rely on direct transfer. Instead, a local calibration period is needed to collect some historical data before the model can make reliable predictions there.

3.4.7 Explainability

The global feature attribution of the case-history regression model is shown in Figure 3.9, which ranks the features by their mean absolute Shapley contribution. The ranking is consistent with the validation findings and with malaria epidemiology. The lagged case count is the single strongest driver, which reflects the autocorrelation of transmission and explains the gain of the case-history regime. The geographic descriptors, namely longitude, latitude, and area, occupy the next positions, which is the attributional counterpart of the spatial dependence exposed by the leave-one-region-out result. The climate variables, in particular pressure and the temperature and humidity lags, contribute a smaller but non-negligible signal, consistent with an environmental modulation of transmission acting on top of a geographic and historical baseline. Notably, among the environmental variables it is the lagged temperature that carries the leading signal, in agreement with the exploratory correlation analysis of Subsection 3.1.3.2 and with the temperature dependence of the vector and parasite development cycles.

Cross-task consistency: To test whether the identified drivers are a property of the phenomenon rather than of a particular target formulation, the native feature importances of the tuned regression and classification models, both gradient-boosted ensembles, are compared in Figure 3.10, which shows the regression model on the left and the classification model on the right, with the features grouped by family. The two models are trained on different targets, the continuous case count and the binary risk class, and optimise different objectives, yet they agree on the dominant structure: the geographic features, namely area, longitude, latitude, and the ecological cluster, occupy the leading positions in both. The principal difference is that the classifier weights the seasonal encoding, the sine and cosine of the month, more heavily, ranking it among its top features, whereas for the regressor the seasonal terms sit just below the geographic block. Both models then rank pressure and the lagged climate variables as the secondary, environmental tier. The reading is therefore one of substantial agreement on the dominant geographic drivers, with a task-dependent difference in the weight given to seasonality: the regression, which predicts a magnitude, leans on the stable geographic baseline, while the classification, which predicts a threshold crossing, draws more on the seasonal timing of those crossings. This convergence should be read as a consistency check rather than as validation that the signal is genuine. Both models are gradient-boosted trees fitted to the same features on the same data, so a degree of agreement on the

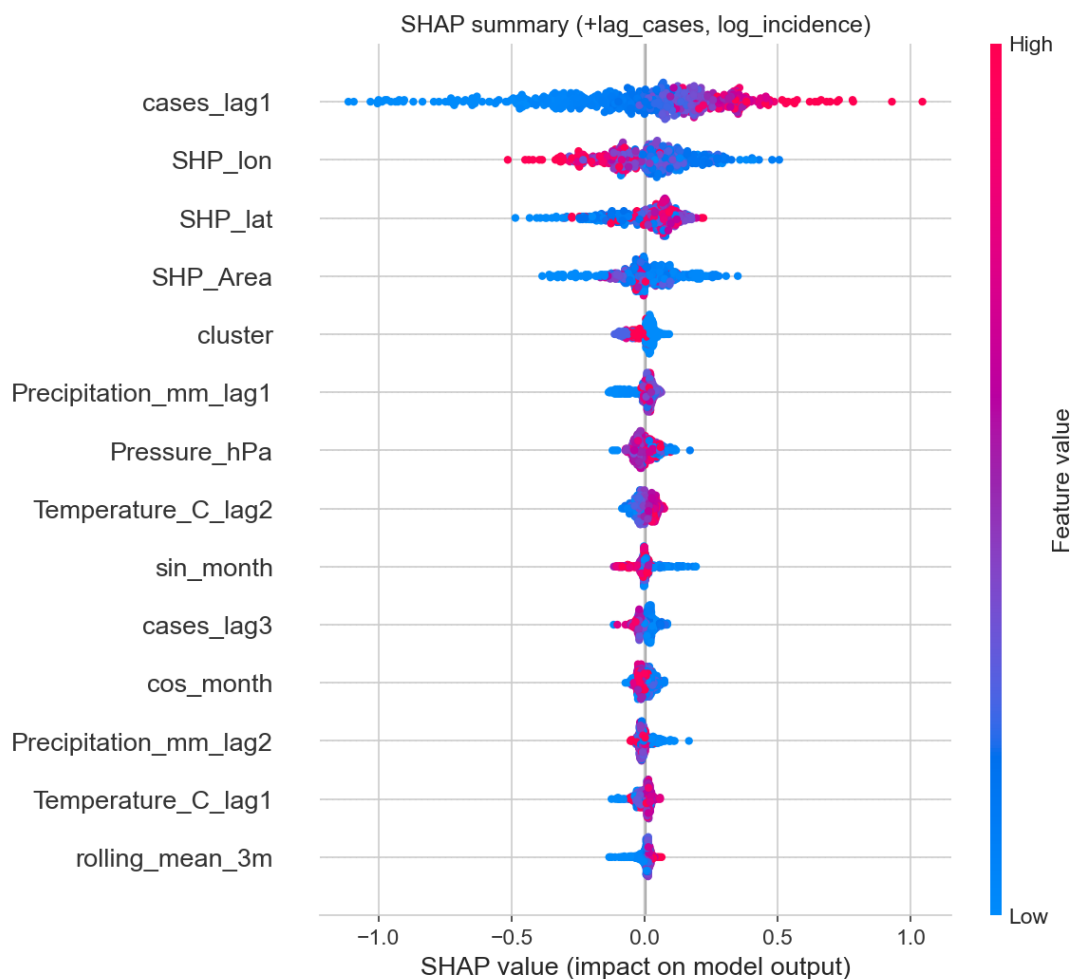


Figure 3.9: Global feature attribution (mean absolute SHAP value) of the case-history regression model

dominant features is expected by construction and does not by itself discriminate a genuine phenomenon from a shared dataset artefact. The comparison here moreover uses native feature importance, which, as noted in the review of methods, is biased toward high-cardinality and continuous features, precisely the geographic variables that dominate the ranking; a comparison conducted in Shapley space, which neutralizes that bias, would be the stronger instrument and is the appropriate form of this check for future work. The convergence is therefore reported as corroborating evidence consistent with a stable geographic-and-seasonal signal, not as independent proof of it.

3.4.8 Recursive Forecasting and Deployment Performance

The case-history models attain their performance with the real case lags supplied, which is not the situation at deployment, where no measured case counts are available. The recursive forecasting procedure, which feeds the model's own predictions back as the case lags, was evaluated on the forward holdout to measure the realistic deployment performance. Figure 3.11 contrasts the oracle, in which the real case lags are supplied, with the Recursive Chain, in which the lags are synthetic, and shows how the error of each behaves across the forecast horizon.

Interpretation: The oracle attains a coefficient of determination of 0.927 on the held-out months, with a mean absolute error of 285 cases, and the Recursive Chain, operating on its own predictions as a deployed system would, attains 0.875, with a mean absolute error of 321 cases, a degradation of

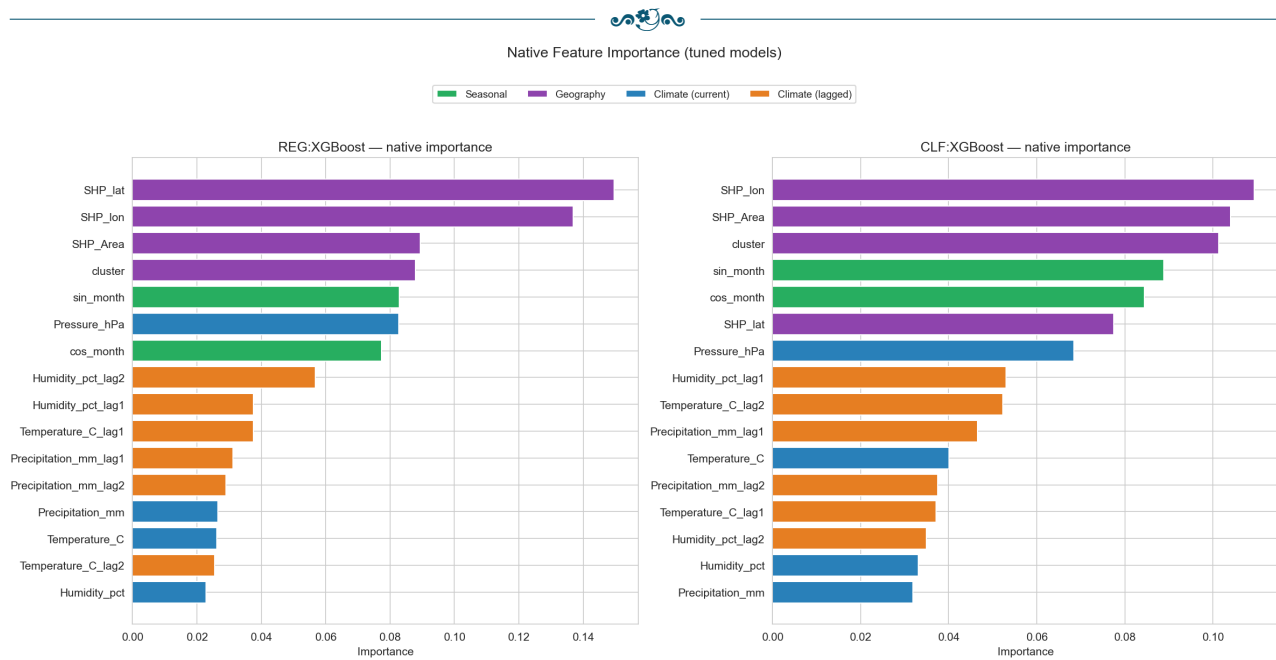


Figure 3.10: Cross-task native feature importance: regression versus classification

about 0.05 in R^2 . The absolute error of both chains falls across the six held-out months, because that window covers the post-peak seasonal decline over which mean monthly case volume drops by close to a half; horizon and calendar month are confounded here, so the absolute error cannot be read as a measure of forecast degradation. Isolating the gap between the Recursive Chain and the oracle removes that confound, and what remains is a penalty for consuming one's own predictions that is positive on average, of the order of fifteen cases of mean absolute error per additional month ahead, but that is *not* monotonic: across the six available horizons the gap is erratic and the trend is weak ($r = 0.27$). The defensible reading is therefore that the chain pays a cost which does not vanish as the horizon lengthens, and that six horizons are too few to characterise how that cost accumulates. The recursive figure, not the oracle, is the honest deployment number for a sensor-only system, and its proximity to the oracle establishes that the recursive procedure is a viable substitute for measured case history over a short forecast horizon.

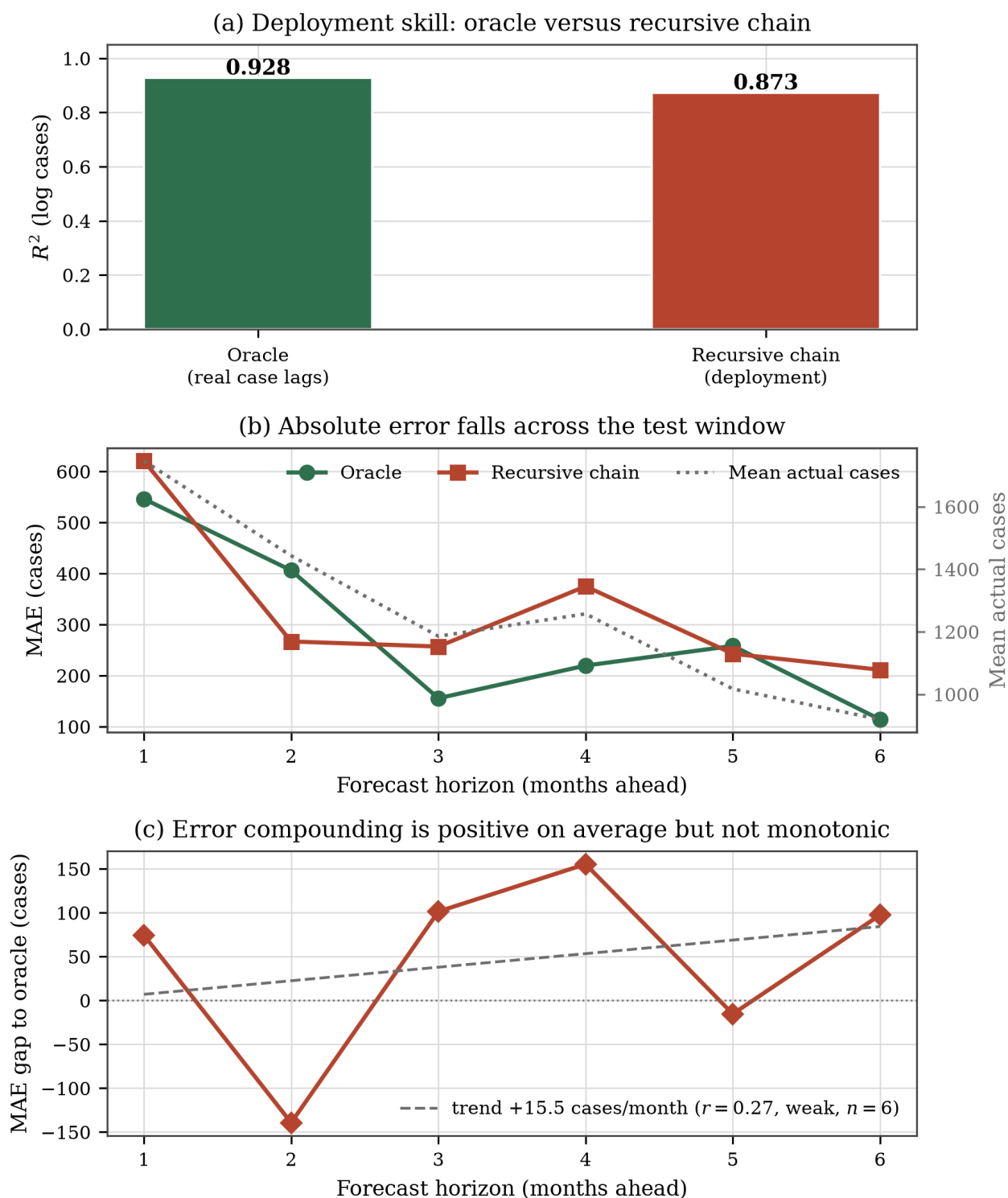


Figure 3.11: Forecast accuracy under oracle and recursive deployment scenarios

3.5 External Validation on Unseen Years (2023–2025)

The results of Section 3.4 are obtained by cross-validation within the 2019–2022 training window. However careful, an internal estimate can still flatter a model that will ultimately operate on data drawn from a later period. The most demanding test of an operational forecaster is therefore genuinely unseen data. This section reports four experiments conducted on real PNLN records from **2023–2025**, entirely outside the training window, that respectively (i) confront the frozen models with real ground truth in a single district, (ii) extend the test to all ten regions, (iii) quantify how a deployed model must be maintained over time, and (iv) verify the single-global-model design choice. The climate covariates for 2023–2025 were retrieved from the same Open-Meteo (ERA5) source used for training, so that the reanalysis product remains consistent; over the overlapping days they agree with the existing series to within 0.57°C (temperature, $r = 0.995$), 1.06 % (humidity), 2.21 hPa (pressure), and 1.61 mm (precipitation).

3.5.1 Single-District External Test: Fouban 2023–2025

The frozen models were scored against 36 months of real PNLN data for Fouban (Ouest region), a district that was itself present in the training set. The results, set beside the internal time-aware cross-validation reference, are reported in Table 3.7, in which the regression R^2 is computed on the log-incidence scale.

Table 3.7: External test on Fouban 2023–2025 against the internal reference

Metric	External (Fouban 2023–25)	Internal (time-aware)
Regression R^2 (log-incidence)	−1.74	+0.628
MAPE	31.4%	30.8%
Mean incidence (observed vs. predicted)	7.59 vs. 9.73 (+28%)	—
Classification $F1$	0.250	0.655
Recall / Precision	0.75 / 0.15	—
ROC-AUC	0.664	0.845

Interpretation: The external test fails on absolute accuracy: the regressor over-predicts mean incidence by 28% (9.73 against an observed 7.59 per thousand) and returns a strongly negative R^2 , while the classifier retains only weak ranking ability (ROC-AUC 0.664) at an operationally unusable precision of 0.15. Crucially, this is not a spatial-transfer failure, since Fouban was in the training set and sat at the 67th national percentile: it is *temporal drift*. As Figure 3.12 shows, Fouban’s mean incidence declined from 9.67 per thousand over 2019–2022 to 7.59 over 2023–2025, a 21.5% fall that closely matches the +28% over-prediction. The left panel of the figure marks both period means (9.7 and 7.6) on the national training distribution, the centre panel shows the monthly incidence series and the real decline between the training and test periods, and the right panel compares the seasonal profile of the two periods. The drift is also largely an artefact of scoring each month from climate alone: supplying the real case history that Fouban actually reports (teacher forcing) cuts the mean absolute error from 664 to 261 cases and lifts R^2 from −3.2 to +0.05. The operational lesson is unambiguous: internal cross-validation, however careful, substantially overstates real deployment accuracy, and a climate-only model must be re-anchored on recent local case history to remain calibrated.

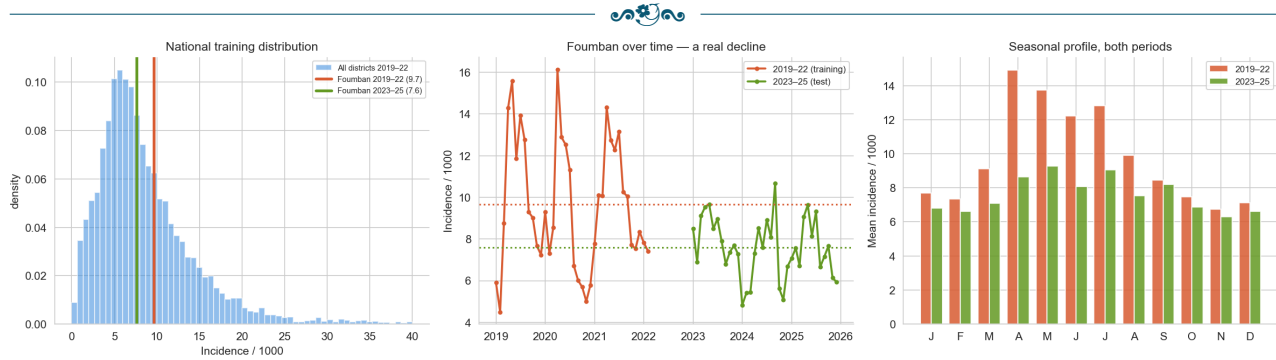


Figure 3.12: Temporal drift at Fouban between the training and test periods

3.5.2 National Rank-Transfer Validation

To test spatial transfer at scale, the frozen models were evaluated against DHIS2 case records for **194 districts across all ten regions** over 2024 and 2025 (388 district-years). Because the available DHIS2 indicator counts confirmed *simple* cases only, roughly half of the total-confirmed definition on which the models were trained (a median predicted-to-observed ratio of $1.80\times$), absolute error is not comparable across the two sources. The evaluation is therefore conducted on *ranks*, which are invariant to a constant scale factor; the rank statistics are reported in Table 3.8 and Figure 3.13. The figure plots, on the left, the predicted against the observed burden rank; in the centre, the within-region rank agreement; and, on the right, the absolute levels of the two sources, whose median $1.80\times$ gap reflects the difference in indicators and is the reason the ranking, not the level, is evaluated.

Table 3.8: National rank-transfer validation across 194 districts

Statistic	Value	Chance baseline
Spearman ρ (pooled)	+0.859	0.00
Kendall τ (pooled)	+0.669	0.00
Top-25% targeting recall	0.679	0.25
Within-region ρ (median)	+0.708	0.00
Between-region ρ	+0.915	0.00
Ground-truth year-to-year stability	+0.971	—

Interpretation: On genuinely unseen data the model ranks district burden well: a pooled Spearman ρ of 0.859, and it places 68% of the truly worst-quartile districts in its own worst quartile, against 25% expected by chance. This materially qualifies the pessimistic reading of the internal leave-one-region-out result (mean $F1$ 0.372, mean R^2 -0.461): that regime measured absolute per-month classification, whereas targeting decisions rest on relative annual ranking, and the latter transfers. The between-region agreement ($\rho = 0.915$) exceeds the within-region agreement ($\rho = 0.708$, ranging from Littoral at 0.94 down to Adamaoua at 0.26), which is the expected signature of geography-dominated importance: the model knows which *region* is worse more reliably than which district within it. Two caveats bound the result. First, the ground truth is 97% stable year to year ($\rho = 0.971$), so simply ranking districts by last year’s reported cases scores higher than the model; the model’s value is not that it beats surveillance but that it needs *no* case reporting at all, which is precisely the situation in under-reporting districts. Second, the annual granularity of the DHIS2 export precludes any test of seasonality or monthly accuracy.

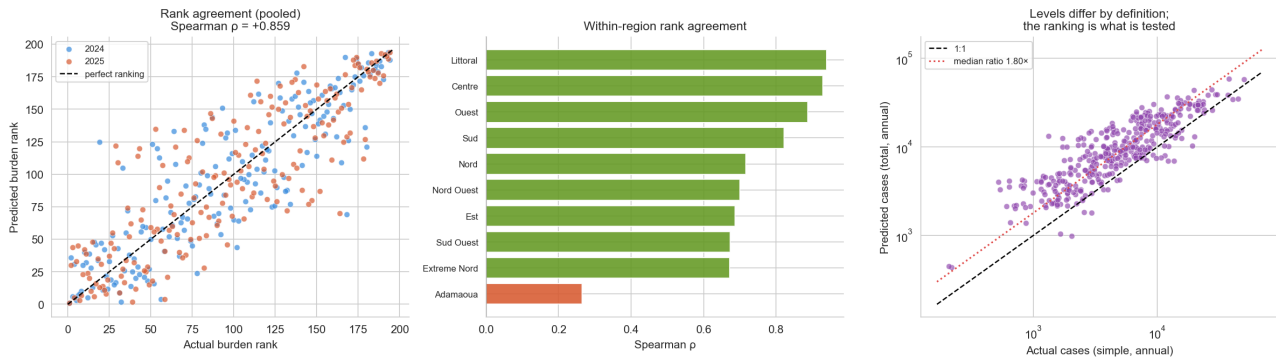


Figure 3.13: National rank-transfer validation on unseen 2024–2025 data

3.5.3 Model-Update Strategy under Drift

The Fouban test shows that a frozen model drifts out of calibration. The natural follow-up question is how a deployed model should be maintained. Five maintenance strategies were compared on Fouban over four successive six-month update cycles, each strategy re-using a different amount of newly arrived local data: leaving the model *frozen*; *recalibrating* only its output scale; *fine-tuning* the existing model on the new window; *local* retraining from scratch on local data; and a *hybrid* retrain on pooled global and local data. The mean absolute error and the absolute systematic bias of each are reported in Table 3.9 and Figure 3.14; for both measures, lower values are better.

Table 3.9: Model-maintenance strategies under temporal drift

Update strategy	Mean MAE (cases)	Absolute bias (%)
Fine-tuning	218	7.7
Recalibration	271	9.7
Hybrid retrain	288	11.1
Local retrain	327	10.0
Frozen (no update)	434	22.6

Interpretation: Every update strategy beats leaving the model frozen, and periodic *fine-tuning* on a short window of recent local case history is the clear winner, halving the error relative to the frozen baseline (mean absolute error 218 against 434 cases, a 50% reduction) and removing most of the systematic bias (from 22.6% down to 7.7%). Figure 3.14 traces the same result: its left panel shows the error across successive updates, its centre panel the mean error by strategy, from fine-tuning (218) to frozen (434), and its right panel the systematic bias, which the frozen model alone fails to correct. This turns the qualitative “local calibration” recommendation that follows from the leave-one-region-out and Fouban results into a concrete operational prescription: a deployed model should never be left unattended, and a light periodic fine-tune, rather than a full local retrain, is both the cheapest and the most accurate maintenance policy.

3.5.4 Global versus Regional Models

A final experiment tests the architectural choice of a single global model against the alternatives of one model per region and one model per ecological cluster, evaluated on the same folds. The three approaches are, for practical purposes, indistinguishable: the global model reaches a regression R^2 of 0.685, the per-region models 0.676, and the per-cluster models 0.689, a largest gap of only 0.012 in R^2 ,

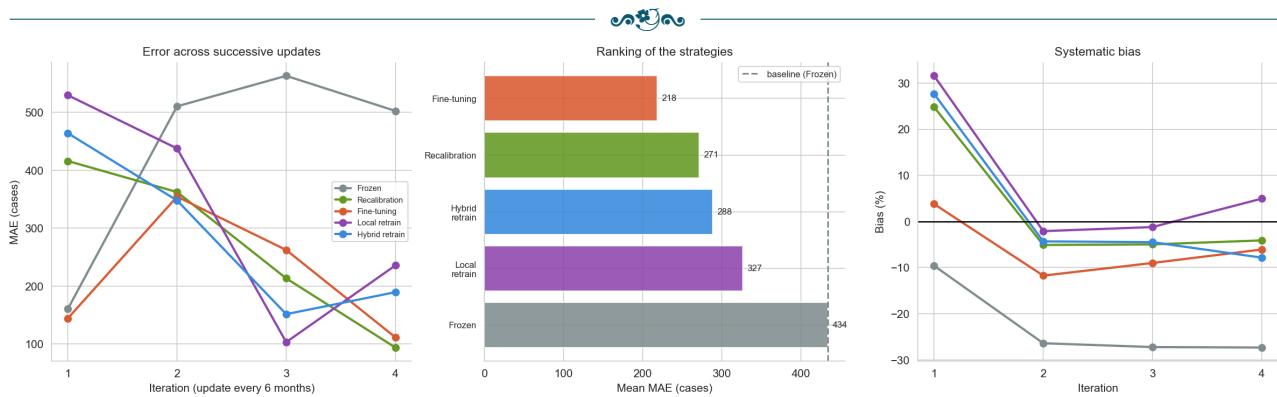


Figure 3.14: Model-maintenance strategies over successive update cycles

as shown in Figure 3.15. The left panel of that figure compares the overall regression R^2 of the three approaches, the centre panel gives the per-region gain or loss of the specialised models over the global model, and the right panel relates that gain to the data volume available per region. Region-specific models win in five regions and lose in the others, most sharply in Est (-0.29 in R^2), and the region-by-region gain does not correlate with the volume of data available per region ($r = 0.12$), which rules out “more local data” as a systematic advantage. The single global model is therefore retained: it matches the accuracy of the specialised alternatives while pooling all districts into one estimator that is simpler to train, tune, and deploy.

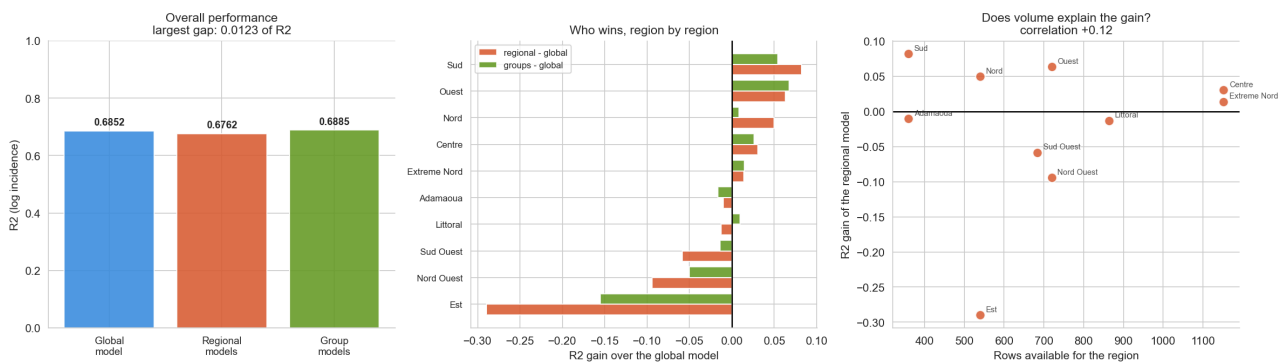


Figure 3.15: Global versus per-region versus per-cluster models

Taken together, these four experiments qualify the internal results of Section 3.4 rather than overturn them. The model’s *absolute* accuracy does not survive contact with unseen years, as the Fouban test shows; its *relative* spatial ranking does, as the national validation shows; the resulting drift is correctable by periodic fine-tuning; and a single global model is the right architecture. These are the honest measures of predictive accuracy, against which the deployment demonstration that follows must be read.

3.6 Results of the Deployment

The trained models, assembled into the deployment bundle, are operated through the inference service and an interactive cartographic dashboard. This section presents the operational behaviour of the system on the live deployment data of the Yaoundé network, for the May–June 2026 forecast horizon. It is important to state at the outset what this section does and does not establish. The PNL records used to *train* the models end in February 2022; although later PNL records enabled the external

validation of Section 3.5, no malaria case data are available for the 2026 deployment window itself. The 2026 figures shown here are therefore model outputs that have not been, and at present cannot be, validated against observed case counts; their correctness can only be confirmed retrospectively, once PNL case data for the deployment period become available. This section is consequently a systems demonstration, establishing that the end-to-end pipeline ingests live sensor data, runs the Recursive Chain, derives the risk class, attaches an explanation, and renders the result, rather than a predictive validation, which was the role of the cross-validated results of Section 3.4 and the external validation of Section 3.5. The accuracy claims of this chapter rest on those internally and externally validated results, computed where ground truth exists; the 2026 figures illustrate operation and explanation, not predictive accuracy.

3.6.1 Per-Prediction Explanation on the 2026 Forecast

Figure 3.16 shows the local feature attribution of the bootstrap (no-case-lag) model for the May–June 2026 forecasts, the model that initiates the Recursive Chain from environmental inputs alone. The attribution confirms, at the level of an individual operational prediction, the pattern established on the training data: the geographic features, namely area, the ecological cluster, and latitude, dominate each prediction, while the environmental inputs, namely pressure and the humidity and temperature lags, contribute the smaller, modulating share. This is the explanatory output that accompanies every forecast the service issues, rendering each risk figure interpretable rather than opaque.

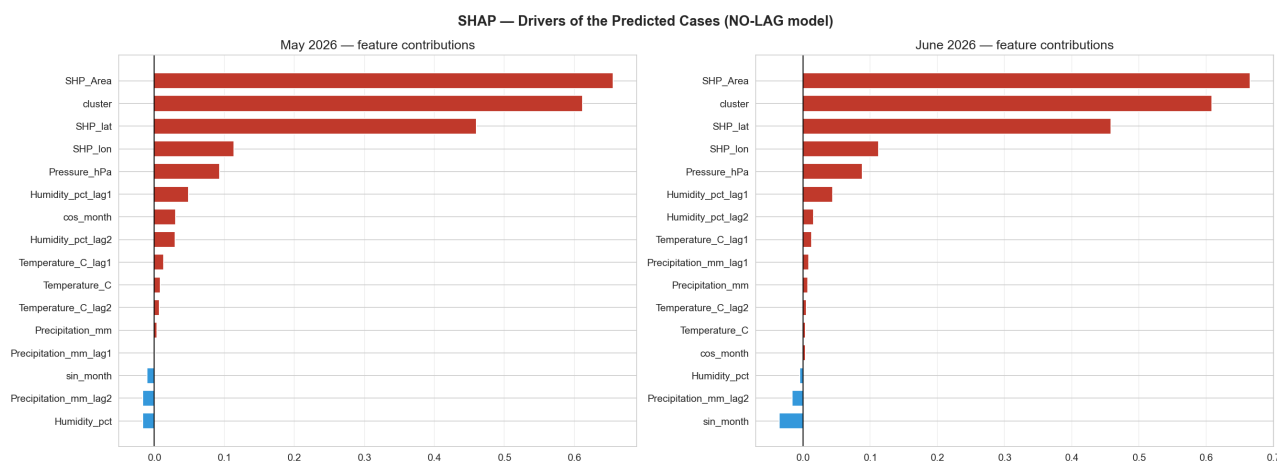


Figure 3.16: Per-prediction SHAP attribution, May–June 2026 forecast

3.6.2 The Early-Warning Dashboard

The end of the pipeline is an interactive dashboard that renders the district-level forecast on a map, shown in Figure 3.17 for the Yaoundé districts over the May–June 2026 horizon. Each district is coloured by its predicted risk class, and a side panel reports, for the selected district, the predicted case count, the derived incidence, the probability of the high-risk class, the environmental inputs, and the ranked feature attribution that explains the prediction.

The Biyem Assi panel in the figure highlights a particularly important operational case because its predicted incidence falls close to the decision threshold. The regression chain predicts 5,529 cases, which corresponds to an incidence of 13.47 per thousand for the district population. This value falls

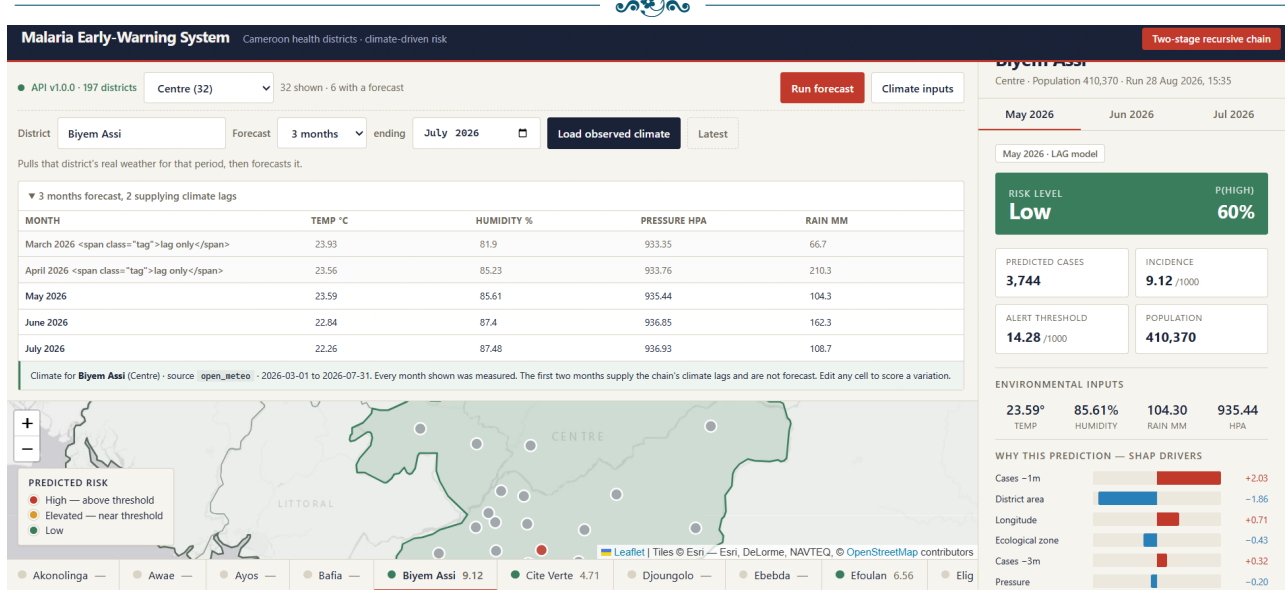


Figure 3.17: Malaria early-warning dashboard, Yaoundé, May–June 2026

slightly below the district threshold of 14.28 per thousand, leading to a low-risk classification from the regression output. In contrast, the classifier estimates a 76 % probability of the high-risk class.

The dashboard resolves this as a *near-threshold* case rather than forcing a confident binary label: Biyem Assi is drawn in the amber *elevated* tier between low and high in Figure 3.17, because the predicted incidence lies within the narrow margin of the threshold where a small change of input, or of the model estimate itself, would move it across. The classifier’s 76% probability of the high class reinforces this reading, indicating that a crossing into high risk is more likely than not.

A district whose predicted incidence sits this close to its threshold should not be given a rigid label. The system therefore surfaces it as an elevated case requiring closer monitoring, combining the magnitude estimate of the regression with the probability of the classifier so that a marginal, uncertain risk zone is made explicit rather than hidden behind a single hard cut-off.

3.6.3 Modes of Operation

Across these scenarios the service exhibits the graceful degradation built into the methodology. When recent case counts are available for a district, they override the synthetic lag and re-anchor the Recursive Chain, and the subsequent predictions are conditioned on observed rather than synthetic history. When they are not, the chain runs autonomously from environmental inputs alone. The system thus operates across districts with differing levels of case surveillance, from the fully observed mode to the fully autonomous mode, which is the property required of an early-warning system intended for uneven operational conditions.

3.7 Discussion of the Results

The systematic evaluation reveals significant differences in performance between the feature regimes, the validation regimes, and the two tasks. Rather than merely noting these differences, this discussion identifies the factors that explain them, verifies the hypotheses that guided the work, situates the contribution against the state of the art, and closes with the limitations of the work.



3.7.1 Explanatory Factors of the Performance Differences

3.7.1.1 Factor 1: The Structure of the Predictive Signal

The predictive signal is not carried by instantaneous climate, which alone is weak, but jointly by geography, seasonality and the delayed effect of climate. Two independent lines of evidence agree on this, the improvement from the raw to the engineered regime and the dominance of the geographic descriptors in the attribution. Malaria risk in this setting therefore has a strong structural and spatial component on which environmental variation acts as a modulation, which is what the epidemiology predicts: the baseline level of transmission is set by stable ecological conditions and varies seasonally around it.

3.7.1.2 Factor 2: Temporal versus Spatial Generalisation

The most pronounced performance difference is between the temporal and the spatial generalisation regimes. The model predicts future months in known districts well, with a time-aware classification $F1$ near 0.70, but transfers poorly to entirely unseen regions, where the mean leave-one-region-out $F1$ falls to 0.372 and the regression R^2 becomes negative. The factor responsible is the same geographic dependence that gives the model its accuracy in known districts: by anchoring its prediction on the geographic identity of a district, the model gains accuracy where that identity is known and loses it where it is not. The per-region breakdown sharpens this: transfer succeeds between the ecologically similar northern regions and fails on the distinct coastal and southern regimes. As noted in the leave-one-region-out interpretation, part of this measured spatial failure may additionally reflect the single-centroid climate representation in the largest, most heterogeneous districts, a confound whose separation is left to future work; the finding is best read as an upper bound on the model-attributable spatial difficulty. Either way, this is not a defect to be concealed but a property to be managed operationally, through local calibration in new regions. The external validation of Section 3.5 sharpens this reading: on real 2024–2025 data across 194 districts the model's *absolute* predictions did not transfer, in line with the leave-one-region-out result, yet its *relative* ranking of district burden did (Spearman $\rho = 0.859$), so a spatially non-transferable model still holds operational value for targeting rather than for absolute calibration.

3.7.1.3 Factor 3: The Cost of Case-History Features

The case-history features are the strongest single driver, but they are not available at deployment, and their apparent benefit is therefore partly an artefact of evaluating with real lags. The recursive forecasting result shows that when the lags are supplied synthetically, as they must be in service, the benefit is largely retained over a short horizon, the recursive R^2 of 0.875 standing about 0.05 below the oracle, with the shortfall against the oracle widening as the horizon lengthens. The factor governing the usable forecast range is thus the accumulation of recursive error, which sets a practical bound on how far ahead the case-history model can be trusted before the bootstrap, climate-only model becomes preferable.

3.7.2 Discharge of the Objectives and Verification of the Hypotheses

The two hypotheses formulated in the general introduction guided the experimental approach and are assessed here against the results obtained. They are preceded by the discharge of the objectives,



which the introduction distinguishes from the hypotheses: an objective is a deliverable, shown to have been built, whereas a hypothesis is a proposition that the evidence could have refuted.

Discharge of the objectives. The end-to-end deployment objective required a running pipeline from raw environmental measurement to a district-level risk alert, and it was delivered. The ingestion layer demonstrated, on a representative sensor export from the Yaoundé deployment, that the raw sub-daily readings are transformed into the exact feature space of the historical training data, namely the four environmental variables in the same units and at the same monthly resolution that the models consume, as shown in the passage from Table 3.2 to Table 3.3; once preprocessed, the sensor stream is indistinguishable in form from the historical reanalysis, so the trained models are served on it without modification. Downstream of ingestion, the inference service runs the two-model Recursive Chain to produce the predicted cases, the incidence, and the risk class, the explainability layer attaches the feature attribution, and the records are persisted and exposed through the dashboard demonstrated on the 2026 forecast.

Three boundaries bound that claim, and are stated here rather than blurred. First, schema compatibility is not empirical validity: that the served predictions are *correct* cannot be established here, since it would require ground-truth case data for the deployment period, which do not yet exist, and it depends on the proxy validation across all four variables that the methodology flagged as outstanding. Second, the lead time is bounded rather than unlimited, because the case lags are unavailable at inference and the alerts are therefore generated by the recursive chain, whose shortfall against the oracle widens with the forecast horizon; within that range the recursive forecast remains close to the oracle, so the anticipation is useful but finite. Third, the IoT network itself is the operational substrate contributed by the concurrent engineering project [16], not a contribution evaluated here. The objectives of predictive modelling and validation and of model explainability are discharged by the dataset construction and the validation architecture of Section 3.4 and by the attribution analysis, whose substantive findings are the subject of the two hypotheses below.

Hypothesis: contextual and lagged features

The predictive signal for district-level malaria risk is carried jointly by geographic context, seasonal structure, and the delayed effect of climate, rather than by instantaneous climate conditions. Ensemble models trained on these engineered features therefore achieve superior predictive performance relative to models trained on raw climate variables alone, under both classification and regression formulations.

This hypothesis is confirmed, in both of its parts. The progression across the feature regimes shows that the engineered features produce the largest single improvement over raw climate, in both tasks: under time-aware cross-validation the regression R^2 rises from 0.134 to 0.641 and the classification $F1$ from 0.466 to 0.666 when the seasonal, geographic, and lagged-climate features are added to the raw climate variables, and the corresponding random-partition figures, 0.183 to 0.774 and 0.503 to 0.736, show the same ordering at a leakage-inflated level. That raw climate alone is the weaker predictor is the second part of the claim, and it holds sharply: out of time, the raw-climate regression has a fold-to-fold deviation as large as its own mean. The attribution analysis identifies the same structure from the other direction, with the geographic descriptors occupying the leading positions and the climate terms entering below the seasonal encoding. The improvement holds under both the classification and the regression formulation, which is the quantitative justification for the feature engineering stage.



Hypothesis: epidemiological coherence of the attributions

SHAP-based feature attributions applied to the final selected model identify environmental drivers that are consistent with established epidemiological theory regarding *Anopheles* breeding and *Plasmodium falciparum* transmission dynamics.

This hypothesis is confirmed. The relevant evidence is the ordering *within* the environmental tier of the SHAP attribution: lagged temperature is the leading environmental driver, ahead of the humidity and precipitation terms, and it acts with a delay rather than instantaneously. This is exactly the relationship established in the exploratory analysis and expected from the temperature dependence of the *Anopheles* and *Plasmodium falciparum* development cycles, in which environmental change precedes the vector-population response by a lag. The attribution therefore recovers biologically interpretable environmental drivers rather than incidental structure. It should be read in context: geography and case history dominate the overall ranking, so the environmental drivers are modulators on a structural baseline rather than the primary signal; the hypothesis, however, concerns whether the environmental drivers that do appear are epidemiologically coherent, and on that question the lagged-temperature result answers in the affirmative. The agreement of the attributions across the classification and regression formulations, examined in the explainability results and with the caveats stated there, provides a supporting consistency check.

3.7.3 Comparison with the State of the Art

This section situates the contribution against existing work on machine learning for malaria prediction, highlighting the methodological specificities and the advances of the present work.

Axis 1: validation versus optimistic reporting : A substantial part of the literature reports malaria-prediction performance under random cross-validation, which, as the temporal-leakage result of this work shows, overstates the performance attainable in operational forecasting. Table 3.10 situates the present work, which reports the full envelope from the optimistic in-sample estimate to the spatial-generalisation limit, against representative works that report a single favourable figure.

Axes 2 to 4: deployability, delivery, and explanation: Three further distinctions follow from the same deployment orientation. Much of the literature trains on, and assumes the availability of, recent case counts, which a sensor-only early-warning setting does not have; the Recursive Chain constructed here forecasts from environmental inputs alone and *quantifies* what that costs. Where comparable studies remain at the level of retrospective analysis, this work delivers a running inference service behind a reusable ingestion layer. And the feature-attribution layer explains every prediction rather than the model in aggregate, with the agreement of the attributions across the two tasks serving as a consistency check on the identified drivers.

Table 3.10: Comparison of validation strategies in representative malaria prediction studies

Work	Validation approach	Main limitation
Nkiruka et al. (2021) [7]	Used random cross-validation and reported strong AUC performance across six countries	Did not distinguish between temporal and spatial generalisation
Mbunge et al. (2022) [8]	Evaluated ward-level prediction accuracy using remote-sensing environmental variables	Future forecasting performance was not assessed separately
Ly et al. (2025) [6]	Applied time-series cross-validation and achieved high R^2 for malaria prediction in Senegal	Performance on completely unseen districts was not evaluated
This work	Combined random, time-aware, leave-one-region-out, and recursive holdout evaluations to measure performance under different conditions	Provides a clearer and more realistic view of model strengths and spatial limitations

3.7.4 Limitations

The results above define the realistic scope of the work, and four limitations bound it. Each is stated with the remedy that the evidence of this chapter points to.

- (1) **Spatial transfer.** Although the models generalise well across future months in districts already seen during training, they perform poorly when transferred to completely unseen regions. The leave-one-region-out evaluation showed that the models rely heavily on the geographic identity of districts, making spatial transfer the main limitation of the system: predicting new locations remains significantly harder than predicting future time periods. *Remedy:* deployment in a new district should begin with a short calibration phase during which local case history is collected, rather than expecting the model to predict an entirely new region without prior information.
- (2) **Absolute accuracy outside the training window.** On real 2023–2025 PNLP records, the absolute accuracy of the models degraded sharply, the single-district Fouban test giving a negative R^2 and a 28% over-prediction driven by a genuine decline in local transmission, while their relative ranking of district burden transferred well across 194 districts (Spearman $\rho = 0.859$). The 2026 forecast itself remains a systems demonstration until the corresponding case records are published, since no ground truth yet exists for that window. *Remedy:* periodic fine-tuning, which halved the error of a frozen model in the incremental-learning experiments, together with validation of the 2026 forecasts once their ground truth becomes available.
- (3) **The environmental proxy.** The climate reanalysis used for training and the live sensor measurements used for deployment are highly similar, but they are not perfectly interchangeable, and the agreement between the two sources remains to be established; until it is, the equivalence of the two sources is assumed rather than demonstrated. *Remedy:* a period of overlapping sensor and reanalysis observation, over which the source differences can be measured directly.



- (4) **The strength of recent case history.** A simple persistence baseline, based on recent malaria history, remained the strongest predictor of next-month malaria. Autocorrelation in case history therefore carries more predictive power than the environmental variables alone, which positions the environmental factors as important modulators rather than primary drivers. *Remedy*: the Recursive Chain, which reconstructs the case-history signal from the model's own predictions when measured counts are unavailable, and re-anchors on a measured count as soon as one is reported.

3.8 Chapter Summary

This chapter presented the implementation of the methodology and the results obtained from it.

The predictive signal proved to be carried by geographic context, seasonality and lagged climate rather than by instantaneous climate, the engineered features lifting the time-aware regression R^2 from 0.134 to 0.641 and the classification $F1$ from 0.466 to 0.666. Against a per-district constant baseline, which had not previously been reported, the classifier retains an advantage of 0.178 $F1$, so the score measures temporal discrimination and not merely spatial sorting. Performance falls substantially on unseen regions, the mean leave-one-region-out $F1$ reaching 0.372 and that figure itself assuming local calibration, which confirms spatial transfer as the binding constraint, with the caveat that part of the gap may be attributable to the single-centroid climate representation in the largest districts. The external test on real 2023–2025 records both confirmed and qualified this: absolute accuracy degraded out of sample (Foumban $R^2 = -1.74$), driven by temporal drift, whereas the district-burden *ranking* transferred nationwide (Spearman $\rho = 0.859$ over 194 districts), and periodic fine-tuning halved the error of a frozen model.

A deliberate baseline analysis established that recent transmission history, through a simple persistence predictor, is the single strongest signal for next-month malaria, exceeding the environment-only model; the environmental model nonetheless improves on a seasonal baseline and, decisively, is deployable where persistence is not. The Recursive Chain preserves most of the case-history advantage without live case reports, reaching an R^2 of 0.875 against an oracle of 0.927. The attribution analysis recovered epidemiologically coherent environmental drivers, with lagged temperature leading among them, and the 2026 Yaoundé forecast demonstrated the operational behaviour of the system while remaining, as stated there, a systems demonstration and not a predictive validation.

CONCLUSION AND PERSPECTIVES

This work addressed the prediction of district-level malaria transmission risk in Cameroon from environmental and climatic conditions, with the aim of building an explainable early-warning system able to run on live measurements from a ground-based sensor network. It formed part of a broader engineering effort: the Internet of Things infrastructure, from the LoRa sensor nodes to the cloud ingestion chain, was developed by a collaborating engineering student, while the present work covered everything downstream of it, from data integration and predictive modelling to the Presentation Layer that serves the models in operation.

A complete end-to-end pipeline was built and evaluated. Malaria case records from the Programme National de Lutte contre le Paludisme were combined with a historical climate reanalysis into a single dataset of **7,092 district-months**, covering 197 health districts in 10 regions over 36 months. Seasonal, geographic, and lagged-climate features were organised into nested feature sets, so that the contribution of each family could be measured. Regression and classification models were then assessed under four validation regimes, from in-sample interpolation to temporal generalisation, spatial generalisation, and out-of-time forecasting, and explained through Shapley additive explanations. Finally, the models were packaged into an inference service with a cartographic dashboard, behind an ingestion layer that maps live sensor streams onto the feature space used in training.

The results support four main conclusions. First, the predictive signal is carried by geography, seasonality, and lagged climate rather than by instantaneous climate: the engineered features raised the time-aware classification $F1$ from 0.466 to 0.666. Second, under time-aware validation the classifier attains $F1 = 0.668 \pm 0.076$ without case history and 0.697 ± 0.059 with it, exceeding a per-district constant baseline by 0.178, and the Recursive Chain recovers most of the case-history advantage without live case reports, reaching $R^2 = 0.875$ against an oracle of 0.927. Third, the evaluation reports, rather than conceals, where the system fails: the leave-one-region-out $F1$ falls to 0.372, with a negative R^2 on eight of ten regions, and on unseen years at Foumban the frozen models over-predicted incidence by 28 percent. Fourth, the relative ordering of districts survives where absolute calibration does not, predicted and observed burden agreeing at a Spearman correlation of $\rho = 0.859$ across **194 districts**. Throughout, the attributions identified drivers consistent with the epidemiology of transmission, and their agreement across the two tasks strengthened confidence in the predictive signal.

These results point to a staged route to operational use by the PNLP, and to several directions for future work. In districts already represented in the training data, the system can serve immediately as a ranking instrument, prioritising where nets, spraying campaigns, and stocks are directed even where its absolute incidence estimates are not trusted. In a district new to the system, deployment should begin with a calibration phase during which local case history is collected, since cold-start transfer is unreliable and the regional risk threshold itself requires local history. The most urgent next step is to



extend the external validation: monthly, total-definition case records after February 2022 would both lengthen the training window and turn the rank-based national test into a like-for-like evaluation, and the 2026 forecasts should be validated once their ground truth is published. The environmental proxy should be validated formally over an extended period of overlapping sensor and reanalysis observation. The drift-monitoring component, implemented but not yet exercised on a live stream, should be run over a sustained deployment, so as to trigger the periodic fine-tuning that halved the error of a frozen model. Finally, sequence-based architectures such as recurrent and attention-based networks, which did not match the gradient-boosted ensembles on the present short series, may become competitive once the network has accumulated longer operational records.

Overall, this work provides a complete and explainable framework for turning environmental observations into district-level malaria risk predictions, reliable for forecasting within known districts and explicit about its limits in spatial transfer. Building on the sensing infrastructure of the parallel engineering effort, it lays a practical foundation for an operational malaria early-warning system in Cameroon.

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Annexe

REPRODUCIBILITY: ENVIRONMENT, SEEDS, AND HYPERPARAMETERS

This appendix collects the information required to reproduce the results of Chapter 3: the computing environment, the random seeds, the exact library versions, the hyperparameter search space, and the configurations selected.

A.1 Computing environment

The hardware on which all the experiments were run is described in Table A.1, and the software tools used, together with the role each plays in the pipeline, are listed in Table A.2.

Table A.1: Hardware characteristics of the experimental environment

Element	Characteristic
Processor	Intel® Core™ i7-1065G7 CPU @ 1.30 GHz (4 cores, 8 logical processors)
RAM	8 GB
Storage	477 GB Micron NVMe SSD
Operating system	Microsoft Windows 11 Home Single Language (64-bit)

Table A.2: Software tools used and their roles

Tool	Role
Python	High-level programming language for data science and machine learning.
pandas, NumPy	Data manipulation, integration, and numerical operations.
XGBoost, LightGBM	Gradient-boosted decision tree ensembles for the regression and classification tasks.
scikit-learn	Cross-validation utilities, evaluation metrics, and scale-sensitive baseline models.
Optuna	Hyperparameter optimisation by sequential model-based search.
SHAP	Feature attribution for model explainability.
rapidfuzz	Approximate string matching for district-name reconciliation.
Jupyter	Interactive development and reproducible execution of the pipeline.
joblib	Serialisation of the trained models for deployment.
FastAPI	Web framework for the operational inference service.
PostgreSQL	Relational and spatial storage of the operational records.

A.2 Random seeds

A single seed, 42, is fixed globally and passed to every stochastic component: the NumPy global generator, the `random_state` of every scikit-learn and XGBoost estimator, the `random_seed` of the CatBoost estimators, and the seed of the Optuna sampler. The cross-validation splitters that shuffle are constructed with the same value, so the fold assignment is reproducible.

A.3 Library versions

The reported results were produced against the versions of the scientific libraries listed in Table A.3.

Table A.3: Versions of the scientific libraries

Package	Version	Package	Version
pandas	2.3.3	optuna	4.8.0
scikit-learn	1.8.0	shap	0.46.0
xgboost	3.3.0	matplotlib	3.10.9
lightgbm	4.6.0	rapidfuzz	3.12.2
catboost	1.2.10		

A.4 Hyperparameter search space

Tuning was carried out with Optuna's tree-structured Parzen estimator over **25 trials** per model, applied to the two best-performing models per task as ranked on the engineered feature set. The search space is given in Table A.4. The objective was the mean cross-validated score over the inner resampling scheme, which was a random partition; the consequences of that choice are examined in Chapter 3.

Table A.4: Optuna search space for the gradient-boosted ensembles

Hyperparameter	Range	Scale
n_estimators	200 – 700, step 100	integer
max_depth	3 – 8	integer
learning_rate	0.01 – 0.2	logarithmic
subsample	0.6 – 1.0	uniform
colsample_bytree	0.6 – 1.0	uniform

A.5 Selected configurations

Table A.5 gives the configurations selected by the search, together with the objective value attained. Two points should be noted. First, LightGBM attained a marginally higher tuning objective than XGBoost on the regression task ($R^2 = 0.798$ against 0.792); the two are separated by far less than the fold-to-fold dispersion reported in Table 3.4, and XGBoost was retained as the primary model on that basis rather than on a demonstrated advantage. Second, the four models actually persisted in the deployment bundle, whose scores are those of Table 3.4, use the fixed configuration given in the final row rather than the tuned values.

Table A.5: Configurations selected by hyperparameter search, and the fixed configuration of the deployed models

Model	Objective	n_est	depth	lr	subsample	colsample
Regressor (XGBoost)	$R^2 = 0.792$	500	5	0.0620	0.715	0.817
Regressor (LightGBM)	$R^2 = 0.798$	600	8	0.0571	0.704	0.691
Classifier (XGBoost)	$F1 = 0.747$	400	5	0.1095	0.936	0.831
Deployed models (all four)	—	400	5	0.0500	0.800	0.800

The classifiers additionally carry a positive-class weight equal to the ratio of negative to positive instances in the training frame, which is 2.87 on the 35-month case-history frame and 2.92 on the 36-month frame.

A.6 Data availability

The complete pipeline is archived alongside this thesis in the accompanying package `malaria_thesis_package/`. It contains the source data, the notebooks used to produce the figures and tables reported in this thesis, the trained model artefacts and associated metadata, and the operational service. The package also includes the district-to-climate reconciliation described in Subsection 2.2.4, together with the curated correspondences, feature definitions, preprocessing parameters, and recorded validation results. A per-district audit of the climate join, including the approximate matches and districts for which climate series were imputed, is also provided in the package.

ALGORITHM BENCHMARK

Chapter 2 states that the choice of gradient-boosted tree ensembles rests on a broad empirical comparison rather than on assertion. This appendix reports that comparison in full: thirteen regression and twelve classification algorithms, each evaluated under both the random and the time-aware cross-validation regimes. The figures are those of the modelling protocol as executed, and they settle the choice of learning algorithm that accompanies the sub-question on predictive features.

B.1 Regression benchmark

Table B.1 reports the coefficient of determination of thirteen algorithms, on the raw and engineered feature sets, under both validation regimes. The target is the variance-stabilised incidence rate. Models are ordered by their engineered, time-aware R^2 , the most realistic of the four columns.

Table B.1: Regression benchmark across feature sets and validation regimes

Algorithm	Raw climate		Engineered	
	Random	Time-aware	Random	Time-aware
XGBoost	0.194	0.116	0.769	0.625
LightGBM	0.217	0.121	0.769	0.608
CatBoost	0.209	0.144	0.703	0.584
GradBoost	0.195	0.097	0.726	0.581
RandomForest	0.175	0.106	0.601	0.487
ExtraTrees	0.162	0.102	0.555	0.472
KNN	0.130	0.046	0.412	0.317
SVR	0.145	0.076	0.438	0.316
DecisionTree	0.103	-0.253	0.493	0.302
ElasticNet	0.027	-0.031	0.149	0.071
Lasso	0.024	-0.028	0.138	0.066
Ridge	0.029	-0.038	0.155	0.013
LinearReg	0.029	-0.039	0.155	-0.015

Two observations bear on the argument of the thesis. The tree ensembles occupy the leading positions under every combination, and the linear models are near-useless throughout, reaching a negative R^2 out of time on raw climate. And the gap between the raw and engineered columns dwarfs the gap between algorithms within either column: moving from raw to engineered features is worth far more than any choice of learner, which is the central claim of the hypothesis on contextual and lagged features.

B.2 Classification benchmark

Table B.2 reports the $F1$ score of twelve algorithms on the engineered feature set with class weighting, under random and time-aware cross-validation. The final column is the amount by which the random partition flatters each model, that is, the temporal leakage it conceals.

Table B.2: Classification benchmark under random and time-aware cross-validation

Algorithm	Random $F1$	Time-aware $F1$	Leakage gap
XGBoost	0.725	0.651	+0.075
CatBoost	0.706	0.643	+0.064
LightGBM	0.725	0.641	+0.084
GradBoost	0.633	0.572	+0.061
DecisionTree	0.598	0.543	+0.055
RandomForest	0.627	0.537	+0.090
MLP	0.605	0.529	+0.076
SVM	0.543	0.496	+0.047
ExtraTrees	0.555	0.494	+0.061
LogisticReg	0.448	0.381	+0.066
KNN	0.450	0.362	+0.088
NaiveBayes	0.414	0.334	+0.080
Mean leakage gap			+0.071

Every one of the twelve scores higher under the random partition, by a mean of 0.071 $F1$. The leakage is therefore systematic rather than model-specific, which is what justifies treating the time-aware figure as the operating point throughout this work.

The two gradient-boosted implementations, shown in bold, lead both tables and are separated by margins far smaller than the fold-to-fold dispersion reported in Table 3.4. They should be read as jointly best rather than as ranked against one another; the extreme gradient boosting implementation was retained as the primary model.